



→ ANNUAL REPORT 2014

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→ FOREWORD

Tribute to Harald Posch (1955–2015)

Harald Posch, former Chair of the ESA Council, passed away on Thursday 21 May 2015 in Wiener Neustadt, Austria. He was 60.

His death left an immense void and was a tremendous loss to his family, to our Agency and to the European space community. His contribution to Europe's efforts in space cannot be overstated. He was a great servant of ESA, of Europe's space community as a whole and, of course, of his homeland, Austria.

Harald Posch's career in the space sector spanned more than three decades, first in industry with Austrian Aerospace and Österreichische Raumfahrt-und Systemtechnik GmbH, then in governmental service. Since 2005 he led Austria's Aeronautics and Space Agency, part of the Research Promotion Agency (FFG). Here, he was key in shaping Austria's engagement in space and represented his country in international forums and organisations. His natural authority, humility, integrity combined with a keen intelligence and a friendly sense of humour, allowed him to always rally support. Under his leadership, solidarity and common interest always won over divergences and uncertainty.

Within ESA, too, Harald Posch held the most prominent of roles, in particular those of IPC Chair and, from 1 July 2014, Chair of Council. In this latter function, he played a central role in the preparation of the Council meeting at ministerial level held in Luxembourg. He was also instrumental in the setting up of the European Space Policy Institute (ESPI) in Vienna, and served as Chair of its General Assembly since its inception.

All those who met Harald Posch will remember him as warm-hearted, dedicated person, with a robust stature, and a voice as rich and melodious as that of an opera singer. For more than 10 years, we heard him defending the interests of the Agency, its Member States and staff in ESA's official meetings.



We knew 2014 would be a year with challenges aplenty and indeed no one could claim to have been disappointed in that regard. It was, in fact, a year which contained some of the most remarkable achievements in 50 years of European space efforts.

If you were to ask me which number best characterised this past year, I would have to say 50: fifty years of achievements, a fair share of crises too, but above all solutions found and some quite remarkable European programmes.

If, on the other hand, you were to ask me which word best characterised the past year, it would have to be 'success'. From Rosetta and Philae to the Council meeting at ministerial level, from January through to December, this was truly a vintage year. The European launcher family – Ariane, Vega and Soyuz – performed ten successful missions between them, the sole exception being the problem encountered placing the Galileo satellites in orbit during the summer. In this latter instance, much credit is due to those who, through great skill, were able to transform initial failure into success. Thanks to the high-

level competence and experience of the ESA teams in Darmstadt, towards the end of the year the two satellites inserted into the wrong orbit were recovered, their missions modified and they were incorporated into the operations of the GNSS constellation.

The Council meeting at ministerial level that took place on 2 December in Luxembourg was the crowning moment of this quite exceptional year. The Member States are to be commended for their terrific effort in subscribing to the new Ariane 6 programme and confirming the continued exploitation of the International Space Station until 2020. In addition to these two important decisions, subscriptions by Council to existing programmes approached €800 million, bringing the total to just under €16 billion for the two ministerial meetings in 2012 and 2014 taken together – an absolute record!

From my vantage point at the Co-Chair's table in Luxembourg, I witnessed at first hand the difficult discussions going back and forth, but my principal memory is of the remarkable cohesion and solidarity displayed by our Member States that ultimately led to the success of the meeting. Also important to stress is the key political role played by those EU Member States that are not members of ESA and the other organisations represented in an observer capacity at the ministerial meeting in lending their unanimous support to the third Resolution on the evolution of ESA – and quite an impressive evolution it is too! In 2015, Hungary and Estonia are expected to complete their accession to ESA, while Lithuania, Latvia, Slovakia, Slovenia, Bulgaria and finally Croatia will join the list of new countries having signed a Cooperation or a European Cooperating State (ECS) Agreement.

To conclude, this year of truly exceptional achievements and landmark decisions has done much to strengthen ESA and foster a very welcome solidarity among its Member States, but it also showed us an ESA enriched by strong partnerships, a scientific community brimming with confidence and a highly innovative industry.

I wish to thank Professor Woerner for his skilful chairing of the ESA Council in the first part of the year and also to congratulate him most warmly on his appointment as the next ESA Director General. I wish him every success as he seeks to continue the excellent work carried out by his predecessor Jean-Jacques Dordain and to write the next chapter in ESA's history as the true Agency of the future.

Harald Posch
Chair of Council

→ YEAR IN REVIEW



At ESA, 2014 will be remembered as the year when we took stock of what has been achieved over the past 50 years of European space cooperation, as well as looking forward towards the future – and taking the necessary decisions to make our vision of that future possible.

Half a century of European space cooperation was celebrated throughout the year in a number of events in several Member State capitals as well as at ESA establishments. Industrial success, scientific achievements and international cooperation were particularly highlighted.

At the end of the year, key decisions needed to secure ESA's future were tabled to the Council meeting at ministerial level in Luxembourg, carrying forward and continuing the rich legacy of the past. The renewed commitment of Member States expressed in Luxembourg, not least for the development of the new Ariane 6 launcher, was a clear indication of the important role space plays in today's society, and of the important role Europe performs in space. Significant reforms of the European launcher sector, including a stronger involvement of industry, are

among the results stemming from that ministerial meeting.

As the future of the European launcher sector was in the process of being mapped out, ten launches from Europe's Spaceport in Kourou took place over the year. The third flight of the most recent launcher, Vega, was also its first commercial flight and a resounding success.

It was a busy year, filled with operational space missions and all the related activities. There were a number of first-time achievements and challenges, some of which will be remembered long into the future.

One mission in particular captured the attention – hearts and minds, I think it is fair to say – of much of the world, through an unprecedented media following for an ESA mission. Rosetta woke up from deep-space hibernation on 20 January, tensely watched from the control room at ESA's ESOC. The spaceprobe arrived at its target comet on 6 August to accompany the comet on its voyage towards the Sun. Images showed an unexpected 'dual' body in great and intriguing detail. On 12 November, Rosetta launched its Philae lander towards the comet's surface, as the world held its breath. This first in the history of space exploration was made even more captivating through the challenges experienced by the lander as it bounced over the comet before finally coming to rest in the shadow of a cliff. The data sent back from Philae before its batteries were depleted (for the time being) added to those collected by the Rosetta orbiter. Indeed the mission has already produced some spectacular scientific discoveries, such as the type of water found on the comet, which proved different from that on Earth.

Another space science mission particularly worthy of note was Venus Express, which performed a successful and daring aerobraking experiment in June and July before finally being declared ended after eight years of service.

Space applications were well represented throughout the year, notably through the launch in August of the first two Galileo Full Operational Capability satellites. Despite the Soyuz launcher placing these into the wrong orbit, the expertise of, in particular, the ESOC/CNES operations teams made it possible to salvage both satellites with the result that they can be used in the navigation system. In November, transmission of the first navigation signal in space from the first Galileo FOC satellite was established. Prior to that, the In-Orbit Validation of Galileo had been achieved, showing the system working well and ready to further refine its performance as more satellites are added to the growing constellation.

The other EU-led flagship programme, Copernicus, got off the ground with the successful April launch and subsequent start of operations of Sentinel-1A. In November, Sentinel-1A and Alphasat linked up by laser for the first Sentinel image downlink via the European Data Relay System, EDRS, demonstrating the viability of this technology. Other highlights from ESA Earth Observation satellites were the first high-resolution results from Swarm released in June, revealing changes in the magnetic field that protects our planet, and further findings on Antarctic ice sheet loss from CryoSat data.

A number of missions linked to the International Space Station were carried out, starting with the launch in May of ESA astronaut Alexander Gerst's 'Blue Dot' mission. In August, he was able to welcome the last ATV to the Station: ATV *George Lemaître*. Alexander Gerst returned safely to Earth on 10 November, but ESA was soon represented on the ISS crew again through Samantha Cristoforetti performing her 'Futura' mission (flying as an ESA astronaut under an agreement between ASI and NASA). So Europe was well present on the ISS, and many Europeans took note, following the astronauts in rapidly increasing numbers, not least through social media.

This was a year that brought a change in the way the world looks at ESA. The ESA 'brand' grew even stronger as a result. The thanks for that are, above all, due to the cooperation between Member States, industry and institutes, and the highly skilled workforce involved in these challenging and sometimes unavoidably risky space missions.

All in all, 2014 saw space write an important chapter in modern history. One might even go a step further and say history was truly written in space.

Jean-Jacques Dordain
Director General

→ HIGHLIGHTS OF 2014



January

20 – ESA's Rosetta spacecraft woke up from deep-space hibernation chasing down Comet 67P/Churyumov-Gerasimenko



February

19 – ESA selected planet-hunting PLATO mission as ESA's third medium-class science mission for 2024



March

17 – ESA astronaut Thomas Pesquet assigned for a long-duration mission on the ISS in 2016
28 – ESA and CERN signed a cooperation agreement



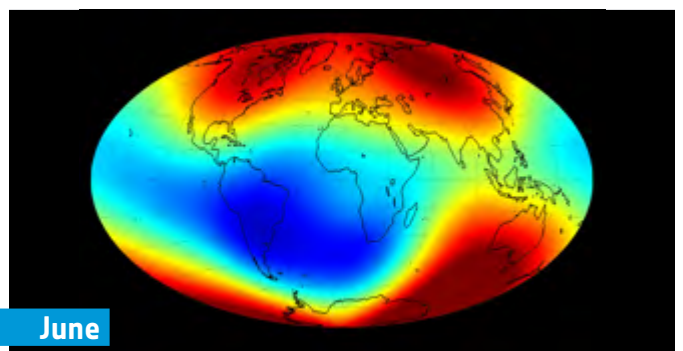
April

3 – Sentinel-1A, first satellite of the Europe's Copernicus environmental monitoring network, launched on Soyuz VS07 from French Guiana



May

28 – ESA astronaut Alexander Gerst launched to ISS (Soyuz TMA-13M) for six months



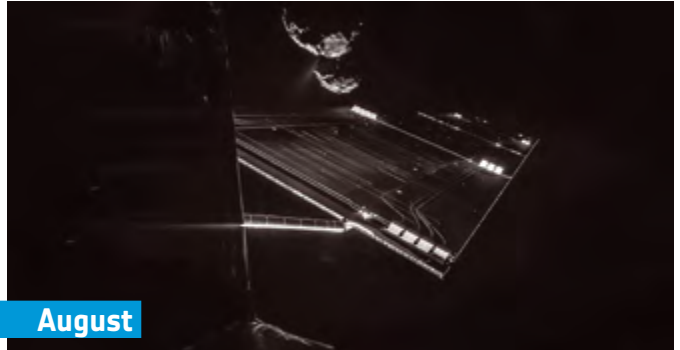
June

19 – First results from Swarm, ESA's magnetic field mission, unveiled at IDA Conference Centre in Copenhagen



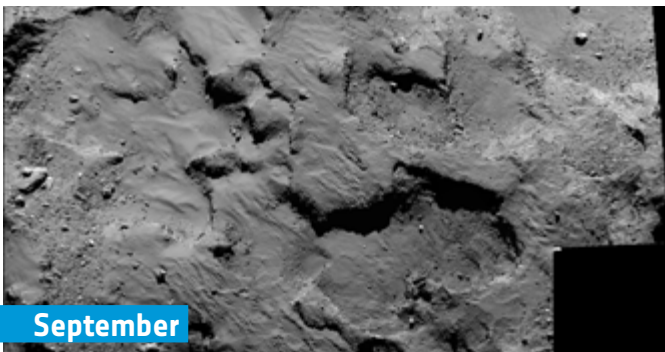
July

- 1** – Harald Posch elected Chairman of ESA Council
- 29** – Fifth and final ATV mission: ATV *Georges Lemaître* lifted off on an Ariane 5 from French Guiana



August

- 6** – Rosetta arrived at Comet 67P/Churyumov-Gerasimenko
- 22** – Galileo 5/6 satellites launched on a Soyuz rocket from French Guiana. The satellites are safely under control, despite having been released on a lower and elliptical orbit instead of the expected circular orbit



September

- 15** – Rosetta primary landing site announced at ESA HQ in Paris



October

- 8** – MetOp-SG industrial contract signed at ESA HQ
- 28** – Agreement signed between ESA and the European Commission to manage Copernicus until 2021



November

- 10** – Alexander Gerst returned to Earth after Blue Dot mission
- 12** – Rosetta mission soft-landed its Philae probe on a comet
- 23** – ESA astronaut Samantha Cristoforetti launched to ISS (Soyuz TMA-15M) for a long-duration mission



December

- 2** – ESA Council meeting at ministerial level in Luxembourg: ministers approved the development of Ariane 6 and Vega C, the European space exploration strategy and ESA evolution, covering the vision for ESA until 2030
- 18** – Council appointed Jan Woerner as Director General, starting 1 July 2015

Tribute to Antonio Fabrizi (1948–2015)

The former Director of Launchers, Antonio Fabrizi, passed away in Rome on Saturday, 16 May 2015. ESA and the European space sector lost an outstanding professional in launchers development and exploitation. He spent 12 years in ESA (2003–15) and will be remembered as a very knowledgeable, loyal, courteous and courageous man. He played a pivotal role in the successful development of the Vega launcher, in the installation of Soyuz at CSG, in building up the reliability of Ariane 5 and in preparing ESA's new Ariane 6 launcher.

→ DIRECTORS



Jean-Jacques Dordain
Director General



Franco Ongaro
Director of Technical & Quality Management



Thomas Reiter
Director of Human Spaceflight & Operations



Antonio Fabrizi
Director of Launchers



Gaele Winters
Director of Corporate Reforms



Volker Liebig
Director of Earth Observation



Didier Faivre
Director of Galileo Programme & Navigation-related Activities



Eric Morel de Westgaver
Director of Procurement, Financial Operations & Legal Affairs



Alvaro Giménez Cañete
Director of Science & Robotic Exploration



Magali Vaissiere
Director of Telecommunications & Integrated Applications



Hans Georg Mockel
Director of Human Resources, Facility Management & Informatics



Giuseppe Morsillo
Director of ESA Policies, Planning & Control





science & robotic exploration

A major highlight was the Rosetta mission, which captured the imagination of the general public and the world's media. But there were also more exciting science results from the fleet of Science Programme missions operated by ESA and its international partners.

Cosmic Vision

The first small mission in the Science Programme, CHEOPS, a small high-precision photometry mission for exoplanets, was adopted in February.

Also in February, the PLATO mission was selected as the third 'Medium' mission (M3) in the Cosmic Vision plan, for a launch in 2024. PLATO will focus on the search for exoplanets down to terrestrial sizes by means of the transit method and studying parent star properties through asteroseismology. Combined with follow-up ground-based observations, it will provide the key information on planet radii, mean densities, stellar irradiation, and the architecture of planetary systems, as well as potential habitability. The industrial Phase-B1 started in October and the mission adoption is planned for mid-2016.

In June, and following a 'Call for the L2 mission', the Athena concept, a large aperture X-ray observatory, was selected for the L2 launch opportunity in 2028. To be implemented under ESA leadership, with contributions from NASA and JAXA, Athena would implement the 'hot and energetic Universe' science theme, involving galaxy formation and evolution, the assembly of baryonic matter into large-scale structures, and the growth and influence of black holes. Phase-0 was completed in October and Phase-A will start in 2015, in parallel with critical technology activities aiming at reaching Technology Readiness Level 6 by the time of mission adoption in early 2020.

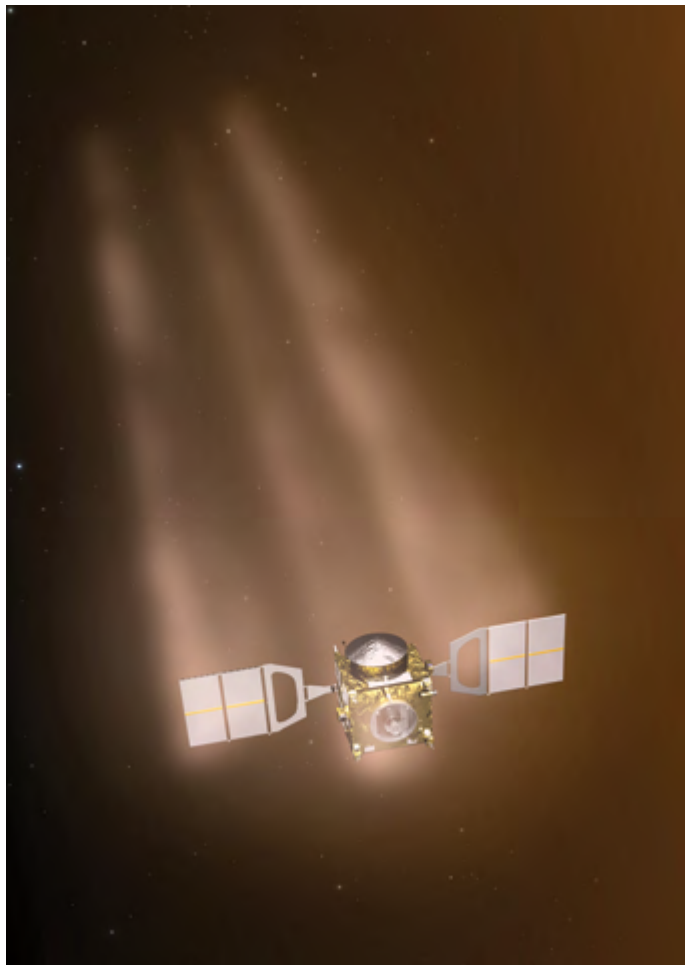
A 'Call for the M4 mission' was issued in August, with a proposal deadline in early 2015.

Following a study phase, the JUICE mission (to Jupiter and its icy moons) was adopted by the Science Programme in November, marking the beginning of its implementation phase. JUICE is the first Cosmic Vision 'Large' mission (L1).

Missions in operation

After a ten-year journey through the Solar System, involving three Earth flybys, one Mars flyby, and encounters with asteroids Steins and Lutetia, Rosetta awoke from deep space hibernation in January, before rendezvousing with Comet 67P/Churyumov-Gerasimenko in August. After just three months studying this extraordinary new world and surveying it to identify a landing site, the Philae probe was deployed to the surface of the nucleus in November. Following an unscheduled journey which saw it bouncing across the comet, Philae came to rest in a shadowed location, and completed its first science

The Rosetta spacecraft snapped a 'selfie' at Comet 67P/Churyumov-Gerasimenko from a distance of about 16 km from the comet on 7 October (ESA/Rosetta/Philae/CIVA)



Venus Express during aerobraking in June and July

sequence before its primary battery was exhausted and it fell into hibernation. The Rosetta orbiter has returned initial scientific results on the physical, chemical, and electrical properties of the comet, and continues to study it from as close as 6 km, witnessing the comet come to life as they move together towards perihelion in August 2015.

Other high-profile missions included Venus Express with its successful aerobraking campaign (June–July) and mission completion (December), together with the transition of Gaia into routine science operations.

Missions in operation at the end of 2014 included astrophysical observatories and surveyors (Gaia, Integral, XMM-Newton, and Hubble Space Telescope with NASA), *in situ* missions to planets and minor bodies in the Solar System (Mars Express, Rosetta, Cassini–Huygens with NASA and ASI), and spacecraft studying the Sun (Proba-2, SOHO with NASA, Hinode with JAXA) and the Sun–Earth connection (Cluster). Post-operational data activities are also continuing for Herschel, Planck and Venus Express.

Science highlights

Cluster has explained the origin of rare, very high-latitude aurorae. These appear as arcs crossing the auroral oval, hence their name, ‘theta aurorae’. By combining results from Cluster and NASA’s IMAGE satellite, it has been possible to discover that plasma is trapped in the magnetosphere and heated by magnetic reconnection.

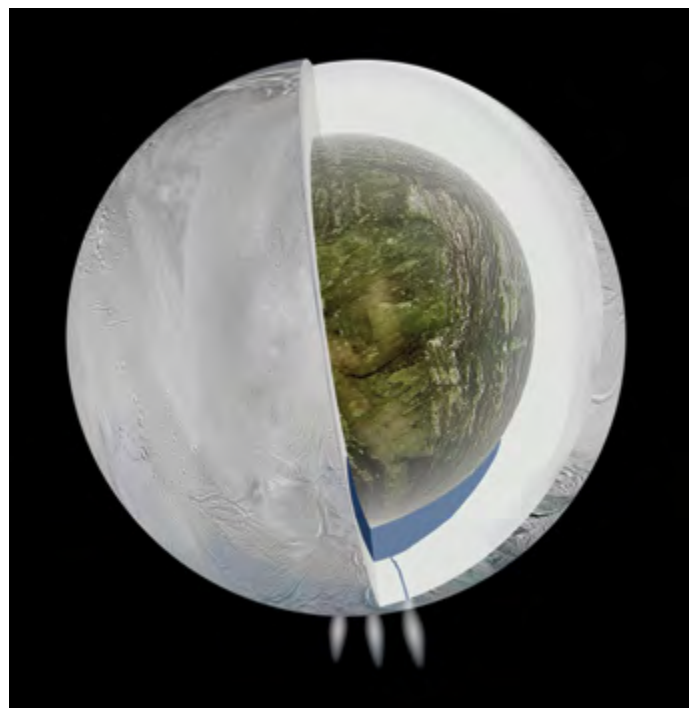
Herschel has revealed that galaxies at the dense core of a massive galaxy cluster called XDCPJ0044.0-2033 were vigorously forming stars roughly four billion years after the Big Bang. In contrast, there is hardly any star formation in the cores of clusters in the nearby Universe today, as only old stars are detected. These results help flesh out our understanding of the formation and evolution of galaxies over cosmic time.

Integral has demonstrated beyond doubt that white dwarfs can explode as supernovae. These Type Ia supernovae have long been suspected of resulting from a white dwarf star blowing up because of an interaction with a companion star. However, astronomers lacked definitive evidence that a white dwarf was involved until now. The ‘smoking gun’ was evidence for radioactive nuclei being created by fusion during the thermonuclear explosion of the white dwarf star, detected by Integral via the gamma rays emitted as they decayed.

Cassini has confirmed that Saturn’s icy moon Enceladus has a regional water ocean in its southern hemisphere. In 2005, Cassini discovered jets of ice and water vapour coming from near the south pole of Enceladus, suggesting the presence of liquid water below its surface. Subsequent detailed measurements of the gravity field and thus the internal structure of Enceladus have now been made via small perturbations in the radio signals received from Cassini during close flybys of this moon. These measurements confirm the presence of a regional water ocean under the icy crust of the moon’s southern hemisphere.

Mission operations

At the November Science Programme Committee meeting, as the conclusion of the biennial mission operations extensions process, approval was given to continue the operations of Cluster, Hinode, Hubble Space Telescope, Integral, Mars Express, Proba-2, SOHO and XMM-Newton until the end of



Observations by Cassini confirm a regional ocean under the icy crust of Saturn’s icy moon Enceladus (NASA/JPL)

2018, all subject to a successful mid-term review in 2016, while Cassini–Huygens was formally extended until the end of 2016 with limited support thereafter until the scheduled end-of-mission in September 2017.

Science archives

The science archives at ESAC provide worldwide, long-term access to the data from over 15 space science missions. Activities focus on maximising the scientific usability of the data, with a significant achievement being completion of a prototype interface to multiple astronomical missions.

Other highlights include full user access to a new Cluster archive, release of a thin-layer client for XMM-Newton and significant improvements to the Herschel, Planck, and Gaia archives.

Research activities

Areas of research in 2014 included Earth's magnetosphere, meteor showers, the terrestrial planets, the formation and evolution of planets, stars and galaxies, fundamental physics, and cosmology.

One research project has been to characterise a newly developed scintillator crystal for the BepiColombo Gamma-ray and Neutron Spectrometer. This new CeBr crystal has a much lower intrinsic noise level than the previous flight crystal and will allow for a much more sensitive detection of the potassium gamma ray line from the surface of Mercury, one of the most important tasks of the spectrometer.

Projects under definition & under development

LISA Pathfinder

Launch: end of 2015

LISA Pathfinder will provide an in-flight demonstration of the key technologies needed to detect gravitational waves. The Flight Model spacecraft is fully assembled and was due for its final test campaign in early 2015.

BepiColombo

Launch: early 2017

BepiColombo will make a detailed study of Mercury, the least-explored planet in the inner Solar System. The Flight Model environmental and functional testing on both ESA-provided modules, the Mercury Planetary Orbiter (MPO) and the Mercury Transfer Module (MTM) are ongoing. The JAXA-provided Mercury Magnetospheric Orbiter (MMO) will be delivered to ESA in mid-2015.

ExoMars

Launch: 2016 and 2018

ExoMars comprises two separate missions under an ESA/Roscosmos cooperation agreement, to provide science return

from orbit around and from the surface of Mars, and to demonstrate technologies for future exploration missions. The 2016 mission includes a trace gas orbiter (TGO), and an Entry, Descent, and Landing Demonstrator Module. TGO will probe the atmosphere for indicators of biological activity and provide a communications link for the 2018 mission, which will land a rover on the surface of Mars. The rover will perform sub-surface investigations and the Pasteur payload will search for evidence of life past and present. The 2016 Flight Models will be completed at the end of 2015. The 2018 mission is at the System Preliminary Design Review stage.

Solar Orbiter

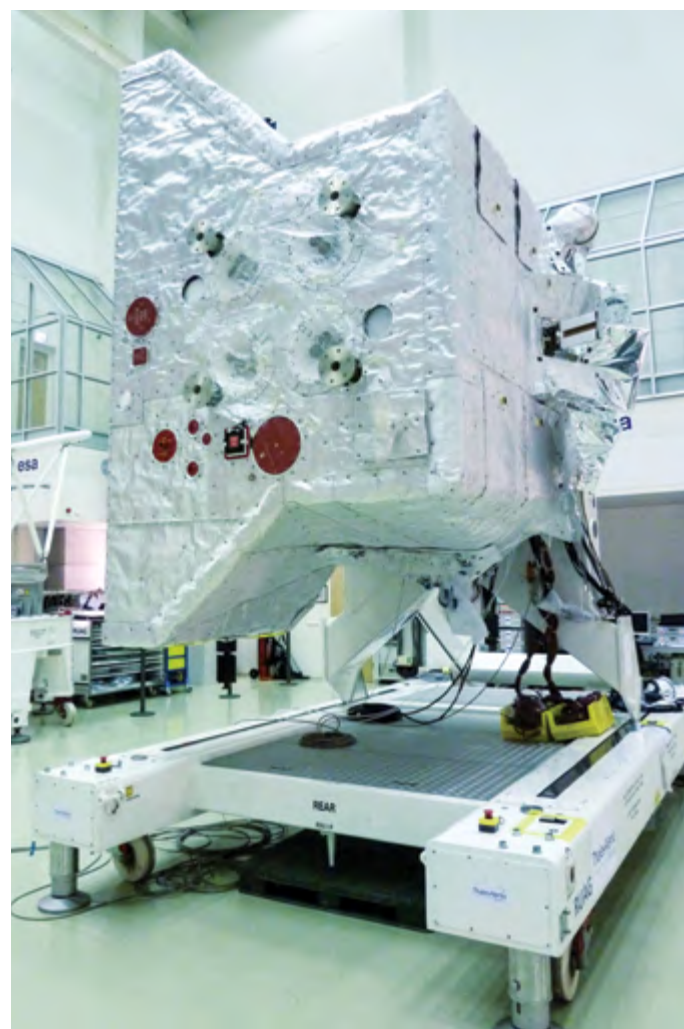
Launch: 2018

Solar Orbiter will provide unprecedented close-up and high-latitude observations of the Sun and inner heliosphere. The spacecraft development is progressing well, with the System Critical Design Review scheduled for mid-2015.

James Webb Space Telescope

Launch: 2018

JWST is the successor to the Hubble Space Telescope, under joint development by NASA, ESA and the Canadian Space Agency. It will search for light from the first objects that formed in the Universe after the Big Bang, study galaxy



BepiColombo MPO at ESTEC for thermal testing



ExoMars heat shield sensor examined by technicians in ESA's ultra-clean microbiology laboratory in ESTEC

evolution, and probe the early phases of star and planet formation. Both European instruments (NIRSpec and MIRI) have been delivered and built into the Integrated Science Instrument Module.

CHEOPS

Launch: 2018

CHEOPS will determine the masses and radii of planets in the super-Earth to Neptune mass range orbiting bright stars via high-precision photometry and provide 'golden targets' for future in-depth characterisation. The mission was adopted by the Science Programme Committee in February, the prime contractor began work in July, and the Mission Preliminary Design Review was completed in October, less than two years after the outcome of the 'Call for S1 mission'.

Euclid

Launch: 2020

Euclid will investigate the nature and origin of dark matter and dark energy. The System Preliminary Design Review will be held in mid-2015. Development of the detectors for the Near-IR Photo-Spectrometer and Visible Imager System instruments has progressed.

JUICE

Launch: 2022

The JUICE mission will explore the Jupiter system and its icy satellites, with emphasis on Ganymede, Callisto and Europa. After a two-year tour of the Jupiter system, the spacecraft will enter orbit around Ganymede. The prime contractor will be selected and design and development work will begin in 2015.

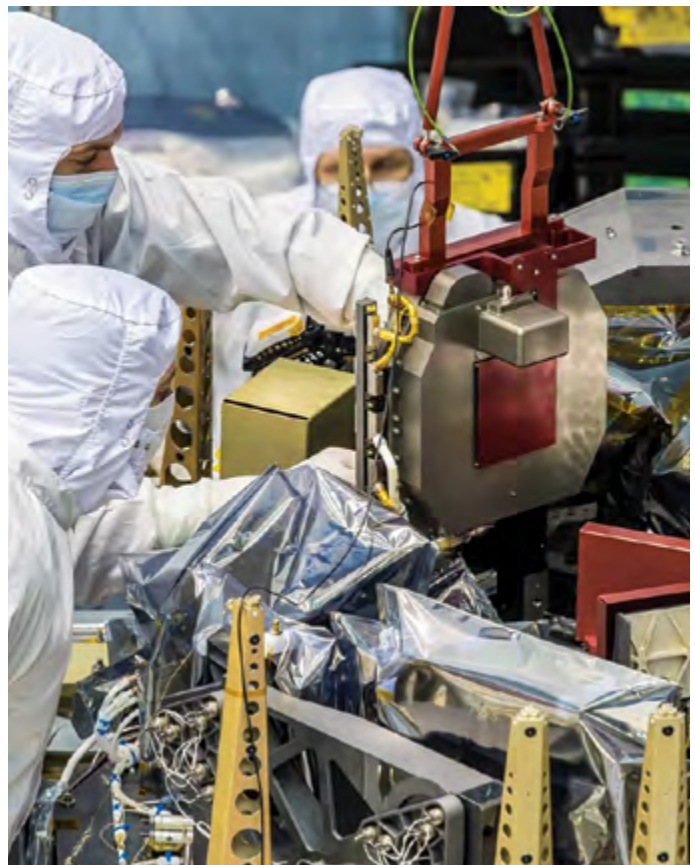
Robotic exploration

To prepare ESA's post-ExoMars robotic exploration programme, a number of candidate missions have been studied for a Mars launch opportunity in 2024 in the context of the Mars Robotic Exploration Programme (MREP). A Phobos Sample Return (PhSR) mission was judged the most promising candidate to prepare technologies for ESA's contribution to Mars Sample Return as well as to deliver important science. A joint ESA/Roscosmos Concurrent Design Facility study confirmed two technically viable and scientifically interesting options for a cooperative PhSR mission. These options will be further explored in 2015 through industrial Phase-A studies, as well as negotiations with Roscosmos.

MREP is also developing technologies relevant to PhSR and for future European participation in Mars Sample Return endeavours. Long-term enabling technologies such as radioisotope heat generation power systems are under development. International cooperation discussions on Mars Sample Return continued in the framework of the International Mars Exploration Working Group.

PRODEX

PRODEX (Programme de Développement d'Expériences Scientifiques) is managing around 200 contracts in its 11 Participating States. To increase programme efficiency, the hardware development contracts have been consolidated into roughly 50 contracts; an additional 15 industrial contracts are in preparation.

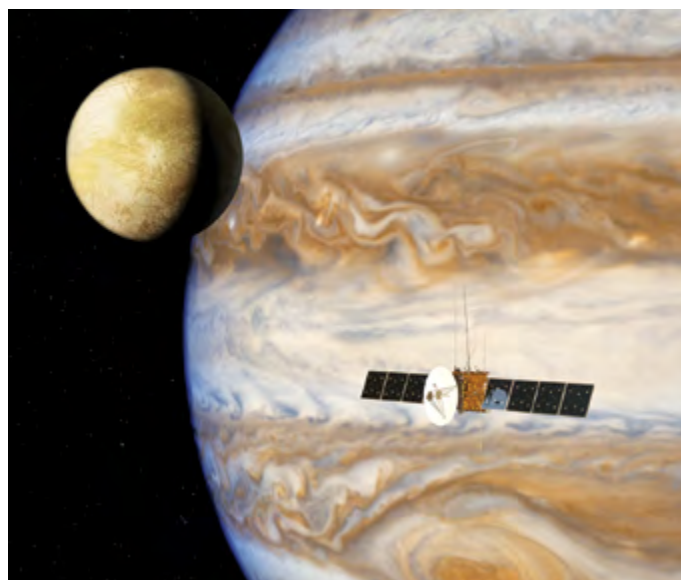


New Micro-Shutter Assembly being installed on the James Webb NIRSpec instrument

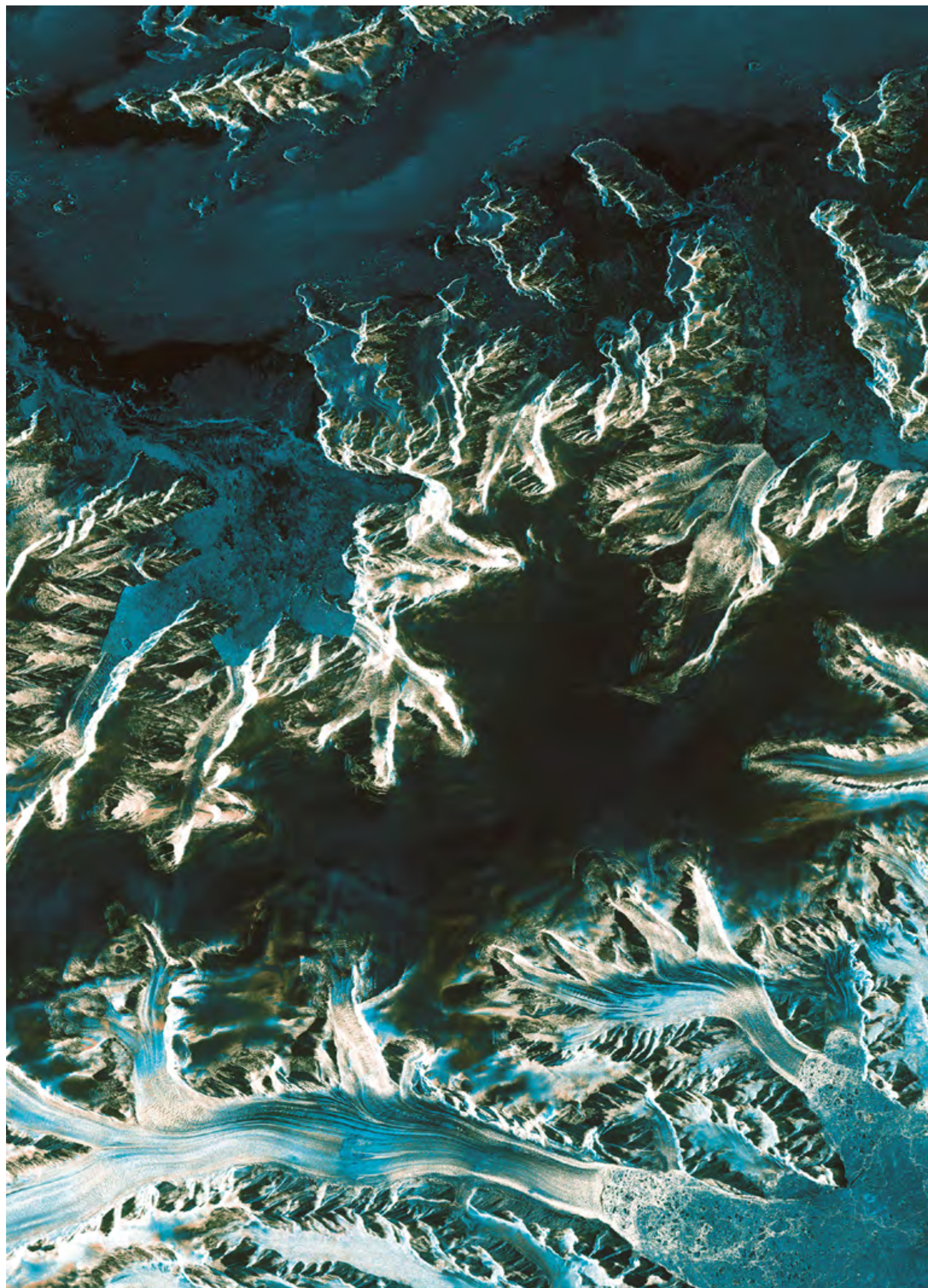
This was a very active year for PRODEX with development activities on major ESA space science missions. PRODEX instrument developments include: for Solar Orbiter, EUI (BE) and STIX (CH), as well as subsystems of RPW, SPICE, METIS, and PAS-SWA, representing 20 contracts in six countries; on ExoMars 2016, NOMAD (Spectrometer, BE) and CaSSIS (stereo and colour camera, CH, PL, IT), with the completion of the NOMAD Flight Model and of CaSSIS subsystems in 2014.

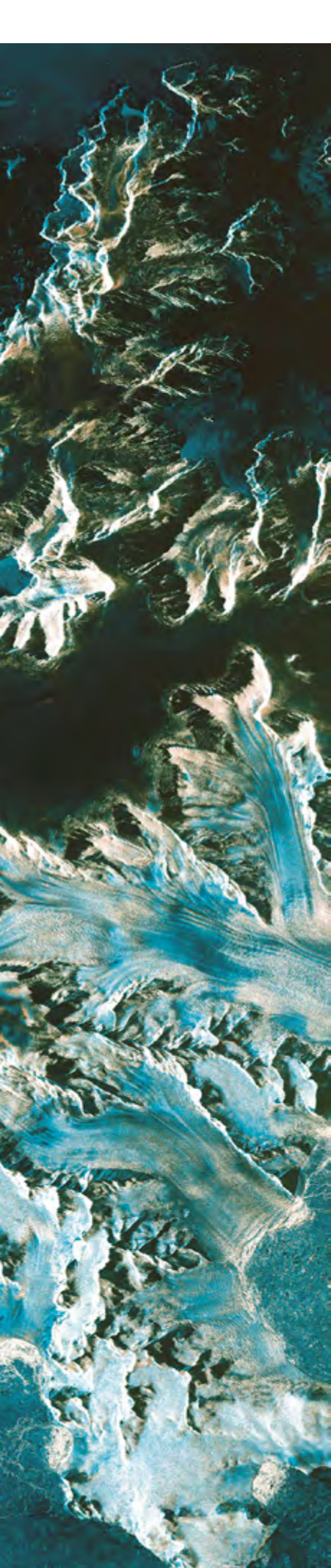
PRODEX also contributes to the NASA Insight mission with the Power, Communication, and Data Processing Unit of the CNES-led seismometer: the Flight Model has been built and the test campaign started.

Other highlights include the PanelSAR consolidation phase (NL), the Clara-Norsat Phase-C/D (CH and NO), preparation of JUICE activities (12 instrument subsystems), the beginning of the CHEOPS (CH) instrument Phase-C, and the continuation of ASIM development (ISS-DK).



JUICE, the JUpiter ICy moons Explorer mission, in the Jovian system (ESA/AOES)





earth observation

Apart from the launch of Sentinel-1A, a highlight for 2014 was the signature of the Copernicus Agreement between the European Union and ESA. Swarm became operational, joining ESA's growing fleet of Earth Explorers.

Copernicus

Copernicus will provide accurate, timely and easily accessible information to improve the management of the environment, understand and mitigate the effects of climate change and ensure civil security.

The main achievement for Copernicus was the signature of the Copernicus Agreement between the European Union and ESA. Besides establishing an administrative and procedural framework for Copernicus operations, this Agreement allocates €3.2 billion to ESA, constituting the financial basis of Copernicus operations until 2021.

Sentinel-1

Sentinel-1 is the Copernicus radar mission based on two satellites, designed in response to user requirements issued by the European Commission and to ensure continuity of C-Band radar observation initiated with ERS-1/2 and continued with Envisat ASAR.

Sentinel-1A, the first spacecraft in the Sentinel-1 mission, was launched on 3 April and, after its Commissioning Phase, has been in operation since 3 October. The second spacecraft, Sentinel-1B, is in the final stages of integration and testing, with all platform and SAR payload equipment delivered.

Sentinel-2

Sentinel-2 is the optical multispectral Copernicus mission ensuring continuity and further development of the SPOT/Landsat missions. The mission is based on two satellites flying in the same Sun-synchronous orbit.

Sentinel-2A underwent its final satellite integration and system testing, with the delivery and integration of the Multispectral Instrument in June. The satellite environmental test campaign began at IABG in Germany in August and was proceeding well and still ongoing at the end of the year. The satellite is due for launch in June 2015.

Sentinel-3

Sentinel-3 is the Copernicus mission delivering global, frequent and near real-time ocean and land monitoring and providing enhanced continuity with Envisat in these areas. Two operational satellites are envisaged to provide a one-day (land) and two-day (ocean) revisit time.

Sentinel-3A completed its integration during the first half of the year and started the satellite testing, completing the reference system tests, the conducted emissions and the mechanical environmental test campaign, which confirmed the compatibility with both the Rockot and Vega launchers to be used for the launch of the A and B satellites respectively. In parallel to the Sentinel-3A activities, integration of the Sentinel-3B Satellite has started.

One of the first images acquired by Sentinel-1A, taken on 13 April 2014, showing part of the northern Antarctica Peninsula



Launch of Sentinel-1A



Sentinel-2A in environmental testing



Sentinel-3 during mechanical testing

Sentinel-4

Sentinel-4 is the Copernicus mission delivering data supporting air quality monitoring and forecasting over Europe. The Sentinel-4/UVN instrument is a high-resolution spectrometer system operating with three designated bands in the solar reflectance spectrum, covering the ultraviolet, visible and near-infrared spectral bands. Two Sentinel-4/UVN instruments will be provided to the Meteosat Third Generation (MTG) programme to be carried on the two MTG-Sounder (MTG-S) satellites.

Since the completion in November 2013 of the Instrument Preliminary Design Review, design activities have continued. All 42 Best Practice procurement activities have started, with 25 of the 30 Subsystem Preliminary Design Reviews completed as well as the first 10 Subsystem Critical Design Reviews.

Sentinel-5

Sentinel-5 is the Copernicus mission providing global data on the composition of the atmosphere. The Sentinel-5 instrument will be carried on the MetOp-SG satellite A series. The service is focused on air quality monitoring and composition-climate interaction.

Phase-B2 of the project began in January. The System Requirements Review was completed by the end of the year. In parallel, focus was on the completion of the industrial team. More than 40% of the subcontractor Invitations to Tender were issued in the course of the year.

Sentinel-5P

The Sentinel-5 Precursor, a single-satellite mission within Copernicus, provides continuity with the atmosphere chemistry mission on Envisat in the period up to the entry into service of the Sentinel-5 operational instrument. It is implemented in cooperation with the Netherlands.

The Flight Model TROPOMI payload and AstroBus platform completed their integration. TROPOMI environmental qualification was achieved and instrument calibration started. Overall Ground Segment Critical Design Review and Launcher Preliminary Mission Analysis Review were held. The launch readiness date is February 2016.

Sentinel-6 /Jason-CS

The Sentinel-6 mission, implemented by the Jason-CS spacecraft, continues on from the high-precision measurement of ocean topography started with TOPEX/Poseidon in 1992 and continued with Jason-1, Jason-2 and Jason-3. There will be two satellites in the series.

Proceeding with Phase-B2, the Satellite Preliminary Design Review was completed and the industrial contract for Phase-C/D has been negotiated for two satellites. The Poseidon-4 altimeter new digital architecture pre-development continued. The launch readiness date is December 2019.

The 2014 ESA Council at ministerial level approved the ESA component of the Sentinel-6/Jason-CS programme and the European Commission approved the procurement of the recurrent satellite.

Ground Segment

During its Initial Operations Phase, the Coordinated Data System has been operationally serving some 70 Copernicus Services, continuously responding to the European Commission's data warehouse requirements. In addition, data are made available to thousands of self-registered users. With Sentinel-1A, the Sentinel Payload Data Ground Segment (PDGS) entered the Ramp-Up phase set to increase gradually in capacity. The Collaborative Ground Segment concept has taken shape through the conclusion of several ESA Member State agreements. The concept has been picked up by the European Commission to enhance Sentinel data access beyond Copernicus Services.

Earth Explorer Missions in orbit

SMOS

SMOS provides the observation of two key variables of the Earth system, namely the soil moisture content over land surfaces and the amount of salt dissolved in the ocean. SMOS is a cooperative project between ESA, CNES (France) and CDTI (Spain).

There was a significant step forward in that scientific products based on SMOS data have migrated into the operational community demonstrating the great utility of SMOS data for operational applications, for example sea ice thickness and soil moisture in NRT products. In the light of its excellent performance, the SMOS mission was extended.

CryoSat

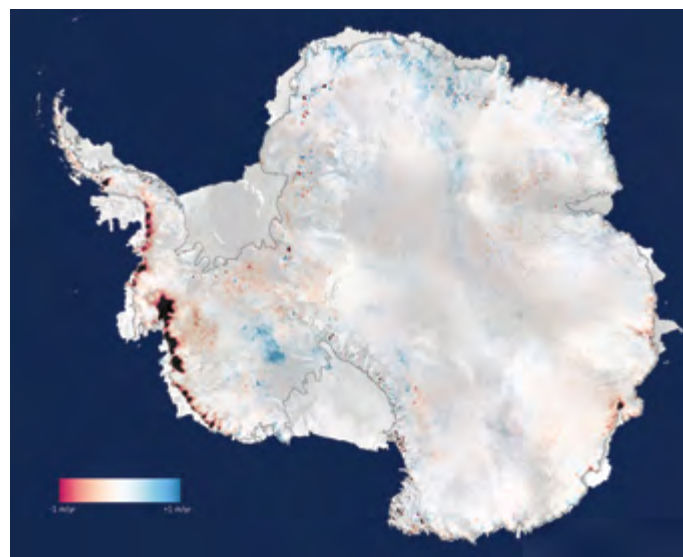
CryoSat is Europe's ice mission, providing multi-year elevation data at latitudes never before reached by a satellite altimeter. It measures thickness changes in ice sheets that overlay Greenland and Antarctica as well as in marine ice floating in the polar oceans.

CryoSat operations have been running smoothly, providing significant insights into how the cryosphere is responding to global warming. The mission delivered the fifth complete map of Arctic sea-ice thickness while three years of CryoSat observations in Antarctica showed that the continental ice sheet is losing 159 billion tonnes of ice each year – twice as much as when it was last surveyed. Thanks to its excellent performance, the mission was extended.

GOCE

GOCE was dedicated to measuring Earth's gravity field and modelling the geoid with unprecedented accuracy and spatial resolution. It advanced our knowledge of ocean circulation, which plays a crucial role in energy exchanges around the globe, as well as sea-level change and processes occurring within Earth's interior.

The final GOCE-based gravity field models, the so-called Release 5 models, were produced. These models confirm that GOCE met all of its ambitious mission requirements with some margins. Consequently, the application domains are now all served with the data products and information that were envisaged at the time the GOCE mission was approved. This represents a major scientific and technical achievement for Europe. The large GOCE User Workshop in November provided an overview of the exceptional achievements of GOCE during its lifetime and, in particular, during the last months of operations.

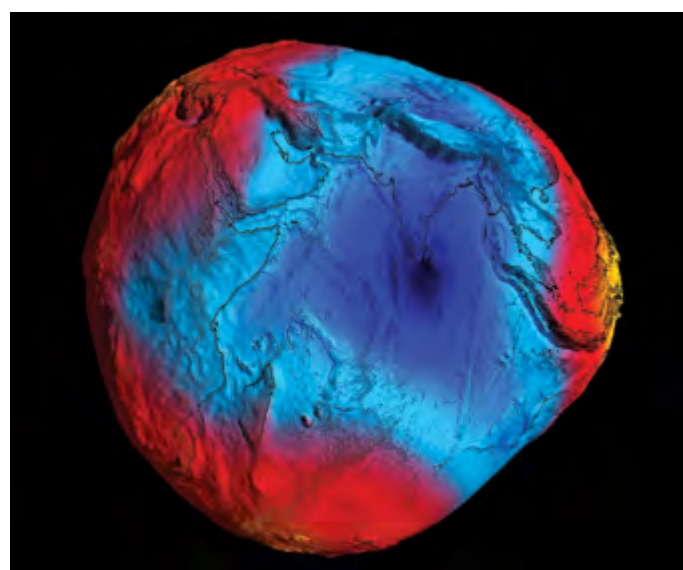


Between 2010 and 2013, west Antarctica, east Antarctica and the Antarctic Peninsula lost 134, 3 and 23 billion tonnes of ice each year, respectively. The average rate of ice thinning in west Antarctica has increased compared to previous measurements, and this area's yearly loss is now one third more than measured over the five years before CryoSat's launch (CPOM/Leeds/ESA)

Swarm

The aim of the three-satellite Swarm mission is to provide the best survey ever of the geomagnetic field and its temporal evolution. It is expected to lead to new insights into the Earth system by improving our understanding of Earth's interior and its effect on the magnetosphere, the vast region around Earth where electrodynamic processes are influenced by Earth's magnetic field.

The three Swarm satellites were transferred into operations in April. Despite the loss of one instrument on one of the three spacecraft, Swarm is delivering a plethora of valuable scientific data. Swarm delivered the observational material for the 12th generation of the International Geomagnetic Reference Field (IGRF) model. This model will be used by huge numbers of scientific, technical and societal users around the globe in the forthcoming five-year period. It is the agreed global reference model.



By the middle of 2014, five gravity field models and corresponding geoids had been generated from GOCE data, each version more accurate than the last. The fifth gravity model includes all gravity data collected throughout the lifetime of the mission – right up to November 2013 when the satellite finally stopped working and succumbed to the force it was designed to measure (ESA/HPF/DLR)

Future Earth Explorer missions

ADM-Aeolus

ADM-Aeolus will provide the first ever measurements of vertical wind profiles from space. These data are expected to advance substantially numerical weather prediction, especially when applied to extreme weather events.

Aeolus made important progress especially with the nominal part of the Aladin instrument completing its end-to-end full fluence tests at Airbus Defence & Space in Toulouse, confirming the UV transmit optics' robustness to laser-induced effects. During these tests the first flight laser transmitter was proven to operate reliably and with good, consistent and repeatable performance. Qualification completion of the second flight laser transmitter encountered a number of setbacks with the result that it could not be delivered for further integration into the instrument. Nevertheless, by the end of 2014, the second flight laser was fully integrated and ready to start a delta qualification campaign.

The target launch readiness of the satellite is end 2016.

EarthCARE

EarthCARE will carry a payload of four instruments that will operate synergistically with the aim of providing a better understanding of clouds and aerosols and their impact on Earth's climatology. It is a joint mission in collaboration with JAXA (Japan).

The EarthCARE Phase-C development continued with the finalisation of all the Platform Unit Critical Design Reviews and the execution of the four-instrument payload Critical Design Reviews. Accordingly, the prime contractor is proceeding with the preparation of the System Critical Design Review. Integration of the protoflight Base-Platform (BPF) has been completed.

Biomass

Biomass mission will provide measurements of forest biomass and forest height to help understand the state of Earth's forests, their evolution over time and to advance our understanding of the carbon cycle. The mission will be implemented through a full polarimetric P-band Synthetic Aperture Radar (SAR) flown on a single satellite.

Two parallel, competitive mission definition studies with consortiums led by ADS and OHB/Thales, respectively, were started in December 2013 and concluded in October. This paved the way to seek final approval from Member States to continue with the implementation of the mission.

Meteorological missions

MetOp

The three MetOp polar-orbiting Meteorological Operational satellites, developed by ESA, constitute the space segment of the Eumetsat Polar System.

MetOp-B has been providing the primary meteorological observations from polar orbit since April 2013. In parallel, MetOp-A continues to operate after more than eight years in orbit. MetOp-C is currently in storage, subject to annual maintenance, with a planned launch in 2018.

MetOp Second Generation

The Second Generation of MetOp satellites (MetOp-SG) are planned to continue and enhance the observations provided by the current, first generation of MetOp satellites and will constitute the space segment of the second generation Eumetsat Polar System. MetOp-SG will consist of two series of satellites (A and B) and will provide the operational meteorological observations from polar orbit for 2020–45.

The evaluation of the industrial proposals in response to the Phase-B2/C/D Invitation to Tender was completed with Airbus Defence & Space France being selected as the prime contractor for Satellite A, and Airbus Defence & Space Germany as the prime contractor for Satellite B. The two contracts began together on 28 May, together with five of the Contractor-Provided Item (CPI) instruments: Microwave Sounder, Microwave Imager, Ice Cloud Imager, Multi-viewing Multi-channel Multi-polarisation Imager, Radio Occultation sounder. The final CPI instrument, the Scatterometer, was selected in accordance with ESA's Code of Best Practices and work began in December. The System Requirements Review got under way in September and was concluded in October.

MSG

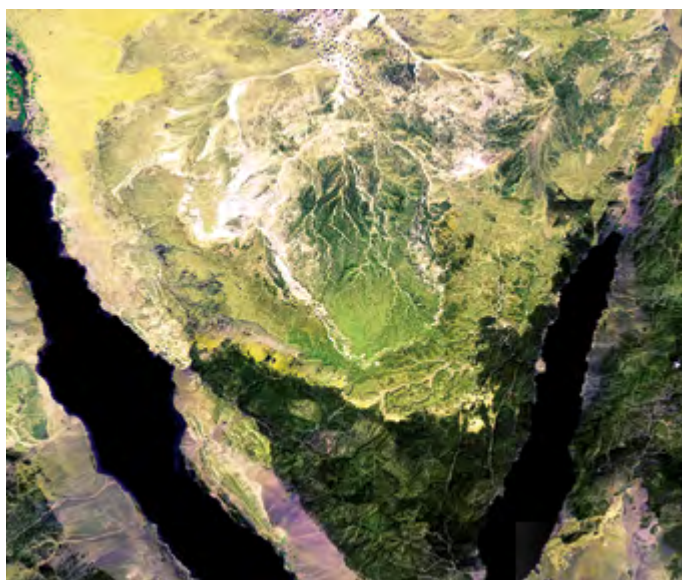
Meteosat Second Generation (MSG) is a continuation and substantial enhancement of the previous Meteosat system. It consists of a series of four geostationary meteorological satellites to be operated until at least 2021, providing continuous imagery of the atmosphere, clouds and surface of Earth.

The performances of MSG-1 (Meteosat-8), MSG-2 (Meteosat-9) and MSG-3 (Meteosat-10) are normal after twelve, nine and two years in orbit, respectively. MSG-1 is the backup for the operational services of MSG-2 and MSG-3. MSG-4 activities concentrated on system tests and launch campaign preparation activities. MSG-4 is ready for a launch planned in July 2015.

MTG

Meteosat Third Generation (MTG) satellites will ensure continuity of operational meteorological and climate data from geostationary orbit after completion of the ongoing MSG missions. The MTG-I (Imager) provides enhanced visible and near-infrared imagery and a new Lightning Imager; the MTG-S (Sounder) provides a new infrared sounding capability from geostationary orbit and will also accommodate the Sentinel-4 UVN spectrometer.

Substantial progress has been made within the MTG Programme. The Best Practice procurement process has been completed and all lower level subcontractors are now in place. The Preliminary Design Reviews at all levels are nearing completion, allowing the start of Development Model hardware manufacture. The Phase-C/D Price Conversion negotiations are complete and have been implemented. These include some optimisation of development philosophy (and recurrent satellite activities) to ensure the predicted cost-at-completion remains within the nominal 100%. Activities at main element level (platform and instruments) are now focused on preparation for the Structural and Engineering Model integration and testing. Predicted performance for both the MTG-I and MTG-S missions remains highly compliant with requirements.



The triangular Sinai Peninsula, acquired by Proba-V on 24 March 2014 (ESA/VITO)

Other missions

Proba-V

The small satellite mission, Proba-V, supplies vegetation imagery for applications such as climate impact assessments, water resource management, agricultural monitoring and food security estimates.

Proba-V was launched at the beginning of May 2013. The In-Orbit Commissioning Review and mission handover from Technical and Quality Management (TEC) to Earth Observation (EOP) took place in December 2013. Throughout 2014, Proba-V data were provided normally. Usage in the main was by the Copernicus Global Land Service and about 300 other users worldwide from almost 70 different countries, for monitoring of global vegetation and its changes over time.

Earth Explorer 8

System-level preparatory studies and supporting scientific study and campaign activities were drawing to a close for the two candidate Earth Explorer mission concepts competing to be selected as the eighth Earth Explorer mission in 2015. Many experimental activities have also been carried out or are in progress to bring all critical technologies to high readiness levels. The two candidates were the Carbon monitoring Satellite (CarbonSat) and the Fluorescence Explorer (FLEX). CarbonSat aims to quantify accurately sources and sinks of carbon dioxide and methane by measuring their distribution in the atmosphere, while FLEX aims to quantify photosynthetic activity and plant stress by mapping vegetation fluorescence.

Heritage missions

ERS-1, ERS-2 and Envisat data, far beyond the end of their active satellite lifetimes, continue to attract considerable user interest. More than 1000 new users were registered in 2014.

SAOCOM-CS

SAOCOM-1b is an Argentinian polarimetric L-band SAR mission expected to be launched in 2017/18. After the approval of Phase-AB1 of the small mission of opportunity SAOCOM-CS

(Companion Satellite) in cooperation with Argentinian space agency CONAE, parallel system studies were initiated, based on re-use of mature technologies. A SAOCOM-CS Workshop took place on 1–2 October, with over 80 experts from Europe, South America and Asia in attendance.

Earthnet/LTDP

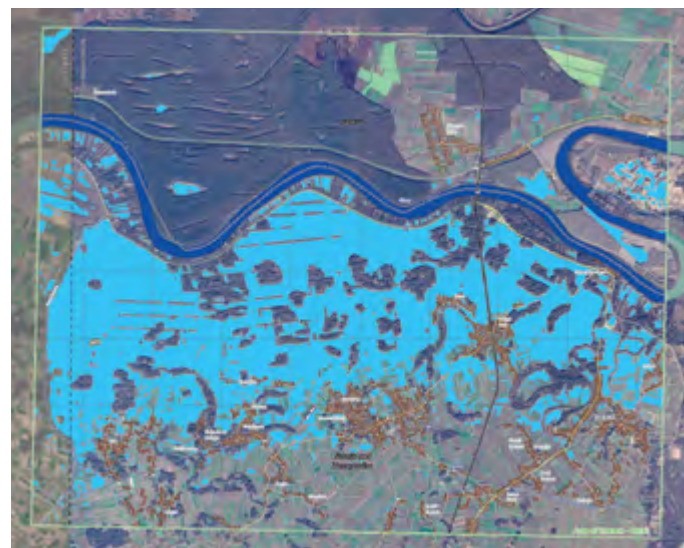
Through the Earthnet programme, ESA is managing access to Earth observation data from around 30 different non-ESA missions and instruments (Third Party Missions, TPMs). This includes, for example, Landsat-8 operations in Europe, as well as preparations for a major joint experiment with CNES before de-orbiting SPOT-5, called SPOT-5 Take 5. In this experiment, CNES will lower the satellite orbit of SPOT-5 for a five day revisit in order to simulate the revisit of Sentinel-2. The Long Term Data Preservation (LTDP) programme continued to ensure that no historical data is being lost, continued with the gradual transcription onto online media and provided increasing accessibility for users.

International Charter 'Space and Major Disasters'

The International Charter 'Space and Major Disasters' continued to provide rapid access to Earth observation data in support of disaster relief worldwide. It was activated 41 times in 2014, with the majority of triggering events being floods.

Climate Change Initiative

Progressing in its aim of providing Essential Climate Variable (ECV) data sets that respond directly to Global Climate Observing System (GCOS) requirements, Phase-1 of the Climate Change Initiative (CCI) programme was completed and Phase-2 started for most projects. Algorithm development, inter-comparison and validation, together with large-scale Earth observation data processing, took place for all 13 ECVs, resulting in prototype ECV data sets that have



Flood delineation map over the village of Balatun in northeastern Bosnia and Herzegovina based on Sentinel-1A data

been made freely available online. The development and testing of sustainable ECV processing systems that took place over the last year is continuing into Phase-2, ready for transfer out of CCI and into the Copernicus Climate Change Service (C3S).

Earth Observation Science Strategy

Following on from its last publication, in *'The Changing Earth'*, back in 2006, ESA's Earth Observation Science Strategy was revised by ESAC with the support of a panel of prominent European scientists in 2014. Published as *'Earth Observation Science Strategy for ESA – A new era for scientific advances and for societal benefits'* (SP-1329), this strategy aims to cover all areas of science to which Earth observation missions from space can make a vital contribution. The overarching goal is to challenge ESA and the scientific community to strive for major advances in knowledge along with the technological capability that can be used to respond to ever-increasing societal needs associated with risks and opportunities in our changing global environment.

STSE/SEOM/DUE/VAE

The Support To Science Element (STSE) continued to be a source of major scientific highlights, such as the first combined use of GOCE and GRACE to retrieve changes in mass lost in Antarctica at high resolution, the discovery of the effects of El Niño on global evapotranspiration or the historical analysis of SAR data to study the retreat of the Pine Inland Glacier and the climatic signal in Arctic lakes.

Within the Scientific Exploitation of Operational Missions (SEOM) Element, four open source scientific toolboxes were released attracting 2000+ users worldwide. The first Sentinel-1 INSAR results, including monitoring earthquakes and volcanoes, have been generated and published. Over 250 young scientists have been trained on advances in Earth systems, land, atmosphere and altimetry remote sensing.

The Data User Element (DUE) continued its engagement with user communities, promoting and demonstrating the benefit of Earth observation. Organising thematic workshops, for example, on permafrost, land surface temperature and water, and participating in dedicated user conferences such as RAMSAR, UNFCCC and UNCBD has strengthened these links and maintained long-term relations with the respective user communities and opened avenues to new ones.

Within the Value Adding Element (VAE), supporting the European Earth observation service industry, the key items to note include a comprehensive review of the information requirements from the oil and gas industry, and the starting of collaboration with the Asian Development Bank (ADB).

Campaigns

ESA airborne campaigns continued to play an important role in preparing future Earth observation satellite missions and helping validate the maps and products of satellite missions

already in orbit. Of special note was the strong international collaboration with NASA that culminated in a major joint ESA/NASA campaign called COMEX addressing atmospheric methane and carbon sensing and involving four aircraft, multiple atmospheric remote sensing instruments and European and US scientists.

Copernicus Master / App Camps

The 2014 edition of the Copernicus Master prize witnessed the largest number of ideas submitted to date. There were 171 proposals from 43 countries, offering an excellent glimpse of the next generation of Earth observation services set to benefit, in particular, from the expansion of digital infrastructures. The winners in eight different categories were given the opportunity to present their ideas at the official Award Ceremony in Berlin.

The Copernicus office contributed to the Space App Camp under the EU-funded project, S2G2M2, which invited 20 software developers to ESTEC from 4–12 May, developing

esrin

ESRIN has been made the primary site for ESA's private cloud storage service, operational since 2014. Named 'esacloud', this is a secure infrastructure storing data on-site in line with ESA security directives.

ESRIN's Energy Management System was recognised as being compliant with the ISO 50001:2011 standard. Building reinforcement works at ESRIN to increase seismic safety proceeded according to plan.

The number of visitors to ESRIN rose from 40 000 in 2013 to 45 000 in 2014, making ESRIN the most visited ESA establishment after ESTEC. The '50 years' event in November attracted numerous visitors and created considerable media interest. In March an Open Day was held with around 1400 students and their teachers.

mobile apps drawing on Copernicus and Galileo data. In the same vein, the office led the planning and implementation of the 'Copernicus App Camp 2014', aiming to encourage, facilitate and foster the development of Copernicus-oriented smartphone applications. Twenty young software developers from all over Europe were selected to participate in September.

CEOS/GEO

The Group on Earth Observations (GEO) and the Committee on Earth Observation Satellites (CEOS) continued to fulfil their role as cornerstones of international cooperation activities. CEOS contributed yet again to international initiatives such as the GEO Global Forest Observation Initiatives (GFOI), the GEO Global Agriculture Monitoring (GEOGLAM) and to preparing the participation of space agencies in 2015 events such as the Third World Conference on Disaster Risk Reduction and the COP 21 Climate Conference. As main coordinator for space matters for GEO, CEOS was very active in various groups in charge of defining the future post-2015 GEO framework to be approved at the 2015 GEO Ministerial Summit.



ESRIN held an event on 26 November to mark the 50th anniversary of European cooperation in space. Left to right, ESA astronaut Luca Parmitano with guests Roberto Battiston, Head of the Italian Space Agency, and Stefania Giannini, Italian Minister of Education, Universities and Research, with ESA Director General Jean-Jacques Dordain





telecommunications & integrated applications

The year saw a number of important announcements as well as further consolidation of various long-term activities. A highlight was the relocation of a major part of our operations to ESA's new ECSAT site in Harwell, UK.

Advanced Research in Telecommunications Systems (ARTES)

ESA's ARTES programme supports the space and satellite communications industry of ESA Member States by promoting research and development activities that lead to innovative satcom products and services. ARTES offers varying degrees of support to projects with different levels of operational and commercial maturity.

The ARTES programme comprises a number of elements, each focusing on specific activities.

ARTES 1 is the strategy element of ARTES. It encompasses preparatory activities covering mission, system and general configuration studies that develop medium- and long-term strategies for the satcoms industry. The industrial activities focus on strategy and market opportunities in the European telecommunication satellite industry, next-generation systems, product development roadmaps and identification of new technologies for space and ground systems.

During the year, 19 new activities were started in ARTES 1 for a value of €4.1 million. The funds available to implement new industrial contracts under ARTES 1 are now almost exhausted and so additional subscriptions are required to ensure the continuation of this element for 2015 and beyond.

The ARTES 3-4 element aims to maintain and improve the capability and competitiveness of the European and Canadian space industries in the global satellite telecommunication market. This element covers the design and development of products as well as demonstration activities in the space segment, including flight heritage opportunities for innovative satellite components, ground segment and user segment areas. An essential characteristic of the programme is the shared funding with industry to develop hardware, software, service or application items in order to be ready for exploitation in the commercial market. Proposals are made by industry to develop products aligned with the commercial imperatives of industry itself.

In 2014, there were 53 new activities started in ARTES 3-4 for a value of €36 million. A number of activities were completed and subsequently resulted in commercial flight orders, thereby validating the industry-driven approach adopted under this key element of the ARTES programme.

The ARTES 5 element focuses on technological innovation in equipment and systems for satellite communications.

Magali Vaissiere, ESA Director of Telecommunications & Integrated Applications, with Khalil Kably, EDRS-A Procurement & Project Control Manager, and Michael Witting, EDRS Project Manager, with the Eutelsat-9B satellite and its EDRS-A payload

Sub-element 5.1 supports activities which are not yet commercially competitive and are 100% funded by ESA. Space, ground and user segments are supported in the element as well as overall system-related activities. Element 5.1 is funded at a stable level of €20 million per year.

Sub-element 5.2 supports industry-initiated activities that are closer to market and are up to 75% funded by ESA. During 2014, ARTES 5.2 continued to grow thanks to enthusiastic responses from industry and ESA Member States. In 2014 there were 30 new activities started for a value of €29 million (ESA price).

European Data Relay System (EDRS)

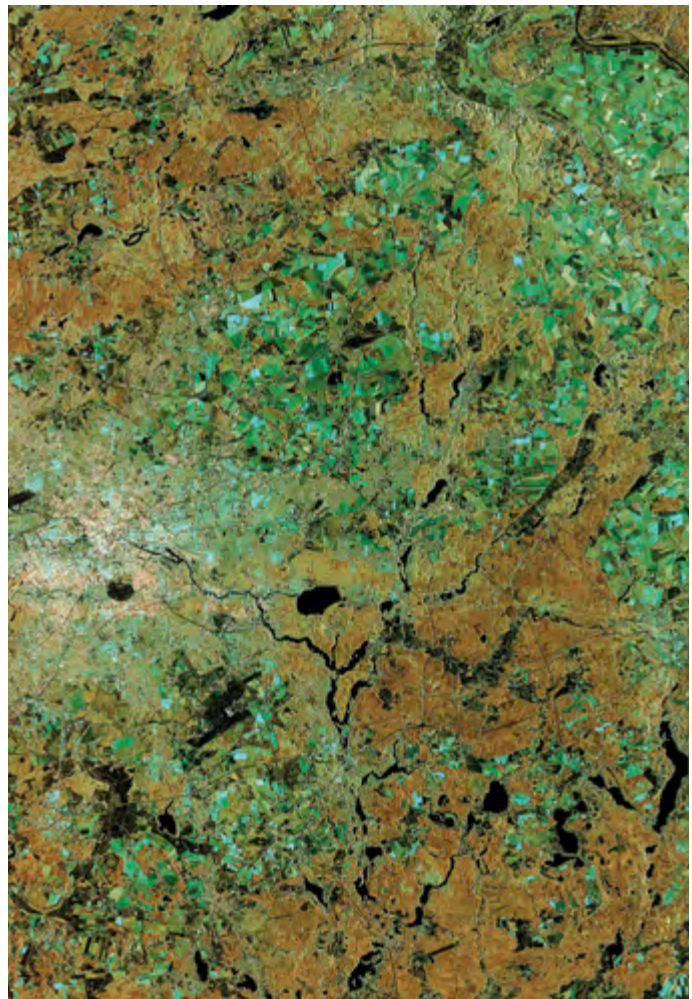
EDRS is being developed under a Public-Private Partnership (PPP) between ESA and Airbus Defence & Space. EDRS consists of:

- the EDRS-A payload and the ASI opportunity payload both integrated on Eutelsat's EB9B satellite to be launched in mid-2015;
- the EDRS-C satellite, based on SmallGEO platform from OHB System AG, to be launched in early 2017 (carrying both the EDRS-C payload as well as Avanti's HYLAS-3 telecommunication payload);
- the EDRS ground segment.

The EDRS-A payload and the ASI opportunity payload have been integrated on their EB9B host satellite and all test activities have been completed. The subsequent EDRS-A Flight Acceptance Review led by ESA and the EB9B Flight Model Completion Review chaired by Eutelsat were concluded in December.



Eutelsat-9B/EDRS-A satellite inside the thermal/vacuum chamber, Toulouse, France, in October (Airbus Defence & Space/SAS)



Berlin, Germany, captured by Sentinel-1A, one of the first images delivered via the precursor European Data Relay System carried on Alphasat (EDRS-A). The image is a result of the two satellites using their optical communications instruments to transfer data via laser for fast delivery (Copernicus data/ESA)

The EDRS-C satellite Critical Design Review (CDR) Board concluded that a number of ongoing design activities would require completion as a prerequisite for conclusion of the CDR and therefore requested that a 'Delta CDR' be held in mid 2015. EDRS Ground Segment activities are progressing on schedule.

Alphabus/Alphasat

ARTES 8 sub-element I, Alphabus, is the ESA programme to develop, in cooperation with CNES, a new large platform for telecommunications satellites in geostationary orbit. This contract has been running since 2005.

ARTES 8 sub-element II, Alphasat, was launched in July 2013 and is based on the Alphabus platform, providing the latter with the necessary first-flight opportunity. It is jointly funded by ESA and Inmarsat. The latter started commercial operation of Alphasat in November 2013.

In January, the three-year nominal operations phase of the four Technology Demonstration Payloads (TDPs) began. All TDPs are being operated nominally on a routine basis.

During the fourth quarter, intersatellite optical communication between Sentinel-1A and Alphasat's TDP1 was successfully demonstrated.



On 28 November, teams at ESOC, in Darmstadt, Germany, watched intently as ESA's Sentinel-1 and Inmarsat's Alphasat linked up using laser signals stretching almost 36 000 km across space (ESA/J. Mai)

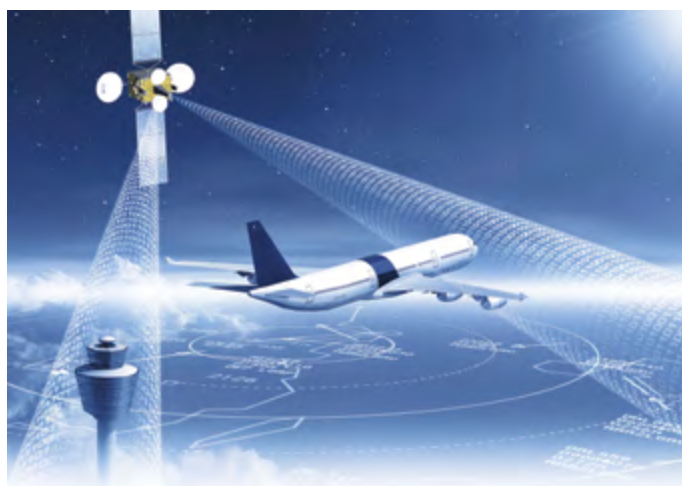
ARTES 8 sub-element III is the Alphasat Extension. It will extend the Alphasat range with the aim of addressing the very high-power end of the communication satellite market. The project is completing subsystem development and CDRs, with the second design review held in June. The qualification phase is now in progress.

ARTES 8 sub-element IV covers the development and qualification of the Alphasat ground and user segment and applications. All the activities have been completed. Final reviews of the maritime and aeronautical services activities were held in December.

Iris

Iris, element 10 of the ARTES programme, is a new satellite-based air traffic management (ATM) and communication system being developed in the framework of the European Union's SESAR (Single European Sky ATM Research) programme.

In October, activities to develop an intermediary system, Iris Precursor, were started with Inmarsat. The detailed definition of mission and system requirements is progressing as planned. The associated certification process is also being addressed.



The Iris Precursor service will provide air-ground communications for initial 4D flight path control by 2018 (ESA-P. Carril)

SmallGEO

SmallGEO, element 11 of the ARTES programme, is a general-purpose satellite platform for telecoms payloads less than three tonnes in mass. SmallGEO is being developed by an industrial team managed by OHB System AG.

ARTES 11 sub-elements 1 and 2 comprise building and launching the AG1 satellite, based on SmallGEO. In August, an important milestone was reached when the payload and platform modules were successfully mated. OHB then embarked on system and performance tests followed by environmental testing.

For Launch and Early Orbit Phase and Mission Ground Segments, development and validation work is progressing in line with the planned launch date. In December, radio frequency compatibility testing with Hispasat Telemetry, Tracking and Command antennas were completed at the Satellite Control Centre.

ARTES 11 sub-element 3 comprises the SmallGEO Extension. In 2014, two new activities were initiated: a modular suitcase study by Antwerp Space (BE) and a Video Board Evolution by Kongsberg (NO). In parallel, OHB and RUAG (AT) finalised an agreement to start a long-planned activity related to a chemical propulsion system. All other activities are progressing.

Neosat

The ARTES 14 elements comprises Neosat, a programme to develop and qualify a new satellite platform for medium-sized satellites with payloads of three to six tonnes of mass. It encompasses development, qualification and in-orbit validation, with Proto-Flight Model delivery in orbit by 2019. Neosat is being undertaken by two leading European satellite integrators, Airbus Defence & Space (FR) and Thales Alenia Space (FR).

In February, engineering activities, critical technology pre-developments and the selection of suppliers for Phase-C/D were initiated. The first invitation to tender for the Neosat supplier selection was issued in December.

Integrated Application Promotion (IAP)

The main goal of IAP, element 20 of the ARTES programme, is to promote the development of new services based on space technology. The IAP element focuses on the use of existing space assets and technologies leading to the increased exploitation of space systems. It also aims to provide input to future space development activities based on knowledge acquired of user needs. During the year, IAP initiated more than 70 activities for a value of approximately €33 million.

SAT-AIS

The Automatic Identification System (AIS) is a short-range coastal tracking system currently used to provide identification and position information to vessels and shore stations.



Signing the Neosat Phase-B contract on 20 February: Bertrand Maureau of Thales Alenia Space with Magali Vaissiere, ESA Director of Telecommunications and Integrated Applications, and Eric Béranger of Airbus Defence & Space

Space-based AIS (SAT-AIS), element 21 of the ARTES programme, will provide AIS data received via satellite, allowing the detection of seafaring vessels equipped with AIS tracking devices. The SAT-AIS initiative is a programme in partnership with the European Maritime Safety Agency (EMSA).

The ARTES 21 design phase was finalised in 2013 and the programme entered into its implementation phase in 2014. There are three sub-elements in the implementation phase:

ARTES 21 sub-element 2B comprises Innovative Medium Performance (MP) Satellite/Technology Developments. As part of these activities, the Novel SAT-AIS Receiver (NAIS) has been developed by Kongsberg Seatex, providing highly advanced signal processing, significantly improving the number of detected AIS messages. The Norwegian Space Centre (NSC) is responsible for the integration of the NAIS receiver in the NORSAT-1 (to be carried out by UTIAS) and the subsequent operation of the satellite. The NAIS receiver contract was signed and began in February. In November, the first Engineering Models were delivered to NSC/UTIAS for testing.

Regarding the European SAT-AIS Micro-Satellites (E-SAIL) activity, the contract for Phase-B2 of two SAT-AIS microsatellites and Phase-C/D of the first of these two satellites began in September. The industrial consortium led by LuxSpace is currently working on the consolidation of the requirements and preliminary design for the two microsatellites towards the system PDR in October 2015. These microsatellites will be launched and operated by exactEarth. The first microsatellite will continue to Phase-C/D directly after closure of the system PDR, while the second satellite Phase-C/D will begin on confirmation from exactEarth.

In ARTES 21 sub-element 2C, ESA supports the development of innovative SAT-AIS applications and services. The PLASMA (Platform for Advanced SAT-AIS Maritime Applications) project established within ARTES 21 2C provides a set of SAT-AIS value-added services including validated/enhanced SAT-AIS messages as well as analysis and warnings of suspicious or anomalous vessel behaviour. The PLASMA services respond to the needs of institutional end-users such as the European Maritime Safety Agency (EMSA) and commercial companies involved in the project. The PLASMA pre-operational service involves more than 170 users across 40 organisations.

Partner

Partner, element 33 of the ARTES programme, is the framework for developing industry-generated PPPs.

ARTES 33 sub-element I comprises Electra, the first PPP developed under ARTES 33. It is being undertaken in cooperation with SES, one of the world-leading telecommunications satellite operators. The goal is to develop, launch and validate in orbit a telecommunications satellite system below three tonnes launch mass that uses full electric propulsion for transfer to geostationary orbit as well as for station-keeping. At the end of 2014, the PDR process started.

ARTES 33 sub-element III comprises Quantum, a new PPP with Eutelsat. The objective is to develop, launch and validate in orbit a new satellite system based on an innovative Ku-band payload flown on a new geostationary small-class platform (3.5 ton launch mass capability). This new sub-element was approved at the December Council meeting at ministerial level.

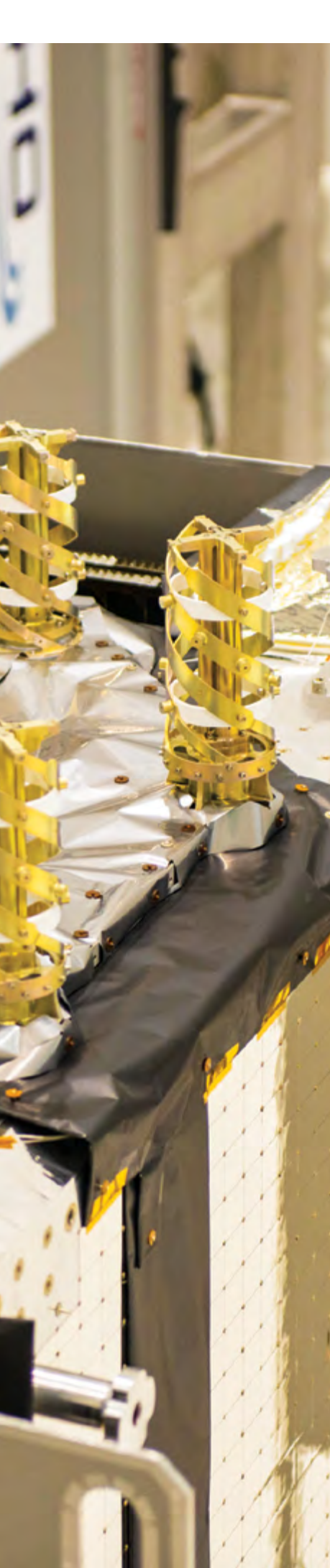


The Space-based Automatic Identification System AIS (SAT-AIS), element 21 of the ARTES programme, will provide AIS data received via satellite, allowing the detection of seafaring vessels equipped with AIS tracking devices

ARTES 33 sub-element IV phase 1 comprises ICE, a PPP between ESA and Inmarsat. The goal of the ICE programme is to support the design and development of the innovative elements that will contribute to an optimised delivery of future mobile satellite communication services. This new sub-element was approved at the December Council meeting at ministerial level.

ARTES 33 sub-element V comprises Indigo, a new PPP between ESA and Intelsat. The objective of the project is to develop, validate and launch an advanced High Throughput Satellite (HTS) system combined with a highly innovative ground segment solution from a European industrial consortium. This new sub-element was approved at the December Council meeting at ministerial level.





galileo & navigation-related activities

Highlights of 2014 were the launch of the first Galileo Full Operational Capability satellites, successfully recovered after being placed in an abnormal orbit, as well as the signature of the Delegation Agreement on the deployment phase of Galileo from 2014 and of the extension of the EGNOS delegation agreement.

The highlight of 2014 was the launch of the first Galileo Full Operational Capability (FOC) satellites within the overall constellation. However, these were unfortunately placed in an abnormal orbit before being successfully recovered. Thus 2014 was a year rich in learning not only with respect to Galileo FOC satellites but also regarding operations on the In-Orbit Validation (IOV) satellites.

2014 was also a turning point for EGNOS with the successful deployment of improved versions, as well as for the European GNSS Evolution Programme (EGEP) with the elaboration of the future of Galileo and EGNOS.

The signature by the European Commission and ESA of the Delegation Agreement on the deployment phase of Galileo as from 2014 and of the extension of the EGNOS delegation agreement were major programmatic milestones.

Galileo In-Orbit Validation (IOV)

Following the completion of the IOV Review, the IOV test campaign was completed with respect to public regulated signal access control and timing. Tests demonstrated that the requirements were fulfilled.

An IOV review close-out report has been prepared. All issues identified by it have been addressed by complementary testing, analysis and the release of additional documentation. Significant effort has been deployed to collect the information required to proceed with formal closure of the IOV contracts.

Close-up view of the first two Galileo Full Operational Capability satellites (Sats 5-6) during preparations for launch on 30 July 2014

Dordain and EC Director General for Enterprise and Industry Daniel Calleja Crespo signed the delegation agreement on the deployment phase of Galileo, and on further implementation of EGNOS on 16 July 2014



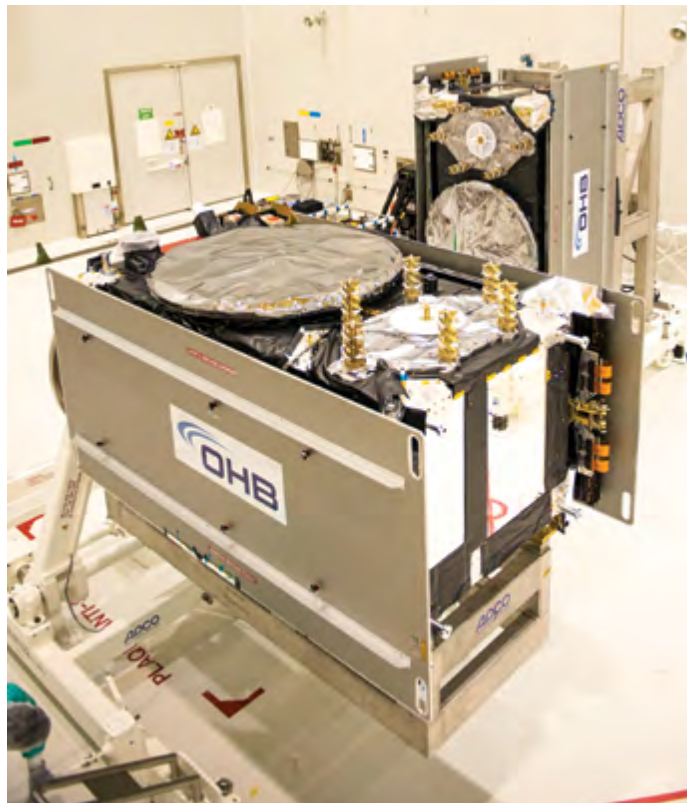
Galileo Full Operational Capability (FOC)

The first two FOC spacecraft, Sats 5 and 6 (FMo5 and FMo6), were launched from Kourou on 22 August, on a Soyuz rocket. They were mounted on a Soyuz ST-Fregat vehicle, and after their separation, the analysis carried out by the LEOP control centre at ESOC revealed that the satellites were not injected into the appropriate orbit. Despite this serious outcome, the ESOC/CNES teams were able to command and control both spacecraft and, after a few days of intense and critical operations, stabilise them in a Sun-pointing safe mode.

Immediately, the ESOC/CNES teams with the support of OHB, Spaceopal and the ESA project team initiated a recovery mission analysis to investigate scenarios that would permit,



Soyuz flight VS09, carrying Europe's fifth and sixth Galileo satellites, lifted off from Europe's Spaceport in Kourou, French Guiana, on 22 August 2014



Galileo Sats 5-6 in the SSA building, Kourou, French Guiana

as far as possible, the achievement of the intended mission objectives and maximise the benefits for the programme.

Orbit-raising manoeuvres for FOC FMo5 started on 5 November, and proceeded according to plan. Platform-commissioning activities were completed before the end of November, leading to the start of payload In-orbit Testing and allowing testing of the navigation payload and first signal-in-space transmission. The mission recovery operations for FMo6 were scheduled to follow in 2015. The integration of both satellites in the operational constellation is a credible option under final analysis.

The testing of follow-on satellites (FMo7 and FMo8) was completed and the deployment of ground segment activities implemented as planned.

EGNOS Exploitation

In 2014, the EGNOS Signal in Space was provided by means of system release V2.3.2 deployed in operations since October 2013. This release brought performance improvements, in particular counteracting the capricious behaviour of the ionosphere. A new communication network was fully qualified and deployed through a seamless transition process.

With the aim of ensuring the continuity of industrial contracts for urgent activities, the EC/ESA Delegation Agreement for EGNOS was extended until December 2016.

A new system release of EGNOS was qualified. This new version represents a major achievement towards performance improvement and obsolescence resolution.

European GNSS Evolution Programme (EGEP)

The EGNOS-related activities carried out within EGEP supported the further definition and development of the third version of EGNOS representing a major improvement with the introduction of a multi-constellation (GPS plus Galileo) multi-frequency service. This major activity is complemented by the associated activities on testbeds and station pre-development.

In support of the detailed definition of Galileo Second Generation (G2G), a System mid-term status review was held supported by the initial results of definition studies of next-generation satellites and of advanced subsystems.

A draft programme proposal for the evolution of the European Navigation programmes and the European Galileo System Evolution Roadmap was issued with a description of the G2G system architecture options, identification of the critical technologies and an introduction to the G2G 2015–20 roadmap.





launchers

Over the past 12 months, there were 11 launches from Europe's Spaceport using Vega, Soyuz and Ariane vehicles – a new milestone since the introduction of this full launcher family. But the highlight was the flight of Vega VV03 on 30 April, marking the launcher's entry into commercial exploitation.

Exploitation programmes

Ariane 5

On 29 July, Ariane 5 ES Flight VA219 delivered the fifth Automated Transfer Vehicle, ATV Georges Lemaître, into orbit on the final mission of the ATV programme. On the same occasion, the low-shock fairing separation system HSS3+ was flown, and the data measured were remarkably in line with expectation in terms of kinematics and shock levels. Besides ATV, five Ariane 5 ECA launchers carried a total of ten satellites into orbit: ABS-2 and Athena-Fidus on 6 February, Astra-5B and Amazonas-4A on 22 March, Measat-3b and Optus-10 on 11 September, Intelsat-30/DLA-1 and Arsat-1 on 16 October and finally DirecTV-4 and GSAT-16 on 6 December.

These achievements are the result of several activities that are required in order to maintain the qualification status of the Ariane 5 Launcher System. A large set of activities was developed in this field, encompassing the qualification of the new Aestus production required for Ariane 5 ES Galileo, the qualification of the Propellant Tank U-ring redesign and the first flight of the new-design Vulcain 2 hydrogen turbopump, which took place in December.

Test campaigns were under way, in particular for the next EAP solid booster stage, the HM7B engine and Aestus. Engine integration for the Vulcain 2 ARTA-10 test campaign, aimed at obtaining qualification of the engine performance when increasing the propellant flow by 3%, was ongoing according to schedule. Finally, obsolescence of critical components, such as avionics components needed for several items of equipment in the next production batch, was being dealt with. In the ground segment, several activities contributing to the maintenance in operational condition of Ariane production facilities were also implemented.

Activities on future missions carried out in 2014 focused on the Ariane 5 ECA Performance Improvement Plan (PIP). The completion of all PIP work will bring the generic performance of the ECA launcher to 10 295 kg into geostationary transfer orbit (GTO) in 2015. Work related to future missions was also carried out on the qualification of the Optimised Payload Adapter System, while the adaptation of the Ariane 5 ECA Upper Part by increasing the fairing height up to 2 m and improving GTO performance was ongoing. Ariane 5 ES Adaptation for Galileo was in its final launcher qualification review phase, with a major achievement being qualification of the dispenser following a comprehensive test campaign.

The third launch by Europe's small launcher Vega marked the start of its commercial exploitation (ESA/CNES/Arianespace)



Ariane 5 flight VA219, which carried ATV Georges Lemaître into orbit, on the launch pad, on 29 July 2014 (ESA—S. Corvaja)

Vega

The highlight of the year was the launch of Vega VVo3 on 30 April, which placed KazEOSat-1 (DZZ-HR) in orbit. Through this launch, Vega entered into commercial exploitation, because the flight was the first to be carried out under the responsibility of the launch service provider.

Intensive activities were carried out aimed at completing the generic qualification, including the closure of reserves, as well as the analysis of the VVo3 flight data. The continuous monitoring of obsolescence and work aimed at reducing recurring launch service costs were major activity drivers in view of the preparation of future Vega exploitation.

The IXV/Vega VVo4 launch campaign, initially planned for a launch on 18 November was stopped on 23 October because of safety issues with the campaign activities. This would be resumed in early 2015 for a launch scheduled for 11 February 2015.

Development programmes

Adapted Ariane 5 ME, Upper Stage and Commonalities and Ariane 6

In line with the decisions taken during the 2012 Council meeting at ministerial level, implementation of the adapted Ariane 5 ME and of the Common Upper Stage projects progressed according to plan. In particular, regarding the adapted Ariane 5 ME project, all set objectives (calendar, performance and recurring costs) were achieved by the results of industrial studies and proposals. Industrial activities mainly

focused on the system performance, cryogenic integrated test bench P5.2 at Lampoldshausen, avionics, the fairing industrialisation process and SylDa. The Qualification Review for the ECA demonstration flights was performed in December.

There were also a number of significant achievements in the Upper Stage and Commonalities project with successful Critical Design Reviews being held for all products: the Vinci Critical Design Review was completed, Upper Stage Bare Tank manufacturing facilities procurement was in its final commissioning phase and PF52 test facilities adaptation was progressing according to schedule.

In the final months of the year, and following the decision at Council, rundown and/or reorientation of Ariane 5 ME-specific activities were initiated, in line with Ariane 6 PHH version needs.

Implementation of the Ariane 6 programme also progressed according to schedule, albeit on the basis of the PPH (three-stage, solid-solid-cryogenic) linear configuration; considerable progress was made regarding design activities covering Phase-A/B1 of the launcher system and launch complex, as well as technology and demonstrator development. The Ariane 6 organisation was set up in several stages. The Request for Quotation covering Phase-B/C/D/E1, based on the PPH concept, was issued to industry in May, for the preparation of the industrial programme file. However, following the unsolicited proposal received from industry in June regarding a new Ariane 6 PHH (three-stage, solid-cryogenic-cryogenic) configuration, activities were reoriented, and the project assessed together with industry the scope of urgent activities to be carried out, introducing commonalities with the development of Vega C. Eventually, a programme declaration including four programme elements (Ariane 6, P120C, Vega C and an element covering initial activities for Launchers Evolution) was subscribed at the Council meeting at ministerial level in November, totalling more than €3.7 billion.

These development activities are completely new on several accounts: they represent a multi-billion programme; ESA is now fully responsible for the programme management, notably through its new responsibility as launch system architect; it is a complex programme given the new governance scheme with industry that needs to be implemented; and the commonalities between Ariane and Vega and the interaction with several prime contractors. In parallel, by mid-2015, the Ariane 5 and Vega launch service cost reductions should be demonstrated following intensive discussions between Arianespace, industry and ESA and a method will be proposed to Council for determining launch service prices for the procurement of Ariane 5, Vega and Soyuz from CSG for ESA missions.

Hence, preparation for development activities, planning, and organisation of the project teams started immediately after the Council meeting at ministerial level in December, not least because of the major decision point on the Ariane 6 Programme Implementation Review to take place in 2016. This review will not only be focused on programme development status but will also cover Ariane 6 exploitation conditions,

risk and cost sharing between the public and private sectors, arrangements to secure launch services for European institutional missions and the preparation of the transition period between development, under ESA responsibility, and exploitation, under private responsibility.

Vega (VECEP)

Phase-A activities for the Vega C configuration decided in September 2013 were completed in June. That configuration was based on the development of a new Vega version aimed at increasing performance by at least 300 kg, with an improved first stage, the P-CV (105 tonnes) and increased AVUM propellant loading, while reusing the other Vega stages without modification. However, following the conclusions of the Scenario Working Group on Ariane 6, which took place during the summer in preparation for the Council meeting at ministerial level, VECEP activities were reoriented in order to converge on a common set of requirements for both the Vega C and Ariane 6 solid rocket motors.

Consequently, the main activities focused on the new Vega baseline architecture, including the development of a P120C, a common solid rocket motor for the Vega first stage and for the Ariane 6 boosters (instead of the P-CV), and on the introduction of the Z40 for the Vega second stage (as a replacement for the Z23). The Declaration on the Programme for Ariane and Vega development subscribed at the Council meeting at ministerial level is based on this new configuration for the Vega C element, in order to benefit from the synergies between the two elements, Vega C and Ariane 6.

IXV

The purpose of the Intermediate eXperimental Vehicle mission is to perform in-flight demonstration of advanced system and technological features necessary to return from space to Earth such as lifting body performance for precision landing, advanced high temperature resistant materials, advanced guidance navigation and flight control techniques, aerodynamics and aerothermodynamics.

There were a number of significant achievements within the programme including the completion of the IXV System Qualification and Acceptance Review. Following this, the IXV launch campaign progressed until 23 October, when it was put on hold because of concerns over the compliance of the Vega trajectory with the near-range safety constraints.

With the aim of maximising the return on technology of investments committed to IXV development, the Programme for Reusable In-Orbit Demonstrator in Europe (PRIDE) was refocused on a new mission scenario based on requirements defined together with the relevant end-users, including multiple payloads for technologies in-orbit demonstration. During the 2014 Council meeting at ministerial level, the number of states participating in the programme increased to now include France, Ireland, Italy, Portugal, Spain, Sweden, Switzerland, Romania and the United Kingdom. The corresponding increase in subscriptions ensures the necessary funding will be available to complete, in timely fashion, the programme step-1 activities, addressing the mission and system definition phase including supporting technology developments, and prepare PRIDE step-2 development, to be submitted for approval at the next Council meeting at ministerial level in 2016.

Preparation for the future

Following the 2012 Council meeting at ministerial level, over 90% of the activities subscribed within FLPP-3 have been contracted out or are under negotiation with industry and major milestones have been achieved for the seven integrated demonstrators structuring the FLPP activities.

In particular, in the area of propulsion, activities in the Cryogenic Upper Stage Propulsion area have progressed on schedule and reached the System Requirements Review in the case of the Integrated Demonstrator and the Preliminary Design Review in the case of the Thrust Chamber. Furthermore, the Manufacturing Readiness Review for the reduced-scale cryogenic tank demonstrator was also performed in September. Regarding the storable propellant technology demonstrator project, as precursor of the AVUM for Vega C, the hot-fire test campaign was completed. In activities related to solid propulsion, the hot-fire test of the pressure oscillation demonstrator (POD-X) was conducted in March, while the POD-X Critical Design Review was started in December. In the area of hybrid propulsion, manufacturing and testing are progressing as planned and the breadboard full-scale motor hot-fire tests started in August.

In the area of technologies, the implementation of integrated demonstrators is progressing with the successful burst test of the aluminium/lithium Cryogenic Tank Demo, while all preparatory steps for Inter Stage Structure technology maturation have been completed with a view to achieving a rapid start in 2015, in line with the Ariane 6 schedule. The System Requirements Review for TTEthernet Demonstration was also performed, and the contract for the European Launcher Localisation Kit has been placed in view of a first implementation on Vega.

FLPP-3 was subscribed still further at the 2014 Council meeting at ministerial level, thus confirming the FLPP stakeholders' interest in and support for the preparation of future launcher evolution.



Rounding off 2014, Soyuz flight VS10 lifts off from Europe's Spaceport during a daytime launch on 18 December carrying four communications satellites (ESA/CNES/Arianespace)





human spaceflight & operations

This was a busy year marked by two main highlights: the spectacular Rosetta and Philae operations on Comet 67P/Churyumov-Gerasimenko, and the celebration of 50 years of European cooperation in space.

Highlights

Rosetta's hibernation exit was performed on 20 January, and the rendezvous with the comet took place on 6 August. The Philae probe landed on the comet on 12 November after a tense wait during its seven-hour descent to the surface. The descent and touchdown were relayed to Earth by the Rosetta orbiter and picked up by ESA's ground stations in New Norcia, Australia, and Malargüe, Argentina. The signal was immediately confirmed at ESOC, and at the DLR's Lander Control Centre in Cologne. Philae survived the expected 63 hours on the comet and performed all its planned activities, downlinking all the collected scientific image data.

Sentinel-1A was launched by Soyuz rocket on 3 April. Following an orbit insertion 7.8 km lower than expected, it was necessary to perform 500 manoeuvres to reach the reference orbit. Commissioning was then completed and the satellite handed over to routine operations on 23 September.

The launch of the first two Galileo Full Operational Capability (FOC) satellites took place on 22 August. When ESOC took over control immediately after separation it became clear that the injection orbit was dramatically wrong. Also, the automatic solar array separation sequence only managed to deploy one of the two array wings on each of the two satellites. The joint ESOC/CNES team managed to take control of the satellites, reestablish the true orbit and manually deploy the remaining solar array wings.



Acquisition of signal from Rosetta was confirmed on 20 January to the delight of the audience at ESOC and those involved in the mission worldwide. Andrea Accomazzo (centre) celebrates with Paolo Ferri, Manfred Warhaut at ESOC (ESA/J.Mai)

The Galileo operations team, joined by Director General Jean-Jacques Dordain, Director of Human Spaceflight and Operations Thomas Reiter and experts from European industry, seen at ESOC on 28 August (R. Solaz)

ESA astronaut Alexander Gerst, NASA astronaut Reid Wiseman and Roscosmos commander Maxim Surayev, were launched to the ISS from Baikonur on 28 May. The Expedition 40/41 crew reached orbit and docked as planned. Alexander's Blue Dot mission came to an end when he and his crew members landed safely in Kazakhstan on 10 November.

ATV *Georges Lemaître* was launched from Kourou on 29 July and docked with the ISS on 12 August. This fifth and last ATV delivered 6602 kg of freight, including 2695 kg of dry cargo and 3907 kg of water, propellants and gases. The Laser Infra-Red Imaging Sensors (LIRIS) operated successfully during fly-under and docking, and

recorded 76.5 gigabits of data. The experiment recorders were dismantled on 29 August by Alexander Gerst, and packed for download on Soyuz TMA-12. In addition to ATV, nine logistics flights were launched to the ISS: four Progress spacecraft, two SpaceX Dragon spacecraft, and three Orbital Sciences Cygnus spacecraft (though one failed to reach orbit).



ATV Georges Lemaitre just before docking with the International Space Station at 15:30 CEST/13:30 UTC on 12 August 2014 (NASA)

On 23 November, Samantha Cristoforetti was launched from Baikonur on a Soyuz together with crewmates Anton Shkaplerov and Terry Virts. The Expedition 42/43 crew reached their destination just five hours and 48 minutes after liftoff and docked safely with the ISS. On her Futura mission, Samantha flew as an ESA astronaut for Italy's ASI space agency under a special agreement between ASI and NASA.



Samantha Cristoforetti greeting onlookers at the launch pad

At the 2014 Council meeting at ministerial level, the Member States participating in the ISS Exploitation programme subscribed to the 3rd Binding Financial Commitment of Phase 2. Through the inclusion of lunar activities within the ISS Exploitation programme, a comprehensive perspective covering activities until 2020 and the first half of the next decade was created.

ATV

All ATV hardware at Europe's Spaceport in French Guiana was returned to Airbus in Bremen by sea transport. First meetings on contract close-out with industry, Arianespace and Russian partners were held. The ATV Control Centre decommissioning plan was agreed. A 'lessons learned' exercise for ATV development and production was initiated and partially concluded. For ATV operations, a dedicated know-how transfer project was started taking account of the fact that no direct continuation project of human spaceflight operations under ESA responsibility is foreseen. Discussion is ongoing regarding ATV data and documentation archiving/recording in accordance with ESA requirements.

Research

European Programme for Life and Physical Sciences and Applications in Space (ELIPS)

After the 2014 Council meeting at ministerial level, the ELIPS-4 Programme is under review to match activities with subscription levels (57.5%). The first prioritisation steps will be presented at the HME Programme Board meeting in February 2015.

ISS Utilisation

In total, 266 ESA experiments have been conducted on the ISS since it has been in orbit; more than 150 experiments have been performed since the launch of Columbus in 2008, with 47 of those in 2014. Since the Columbus launch, ESA's ISS experiments involved a total of more than 692 scientists (including about 100 international science team members).

In life sciences, seven different astronauts were involved in ESA activities on the ISS as test subjects. This included research into improved understanding of the effects of long-duration spaceflight on: alterations in circadian rhythms; neurological alterations related to 3D visual perception; energy requirements of astronauts; as well as research into skin-ageing mechanisms, and headache incidence. Two plant biology experiments were performed related to gravity response in lentil seedlings and light/gravity response mechanisms in arabidopsis seedlings. The Expose-R2 astrobiology facility was installed outside the ISS, hosting experiments that could help explain the origin of life on Earth and the survivability of samples to conditions on the Moon and Mars.

In physical sciences, the Electromagnetic Levitator was installed and research continued in the Materials Science Laboratory. These experiments will help to optimise industrial casting processes. Three fluids experiments concluded which will help lead to more efficient extraction of oil resources and improved insights into emulsion stability while complex plasma research was expanded with the launch of the PK-4 experiment. The

Solar and DOSIS-3D payloads continued to monitor solar/radiation environments while the Magvector experiment started research into how magnetic fields impact conductors.

In technology, the Vessel ID System continued to track global maritime traffic from space while the WiSe-NET experiment is evaluating the possibility of 'energy harvesting' on the ISS.

The International Life Science Research Announcement (ILSRA) in coordination with ASI, CNES, CSA, DLR, JAXA and NASA, received 112 European proposals for future ISS experiments. The science peer panel reviews of the European proposals were concluded with 45 proposals passing the science merit score threshold for further feasibility assessment.

Non-ISS Utilisation

- Two parabolic flight campaigns were performed with a total of 25 experiments. The first test flights of the Airbus A310, replacing the A300, were performed.
- Nine drop tower campaigns were completed including a 'Drop Your Thesis' student campaign.
- A 21-day bed-rest study comprising eight investigations was completed at MEDES in France.
- A nine-month Concordia isolation campaign in Antarctica comprising nine experiments was completed.

Exploration

Multi-Purpose Crew Vehicle/European Service Module (MPCV-ESM)

The contract between ESA and Airbus Defence & Space for MPCV-ESM Phase-C/D was signed on 17 November in Berlin. An agreement on the integrated schedule with NASA should lead to an ESM-1 delivery to the US on 29 January 2017. NASA's Orion Exploration Test Flight was successfully conducted on 5 December.

Lunar Exploration

The European Lunar Exploration Programme concentrates on developing exploration products for robotic lunar missions, through cooperation with Roscosmos on their sequence of robotic Luna missions, with a focus on the Luna-Resource Lander mission (2020), potentially followed in the longer term by the Lunar Polar Sample Return mission (mid-2020). Industrial activities have been initiated for PILOT (precise and safe landing), PROSPECT (drilling and sample analysis), and SPECTRUM (communications and operations) and are ongoing.

International Berthing Docking Mechanism (IBDM)

The Preliminary Design Review of the IBDM flight version in line with the new International Docking Standard took place in the autumn at ESTEC. Testing of the IBDM in the SIRRI robotic facility (BE), showed successful soft docking under all conditions as prescribed by the International Docking System Standard (IDSS). The Sierra Nevada Corporation submitted to NASA their bid for ISS Commercial Resupply Services-2 (CRS-2) with the Dream Chaser space vehicle, which might adopt the European IBDM docking system.

European eXperimental Re-entry Testbed (EXPERT)

The EXPERT aerothermodynamic flight testbed was completed and stored in the clean room of Thales Alenia Space Italy in

Turin, awaiting the integration of the Descent and Landing System (DLS).

Multi-Purpose End-To-End Robotic Operation Network (Meteron)

The OPSCOM-2 experiment was executed on 7–8 August, when Alexander Gerst controlled the Eurobot rover, located in ESTEC, from the ISS. This experiment was the second in a series of Meteron experiments, which aims to demonstrate communications and operations concepts and technologies for future human exploration missions with human robotic elements. The HAPTICS-1 investigation, which is part of the Meteron project, took place on the ISS in December as the first part of a suite of Exoskeleton experiments to demonstrate full control interaction between ground and space.

Strategic planning

The International Space Exploration Forum (ISEF) was held in Washington DC on 9 January. A total of 33 countries were represented (six at ministerial level) and two European entities (ESA and EU). The US announced their intention to extend ISS to at least 2024.

The ESA-wide internal strategic planning process for space exploration was reinforced. The ESA Strategy for Space Exploration was consolidated leading to the adoption of a Resolution at the 2014 Council meeting at ministerial level on 'ESA's Space Exploration Strategy'. Internal work on developing an ESA-wide approach to obtaining and communicating benefits resulting from investments in space exploration has been initiated.

International Space Exploration Coordination Group (ISECG)

An ISECG meeting was held from 27–29 May in Beijing, China. The Chinese National Space Agency was introduced to the work of the various ISECG working groups.

At the Senior Agency Managers meeting in Toronto, Canada, during the International Astronautical Congress (29 September–3 October), ESA took over the chair of ISECG from the Canadian Space Agency, and chaired the next ISECG workshop in Strasbourg, France, on 27–29 October.

Space Situational Awareness (SSA) programme

Space Weather (SWE)

The SSA SWE Coordination Centre (SSCC) in Brussels continued to provide a tailored space weather service to support Gaia operations in January and in particular the manoeuvring of Gaia to the L2 orbit.

The SSCC also provided a daily space weather bulletin to the Venus Express ground controllers to support the aerobraking campaign that started in the second half of May.

Near Earth Objects (NEOs)

During a regular survey observation with ESA's Tenerife telescope on 1 February, the SSA/NEO team discovered its first comet. It was



Detecting space hazards

named TOTAS (Teide Observatory Tenerife Asteroid Survey) after the team, in recognition of their achievement.

ESA hosted the first meeting of the Space Missions Planning Advisory Group (SMPAG) meeting, on 6–7 February at ESOC. Fourteen space agencies joined SMPAG. The second SMPAG meeting took place on 12–13 June in Vienna, where ESA was confirmed as a chair.

Monostatic and bistatic breadboard surveillance radars

The initial test and validation campaign using the monostatic breadboard radar in Spain was devoted to the assessment of the radar performance. It resulted in the detection and follow-up of more than 300 different objects.

Space Surveillance and Tracking (SST)

An augmented system for front-end SSA product dissemination was developed and accepted. New features include reentry predictions and fragmentation detection in addition to improved conjunction predictions.

Ground segment and flight operations

In 2014, Human Spaceflight and Operations operated 15 spacecraft from ESOC, three from Redu and managed the four Galileo In Orbit Validation spacecraft operations performed from Fucino and the German Space Operations Centre (GSOC).

Gaia has been in its final L2 orbit since January and full science operations started after completion of its commissioning in July. After one year in orbit all parts of the sky have been covered at least once. The mission is producing about twice the amount of science data predicted before launch.

All three Swarm satellites reached their final orbits, commissioning was completed and their full science phase started in April.

The Venus Express spacecraft performed aerobraking activities between the end of May and 11 July. Critical operations were performed with the pericentre altitude decreasing by several kilometres per day until the 'plateau' was reached in early June. The spacecraft continued science operations in the new orbit

until the fuel was fully consumed at the end of November. The spacecraft signal was recovered and, after confirmation of fuel depletion, the mission was declared terminated in mid-December.

The test campaign for the commissioning of the inter-orbit optical communications link between Sentinel-1A and Alphasat was completed. It included the end-to-end downlink of payload data from Sentinel-1A to the ground.

The Collaboration Agreement on the European Ground Systems Common Core was signed by ESA, CNES (CST), DLR (GSOC), Airbus Defence and Space, Thales Alenia Space and OHB System in November, setting the basis for the joint development and exploitation of the system.

On 23 June the Space Debris Office hosted an event for internal and external users to showcase their tools and present the new Space Debris User Portal. This new website will enable rapid updates for experts using ESA's space debris software tools for risk assessment and analysis of debris mitigation actions.

Education & outreach

The Zero Robotics Challenge 2013 finals with 170 participants from various ESA Member States took place on 17 January at ESTEC. The activities included robotics lectures and demos,

eac

EAC supported the launch of two ESA astronauts in 2014: Alexander Gerst and Samantha Cristoforetti. With these back-to-back missions of two ESA astronauts, Europe is represented for an entire year on the ISS, a first in the history of the ISS programme.

Another first was the direct transfer of Alexander Gerst back to Cologne after his landing in Kazakhstan instead of flying to Houston, as was the procedure before. Implementation of this 'Direct Return' procedure was carried out in cooperation between EAC and the German Aerospace Center (DLR) and its EnviHab facility. 'Direct Return' further strengthens Europe's capabilities in human spaceflight, in particular in the area of post-flight rehabilitation. The EAC rehabilitation facility was upgraded accordingly.

Astronauts Andreas Mogensen, Timothy Peake and Thomas Pesquet continued their training and medical preparations for their upcoming missions. With their assignments to their missions in September 2015, December 2015 and November 2016 respectively, all ESA astronauts of the class of 2009 have now been assigned to an ISS mission. Never before have so many missions been handled simultaneously at EAC.

and a two-hour live connection with the ISS and the other organising partner (Massachusetts Institute of Technology) to witness the finals happening live on the ISS. The winning alliance comprised teams from Italy, Germany and Spain.

The Mission X 2014 Train like an Astronaut project began on 13 January. The opening event took place at Space Expo, Noordwijk, the Netherlands, on 31 January. A number of primary schools participated in nutrition and exercise activities for nine weeks.

The production of the ATV series Education Kit 'Visionary Science Concepts that Made History' was completed in time for the ATV Georges Lemaître launch. The kit contains educational videos on the namesakes of the five ATV missions, and collectable pins and patches.

A number of videos were produced and published:

- 'Alexander Gerst's Earth Timelapses', an Ultra High Definition (UHD or 4K) video made from sequential photos taken on board the ISS was published on 22 December. It received more than two million views on ESA's YouTube channel, the highest number ever for a video produced by ESA.
- 'Destination Moon', providing the scientific rationale for future Moon South Pole exploration missions.
- 'Samantha talks Futura' and the 'Blue Dot Mission Summary' were published on the ESA YouTube channel in November.



'Direct Return' for Alexander Gerst as he arrives at Cologne's airport in Germany within 12 hours of landing in Kazakhstan after spending almost six months in space

The exchange between EAC and the Astronaut Centre of China (ACC) was continued in meetings of the two working groups on operations and on space medicine.

EAC received more than 7000 visitors in 2014 including politicians, business people, students and school classes. EAC remains a major outreach factor for human spaceflight activities.

ESOC and the Stations

The first optical inter-satellite link between Sentinel-1A and Alphasat was demonstrated in November paving the way for the future European Data Relay System (EDRS) for faster Earth observation imagery.

ESOC conducted the highly successful critical Rosetta operations culminating in the Philae landing on 12 November. These events sparked exceptional interest across the world with hundreds of guests and media representatives present at ESOC and unmatched worldwide media coverage.

The stations

The Estrack ground stations again provided highly reliable Telemetry, Tracking and Command and science payload data services throughout 2014 to ESA missions.

In addition, cross-support was provided to ESA's partners such as NASA (for example, Juno flyby, Stereo-A/B and Maven operations), CNSA (Chang'e-3 lunar rover lander and Chang'e-5 reentry operations), Eumetsat (MetOp-A/B) and DLR (Galileo).

A new deep space Ka-band uplink capability was demonstrated from Malargüe, Argentina, in July in cooperation with NASA/Juno, as a preparatory step for a future radio science capability.





technical & quality management

Continuing to drive the performance of the Agency, the Directorate has expanded its scope to include advanced concepts, strategic technologies and other technical programmes, enabling ESA to move ahead of the emerging technological curve.

ESA provides European space missions with the novel technologies, technical infrastructure and quality control required to make them a tangible reality. By way of example, ESA's long-term fostering of innovation has given rise to the low-intensity low-temperature solar cells powering the Rosetta comet chaser, the high-performance mirrors and cooling systems enabling the Herschel and Planck observatories and the high-accuracy APS star trackers serving the Sentinels and BepiColombo, along with the optical bench at the heart of LISA Pathfinder – engineered to map the very slight curvature of local space time.

Continuing innovation is also an essential ingredient in the ongoing competitiveness of Europe's space industry in global markets, creating new jobs and stimulating growth – the key focus of the Agency's 2012 mandate.

The various technology development programmes can be considered a portfolio of tools designed to enable different classes of innovators to get to work with ESA, from scientific teams to established companies (with a special emphasis on small and medium-sized enterprises (SMEs) to entrepreneurs leading new start-ups.

But the pace of innovation continues to quicken across all technology sectors. The timespan within which a given technology is investigated, qualified, put to operational use then succumbs to obsolescence is continually shrinking.

ESA has responded by emphasising timeliness in reacting to new ideas and needs – including a shorter two-year planning cycle for its flagship Basic Technology Research Programme, and streamlined procurement procedures – by increasing coordination with other technology preparatory programmes and by applying increased innovation to the research and development process itself.

Moving ahead of the curve: GSP, ACT & ESTT

In 2014, the overall scope of the Directorate was enlarged to include responsibility for the General Studies Programme (GSP) with the associated Advanced Concepts Team (ACT) and the Emerging and Strategic Technologies Team (ESTT). With this change, ESA is able to move ahead of the emerging technological curve, starting with the fundamental 'blue sky' thinking performed by the ACT Research Fellows, along with those of the Networking/Partnering Initiative, looking at themes of long-term importance to space.

Preparing a vacuum chamber in ESA's Propulsion Laboratory, one of a suite of technical laboratories at ESTEC, Noordwijk, the Netherlands (ESA/G. Schoonewille)

Technology Research Programme (TRP)



The 'dropship' quadcopter and mockup rover used for testing ESA's latest StarTiger project, Droptor (Airbus Defence & Space)

The long-running TRP remains ESA's innovation factory, serving all programme domains in the area of lower technology readiness level R&D. Besides its adoption of a faster tempo in 2014, the TRP continues to operate more efficiently, further shrinking its overheads to channel more funding to industry. The TRP is also notable as the programme that supports the most SMEs across all Member States, accounting for around a quarter of its yearly budget.

In 2014, some 248 TRP activities were initiated, while 170 were closed. Within the TRP, the Networking/Partnering Initiative, StarTiger and ongoing Innovation Triangle Initiative have continued to be supported – notching up some impressive results, such as StarTiger's 'dropship' precision-landing system for Mars. The new TRP plan for 2016–17 is in active preparation, also including technology developments in support of the Cosmic Vision Plan jointly funded with Science.

General Support Technology Programme (GSTP)

The higher technology readiness level sixth GSTP has similarly seen its overheads decreasing to its 20% target, channelling more of its funding to industry. Furthermore, in an indicator of its popularity, the optional GSTP-6 has received almost double the contributions of its predecessor, GSTP-5, at around €600 million. 2014 saw the initiation of 104 new GSTP activities – from a demonstrator of a multi-frequency high-resolution geostationary sounder to an improved germanium wafer technology for multi-junction solar cells – while 74 were

closed, notably including a CCSDS (Consultative Committee for Space Data Systems) compression ASIC, critical for missions generating high volumes of onboard sensor data.

GSTP Element 1 Core technology activities for projects and industry made progress, covering such significant projects as the European Ground Systems-Common Core and SAVOIR (Space Avionics Open Architecture), both set to make major changes in the way space systems are developed and delivered. GSTP Element 2, a new permanently open invitation to tender, otherwise known as the 'Competitiveness Element' was launched in 2014 with a streamlined application process to boost the frequency and quality of industry proposals. Over 30 proposals were received within the first few months of launch, and more funding than in the previous four years of its predecessor.

Under GSTP Element 3, Technology Flight Opportunities aim to make more opportunities available for in-orbit demonstration of technology through setting up a database of flight opportunities and identifying candidates for flights, building on previous experience such as the technology-proving Proba family of minisatellites described below, and the AIS (Automatic Identification System) receiver demonstrated on the ISS, with an updated version flown to the Space Station in December.

Dedicated CubeSat missions are also in preparation: due to their high degree of modularity and extensive use of commercial off-the-shelf subsystems, CubeSat projects can be readied for flight on a much more rapid basis compared to traditional satellite schedules, typically within one to two years. The missions being supported comprise SIMBA (solar-irradiance-measuring), KARMAN (testing novel reentry technologies), Picasso (surveying properties of the upper atmosphere) and GOMX-3 (evaluating aircraft ADS-B reception and geostationary comsat spot beam quality using a software-defined radio).

A final Element X for Breakthrough Missions is dedicated to small missions demonstrating breakthrough concepts and approaches as a way of preventing promising new technologies falling into the 'Death Valley' between initial development and subsequent project take-up. The most advanced example is precision formation-flying mission Proba-3, whose Phase-C/D was initiated in late 2014. The Asteroid Impact Mission is in the preparatory stages with NASA, intended to demonstrate technologies for small science and exploration missions, with an emphasis on deep-space telecommunications – and the opportunity to fly Europe's first deep-space CubeSats.

ESA's previous in-orbit demonstration missions, developed through earlier incarnations of GSTP, all had busy years. The hyperspectral-imaging Proba-1, launched in 2001 and subsequently harnessed as an Earth Observation mission, chalked a remarkable 14 years in space. The Sun-monitoring Proba-2, meanwhile, surpassed five years in orbit, now serving as a space weather platform for ESA's Space Situational Awareness programme as well as a scientific tool. Proba-V's first year in orbit was marked by its official takeover of the ongoing task of monitoring all terrestrial greenery from the Vegetation instrument on the full-sized Spot-5 satellite. Proba-V has also won distinction as the first satellite to pick up ADS-B signals from orbit, and for the first use of gallium nitride technology in its communication subsystem.

Programme quality control & procurement

In 2014, for the first time, Product Assurance and Safety (PA&S) engineers assumed oversight of technology programme developments up to Technology Readiness Level (TRL) 4–5 seeking to increase confidence in their reliability. In another innovation, procurement rules have been streamlined for purchases under €2 million so as to cover 90% of technology activities.

Cross-cutting initiatives

The Directorate's technology programmes are complemented by the multidisciplinary approach embodied in the cross-cutting initiatives, the first four of which were introduced in 2012 to address new technical challenges. Clean Space, the most mature of the quartet, has succeeded in putting ESA at the forefront of technology to meet worldwide requirements for a more sustainable space industry, with a total of 44 commitments in 2014, variously implemented through the GSP, TRP and GSTP and representing a budget of €14 million.

Innovation is stimulated when different disciplines meet: out of the green technologies branch of Clean Space and exploratory activities in the metallurgy and metal domains, it was decided to start a cross-cutting initiative on Advanced Manufacturing, based on new techniques set to revolutionise traditional approaches to parts design, development and verification. A successful Additive Manufacturing workshop in October served to elaborate a roadmap and workplan on this sector, harmonised with industry, Member States and other stakeholders. In addition, a new programme is being developed from Clean Space: the e.Deorbit mission will be proposed to the Council meeting at ministerial level at the end of 2016.



Advanced Manufacturing: this prototype instrument casing demonstrates the potential advantage of using 3D printing to create very complex space parts in a single piece

Clean Space

The implementation of Clean Space activities through the GSP, TRP and the GSTP continued with a total of 44 commitments representing a budget of €14 million.

- 2014 also saw the completion of a lifecycle assessment (LCA) study analysing the environmental impacts of two entire space missions, one for a telecommunications satellite and one for an Earth observation satellite. Furthermore, two more studies were launched: an LCA of space materials and processes and an LCA of space propellants. These will help build an internal database, to be integrated in the Concurrent Design Facility in order to allow the environmental performance of a design to be considered during the Phase-0.
- The Preliminary Requirements Review for e.Deorbit concluded the three parallel Phase-A studies committed to in 2014 for a budget of €2.25 million. The recommendations of the board of senior ESA project managers were implemented and the e.Deorbit tender for Phase-B1 prepared. Consultations took place with delegations on funding and partnerships. An e.deorbit Workshop was held in May to discuss the system and technical challenges of such a mission.

The other cross-cutting initiatives are progressing well. The Technology Breakthroughs for Science aim to develop technology for major advances in different scientific domains, guided by the recommendations of the cross-sectorial Future Technology Advisory Panel, beginning with cold atom devices and large deployable ultra-stable structures as key enablers, and in-orbit propulsion and radiation protection as key challenges, for which roadmaps are being created.

The Space and Energy initiative has made an effort to consolidate relevant technology roadmaps and identify the most prominent fields for joint research with the energy sector. In April, the ESA-organised European Space Power Conference was held, while ESA also had a major presence at September's European Photovoltaic Solar Energy Conference and Exhibition.

Meanwhile the development roadmaps produced through the Technologies for Exploration initiative have become international points of reference, not just for European industry but for the International Space Exploration Forum in Washington DC in January, and as input by the International Space Exploration Coordination Group.

Technology Transfer Programme

Looking to terrestrial sectors, ESA's Technology Transfer Programme has proved a powerful means of fulfilling the Agency's 2012 jobs and growth mandate. At the same time as the number of technology transfers have increased, 2014 saw 89 new start-up companies enrolled in the incubation programmes of the nine existing ESA Business Incubation Centres (BICs), bringing the total number of incubated start-ups during the last decade to 290. New BICs were opened in Portugal and Spain, while the TEC Directorate also took over management of ESA's patents and licences portfolio.

Coordination and harmonisation

Coordination efforts are increasing, multiplying the effectiveness of every euro invested. A regular cycle of presentations to the Technology Advisory Working Group provides added visibility on technology programmes and coordinated activity, while GSTP Element 1 planning has been better aligned with the 8–10 roadmaps produced annually by ESA's Technology Harmonisation process with industry and Member States, joined for the first time in 2014 by the European Commission.

The release of the eleventh edition of the European Technology Master Plan was accompanied by the publication of an authoritative research and development blueprint designed to make best use of the resources and competencies of all actors – ESA, EU and Member States.

Standardisation & industrial support

A comprehensive set of training materials was introduced on the major elements of the ESA-supported standardisation systems: European Cooperation for Space Standardization (ECSS) and European Space Components Coordination (ESCC). It has already been used extensively in support of new Member States, with a number of dedicated training sessions performed aimed at facilitating their participation in ESA projects. A new Space Debris Mitigation Policy was issued in early 2014 specifying a number of high-level requirements for use in all new ESA programmes, together with implementation and review procedures. It has already been implemented by a number of new projects and has proven its worth as a pragmatic regulatory reference in real-world applications. An accompanying book of guidelines assists with straightforward interpretation and implementation.

In a milestone for readying the Controller Area Network (CAN) vehicle bus protocol for space, the CAN ECSS standard was submitted in November. A second major breakthrough was the close cooperation with three major electronic component manufacturers that resulted in availability of a radiation-hard Qualified Manufacturers List (QML) level and ISO-compatible CAN transceiver. In addition, after a decade of effort, a complete toolbox has been made available for CAN in space, allowing smooth adoption by projects. The recently published ECSS standard on welding will support the maturation of friction-stir welding within the Advanced Manufacturing cross-cutting initiative.

The European Components Initiative (ECI) marked its first decade of operation. The ratio of European to non-European EEE parts used in a typical European telecom satellite has significantly increased, with 50% of part types (in terms of value) now procured from European suppliers compared to just 15% in 2000. This improvement has been achieved one year earlier than the original target set in 2010. Recent achievements include a European gallium nitride (GaN) 0.5-micron foundry process. This GaN technology has performed flawlessly since 2013 as the primary transmitter of the Proba-V satellite.

Support for Thales Alenia Space Italy's recovery of operations and maintenance of customer confidence for its L'Aquila plant, devastated in the 2009 earthquake, culminated in the formal opening of a new custom-made facility. The company honoured ESA's recovery team during a special event in August.

Engineering support

Hands-on engineering skill and specialist expertise is provided to all project teams across the Agency. Specialists typically support several projects at around 30% of their man-hours, along with managing R&D projects, while others are embedded within mission teams. This year saw the support of the final preparation, launch and follow-on commissioning of the final ATV, first two Galileo Full Operational Capability satellites, and pioneering Sentinel-1A, along with the commissioning of star-mapping Gaia, launched at the end of the previous year. ATV *Georges Lemaître* was notable for the rigorous 'disinfection, cleanliness and contamination control' campaign required ahead of launch.

Support also extended to in-orbit troubleshooting when called upon, such as for the salvage effort necessary when a launcher injected a pair of Galileo satellites into the wrong orbit, resulting in a solution that made them usable for navigation purposes. Optical specialists were called in to perform stray light analyses for Gaia, assessing the consequences of unexpected outgassing products that had apparently frozen in place around the optical assembly and sunscreen. Meanwhile the European High-Power

estec

With ESA enlargement to include new Member States, ESTEC is providing even more support to companies both large and small, as well as access to unique laboratory equipment and specialised expertise from all Member States. This pooling of valuable resources for the benefit of all has been key to the last 50 years of European successes, discoveries and ongoing industrial competitiveness.

In those 50 years many things have evolved: today 20 of the former ESTEC laboratories form a coordinated network across various Member States outside the establishment. In parallel, ESA continues to innovate, developing the tools, processes and ways of working with academics, research institutes and industry to keep Europe at the forefront of the global space sector and its evolution.

The ESTEC Test Centre enjoyed a busy 2014. The Large Space Simulator performed major 10 solar constant test campaigns for both BepiColombo's Mercury Planetary Orbiter and an engineering model of Solar Orbiter's heat shield. Environmental verification of the first four Galileo Full Operational Capability satellites was also completed, while the Intermediate eXperimental Vehicle spaceplane underwent a full test campaign ahead of its mission. The VTC 1.5 vacuum chamber was used for long-duration

Radio Frequency Space Laboratory in Barcelona, Spain, played a major role in replicating an anomaly suffered by the fourth Galileo In-Orbit Validation satellite's navigation antenna.

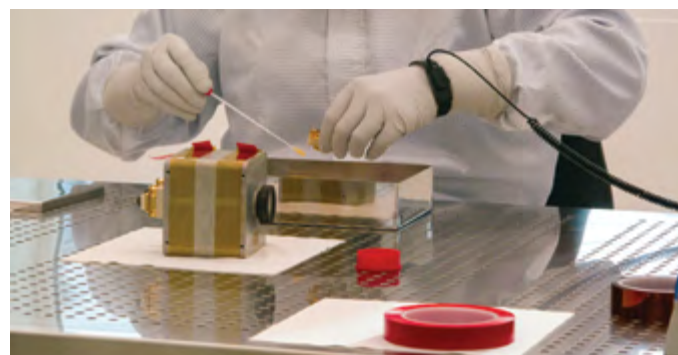
Laboratories

The Concurrent Design Facility (CDF) carried out 14 assessment studies, including AIM3P, the Asteroid Impact and Deflection Assessment (AIDA) with NASA, the Next Generation Cryogenic IR Telescope (NGCCryoIRTel) with JAXA and next-generation X-ray telescope Athena. The CDF also made its software available in an open source version to ESA Member State industry and institutions, for wider use by the concurrent design community.

A large number of long-duration tests was performed by the Mechanical Systems Laboratory, including novel mirror technology, Galileo antenna patches and high-temperature BepiColombo and Solar Orbiter coatings. The Life, Physical Sciences and Life Support Laboratory contributed to the cleaning and sterilisation of ExoMars 2016 elements, in support of planetary protection requirements.

The Optics and Opto-Electronics Laboratory assessed solid immersed gratings for CarbonSat and Sentinel-5.

ESA's Navigation Laboratory performed the first testing of mass market Galileo chipsets as part of its wider Galileo system validation activities. The Advanced GPS Galileo ASIC (AGGA4) multi-GNSS baseband processor was also produced and tested for use in many missions including Proba-3 and MetOp Second Generation.



Final tests of the ExoMars 2016 Descent Camera flight unit in ESTEC's bioburden-controlled Life, Physical Sciences and Life Support Laboratory

thermal testing of BepiColombo's solar cells as well as early thermal qualification of Solar Orbiter instruments. Along with significant non-space testing – with Hydra subjecting large electrical transformers to simulated seismic loads – in-house engineering and industrial support was performed to maintain the reliability of Test Centre facilities.



ESA's Intermediate eXperimental Vehicle removed from its protective container, safely inside the cleanroom environment of ESTEC's test centre in July





resources management, corporate activities & external relations

Human Resources, Facility Management & Informatics

Human Resources Policy

In the framework of the Organisation Renewal Programme (ORP), the Human Resources Department has:

- reviewed the roles and organisational setup of the department with the aim of moving closer to agency core business, enhancing support to individual staff, strengthening the HR capability to provide HR policy and organisational development (OD) support, and streamlining HR operations. Implementation of the new structure began on 1 April;
- initiated and managed projects addressing the staff life cycle, with the aim of attracting and selecting the best candidates, allowing staff to develop their full potential, fostering exchange and cross-fertilisation through mobility, supporting managers and staff to give constructive feedback, and developing leadership skills at all managerial levels (2014 output of these projects included the introduction of a video tool for pre-selection of candidates, the start of a mentoring scheme for newcomers, a revision of the ESA staff contract policy so that A and L categories of staff with standard ESA contracts can be granted an indefinite contract after four years of service, and the delivery of a job family framework to support career management).

Learning and Development

The ESA Learning and Development portfolio covers fundamental knowledge and skills supporting both organisational and individual needs at ESA: Business Efficiency and Personal Development, ESA Environment, Leadership and People Management and Technical and Project Management. After a successful first delivery in 2013, the Project Manager Training targeting current and future ESA project managers of medium to large space projects was delivered to a second group of selected participants in 2014, and has contributed to building a Project Manager Forum where ESA project managers can share experience and best practices.

In the framework of the reorganisation of the Human Resources Department, Learning and Development (including the ESA Internal University, Technical Training at ESTEC and locally delivered in-house training) was integrated into the new HR Competence and Policy Centre. While keeping the broad range of courses and seminars, the overall budget was contained at a level of €2.2 million.

Trainee and Internal Research Fellows Programmes

The ESA Young Graduate Trainee (YGT) programme provides hands-on experience to young graduates from ESA Member States and Cooperating States. In 2014, of the 2089 applicants to the YGT programme, 74 candidates of 18 different nationalities

Alexander Gerst's patch celebrating 50 years of European cooperation in space floating in the Cupola of the ISS in September (ESA/NASA)

received an offer after having conducted successful interviews. In total, 145 YGTs were in post at ESA during the year.

The Internal Research Fellow Programme offers scientists and engineers the opportunity to carry out advanced research in a variety of disciplines related to space science, space applications and space technology. In 2014, the Internal Research Fellow programme welcomed 16 new recruits of seven different nationalities. In total, 61 Postdoctoral Research Fellows were in post at ESA during the year.

National Trainee programmes

The National Trainee programmes for Germany, Luxembourg, Belgium and Portugal continued their activities successfully during 2014, while a new programme for Switzerland got under way.

Statistics

Staff

At the end of 2014, the Agency had 2233 staff in post (including 61 Supernumerary), compared with 2234 (including 65 Supernumerary) at the end of 2013. There were 60 external recruitments (Advance Recruitment, Permanent and Supernumerary) in 2014.

Pensions

At the end of 2014, there were 1341 pension recipients, compared with 1299 at the end of 2013.

Member States	Hors Classe	A	L	B	C	Total Staff
Austria		36		2		38
Belgium	1	84		5		90
Czech Republic		3				3
Denmark		17				17
Finland		14		5		19
France	3	415	9	79	4	510
Germany	3	375	5	42		425
Greece		18		2		20
Ireland		22		8		30
Italy	3	397		23		423
Luxembourg		2				2
The Netherlands	1	64		15		80
Norway		20		2		22
Poland		4				4
Portugal		23		2		25
Romania		2				2
Spain	1	191		4		196
Sweden		38		3		41
Switzerland		25		3		28
United Kingdom		212	3	18		233
Total	12	1962	17	213	4	2208
Non-Member States						
Canada		24		1		25
Total		24		1		25
Grand Total	12	1986	17	214	4	2233

ESA staffing as at 31 December 2014

Facility Management

ESA Facility Management has continued to drive forward a coherent strategy for FM services across the Agency's establishments and centres. At Redu, the preferred model of a results oriented, KPI-driven, outsourced FM contract was implemented. In ESAC, a similar, fully competitive tender was issued and at EAC a full review of service delivery was completed, taking account of the centre's presence within the DLR campus. Lessons learned in this context were also applied to the evolving strategy for support at Harwell, where the UK-financed ECSAT building was under construction.

ESA FM has a broad remit, with responsibilities including industrial health and safety, the maintenance of general purpose infrastructure, the provision of business support activities, physical security services and the management of business travel.

The need to drive ever greater efficiency in response both to budgetary pressure and evolving environmental regulation required considerable tightening of FM internal management processes. To this end, a business analysis framework (based on the balanced scorecard approach) has been fully integrated and used to guide stringent, twice-yearly reviews. The overall aim has been to ensure optimised allocation of resources across a geographically and functionally disparate family of centres. Internal and external benchmarking techniques are also applied as a control mechanism.

A key feature of a high-performing FM service is that it should be little noticed. ESA's core business should be facilitated in fit-for-purpose real estate with high quality business and staff services provided within the allocated resources. Annual surveys of customer satisfaction demonstrate that this has largely been achieved but areas for improvement will always exist and be the focus of attention for the Agency's FM professionals.

Information Technology

Four main areas were in focus in 2014: improving the Agency's efficiency and implementing cost-saving measures; supporting secure solutions for daily work and securing ESA's assets; supporting ESA's space activities; and innovation and transformation of IT services.

In 2014, the IT Department transitioned to the new service provider consortium. The handover, after a dedicated preparation phase, began in July 2014. Users now benefit from improved support, including new face-to-face local service desks, and greater security measures available to all staff and contractors. More new services are being rolled out on an incremental basis.

The ESA IT Department is upgrading the technology and efficiency of the computing hardware infrastructure currently available to the Agency. Virtualisation technology has allowed consolidation and rationalisation of computing needs and the adoption of a private cloud infrastructure has greatly improved availability and flexibility in the provision and use of storage and computing power.

The IT Department has been working to update substantially the corporate IT unified financial and procurement system,

esa-p, which enables staff to extract and process financial data. Timesheet Accounting System upgrades were released in 2014. The benefits for ESA include a uniform and consistent view of and reporting on manpower usage and measurement of manpower utilisation. The Electronic Tender Management System, the Entity Management System and the Project, Programme and Strategic Planning project, as well as the virtualisation and upgrade of HR systems (such as the HR Management System and HR Personnel File System) reached major milestones in 2014. Deployment of the planning tool will contribute to the improvement of the Agency's controlling model and to efforts to achieve increased efficiency across Directorates.

Operational IT has been instrumental in supporting many crucial events for ESA in 2014. The Rosetta landing and its subsequent visibility worldwide relied on ESA IT infrastructure. In addition, Sentinel-1A, and the Galileo launch and critical early orbit phases all had the underpinning infrastructure and the technical IT support by ESA IT.

A new social collaboration platform, esaconnect, was introduced, along with a new corporate-document-centric collaboration platform, Sharepoint, and the Polaris project management system was consolidated, ensuring a robust set of tools to enhance ESA's inner workings.

Procurement, Financial Operations & Legal Affairs

Industrial Policy & Auditing

Geographical return matters

On 13 June 2013, Council confirmed its decision of June 2010 on the discontinuation of geographical return statistics at end-2014, under the condition that a 0.96 lower limit for the cumulative return coefficient would then be reached for all Member States (excluding those still undergoing transitional arrangements in line with their Accession Agreements). A Strategic Initiative (StrIn) was also decided on to facilitate the achievement of targets for critical countries. At the end of 2014, the Formal Review of the 2010–14 period thus had a particular relevance and a significant effort was made by the Executive to ensure its success.

At the end of 2014, the geographical status for each state is shown in the table below. All Member States are well above the threshold of 0.96, with all countries above 0.98 and all but two countries above 0.99. The discontinuation has thus been confirmed and a new set of return statistics were to be started on 1 January 2015. This discontinuation after 15 years of statistics represents a huge success for the Agency and could only be

Country	Contract value - Unweighted (millions)	Contract value - Weighted (millions)	Ideal amount (millions)	Return Coefficient
Austria	383	360	358	1.01
Belgium	1779	1544	1538	1.00
Czech Republic **	37	37	38	0.98
Denmark	229	200	205	0.98
Finland	151	150	152	0.99
France	8072	6260	6195	1.01
Germany	5667	4928	4886	1.01
Greece	44	43	41	1.06
Ireland	112	109	109	1.00
Italy	4016	3344	3292	1.02
Luxembourg	72	71	70	1.00
Netherlands	1544	744	655	1.14
Norway	344	326	330	0.99
Poland *	21	21	28	0.73
Portugal	110	104	102	1.03
Romania *	17	16	22	0.75
Spain	1606	1372	1363	1.01
Sweden	607	562	570	0.99
Switzerland	947	881	889	0.99
United Kingdom	2466	2083	2112	0.99
Subtotal	28224	23157	22952	
Canada	187	180	181	0.99
ESA Total	28411	23337	23134	

Estonia, Hungary and Slovenia do not appear in the table

* Member States under transitional arrangements

** Member States ending its transitional arrangement on 31/12/2014

Geographical distribution of contracts at end of 2014 (without ISS exploitation)

achieved thanks to close cooperation between the Executive and the Member States, and among the Executive as a whole.

Evolution of Industrial Policy

The Council decided in October on the following industrial policy rules:

- a) The lower limit for the cumulative return coefficient for the period 2015–19 is 0.91.
- b) For mandatory programmes and activities, a minimum return coefficient of 0.9 is established for each Member State for the formal review taking place at the end of 2019.
- c) A lower limit for the return coefficient at optional programme domain level of 0.9 is also established for each Participating State at the formal review taking place at the end of 2019.

In addition, Council decided on measures to ensure the fulfilment of these rules, with the creation of a steering committee at Director level for mandatory programmes and activities, and the reinforcement of the proactive management for industrial policy matters in all optional programmes.

The 2014 High Level Forum with industry

In June, the second High Level Forum, created by the 2012 Council at ministerial level, took place at ESTEC. Two round tables were organised. The first: 'How can ESA technology, procurement, industrial policies better meet the needs of industry?' led to industry recommendations inviting ESA to give more responsibility to industry, to better support upstream sector competitiveness, and to have, in addition to its customer role, a partner role for industry. The second: 'How can ESA better contribute to growth & employment in the European space sector?' led to industry expressing its support for a role for ESA in ensuring that European industry will reap in space services and applications the full benefits of European investments in space research and infrastructures.

ESA socio-economic impact assessment activities in 2014

The ESA Socio-economic studies Steering Group (SES-SG) with Member States was established in 2014 to coordinate socio-economic assessment activities, methodologies and best practices across Member States. A report on the socio-economic impact of space activities was delivered to the 2014 Council meeting at ministerial level and included the main results of an extensive impact assessment in the launcher sector.

Industrial Auditing

In addition to activities on industrial rates and overheads, the initial audit activities relevant for the LEAP Ariane Supplementary Exploitation programme were undertaken, the results being used for the calculation scales presented for the LEAP programme at the 2014 Council meeting at ministerial level.

Following the conclusion in 2013 of the trilateral arrangements with, respectively, CNES/DGA for France and ASI for Italy on cooperation in industrial auditing activities, a similar arrangement was concluded with DLR/BMWi for Germany.

The opportunity savings, that is to say the difference between demands from industry and actual agreements on industrial

rates after audit, fluctuate between 2% and 4% of the annual value contracted to industry.

New Members and Cooperating States

Transitional activities to support new ESA Member States – Czech Republic, Poland and Romania – through dedicated 'Industry Incentive Schemes' continued through the placement of competitive calls and contracts to create and develop space-related industrial capabilities. Specific courses on ECSS (European Cooperation for Space Standardization) and ESA procurement were held in these new Member States.

Negotiations with Hungary and Estonia for their accession to the ESA Convention, including a joint assessment of space capabilities, were successfully concluded.

In the framework of the Plan for European Cooperating States (PECS), competitive calls were placed in Estonia, Hungary and Lithuania. A joint assessment on the industrial capabilities of Bulgaria and Slovakia was carried out in the framework of European Cooperating State negotiations.

Procurement

Launchers

Activities were focused on the preparation of the Council at ministerial level in December. Discussions addressed the configuration of the new Ariane 6 launcher and the new governance model, which was then reflected in a Memorandum of Understanding agreed between Airbus Safran Launchers (ASL), Airbus Defence & Space, Safran, CNES and ESA signed on 18 December.

Science and Robotic Exploration

Major efforts were devoted to the adoption of the procurement approach for the JUICE (Jupiter Icy moons Explorer) mission and subsequent release of the Invitation to Tender for the implementation Phase-B2/C/D and -E1, and to the signature of the Rider covering the implementation phase of the CHEOPS (Characterising Exoplanet Satellite) mission.

Earth Observation

With the signature of the Copernicus Delegation Agreement between the EU and ESA, dedicated procurement actions were initiated such as those for the C and D satellites in the Sentinel-1, -2 and -3 satellite series, the Dataware House and the extensions of the GMES Earth Observation Ground Station contracts.

Procurement actions were brought to completion or initiated, relating to MetOp Second Generation (MetOp-2G), the Phase-C/D price conversion of Meteosat Third Generation (MTG), the Contract for Sentinel-5 Phase-B2/C/D/E1, and Jason CS (Sentinel-6).

Human Spaceflight

The Ceiling Price conversion proposal for the Multipurpose Crew Vehicle/European Service Module (MPCV-ESM) was negotiated, leading to the contract signature on 17 November in Berlin. The ISS Exploitation and utilisation services were further extended beyond 2014.

Telecommunications and Integrated Applications

On the European Data Relay System (EDRS), negotiations took place in relation to the Service Level Agreements for Copernicus and for complementary services.

The contracts for the definition phase of Neosat and of Electra have been signed as well as the Phase-C/D contract for SAT-AIS with Luxspace (LU) under a Public-Private Partnership (PPP) with ExactEarth (CA).

Navigation

With respect to the Galileo Full Operational Capability (FOC) Phase, the contractual baseline of the different Work Packages was revisited in line with a realigned schedule.

The Delegation Agreement for Galileo has been signed and led to the initiation of several additional procurement activities in support of the main Work Packages and security-related activities.

Technical and Quality Management

With regard to In-Orbit Demonstration activities, the two Proba-3 contracts, namely the Mission Contract with SENER (ES) and the Instrument contract with Centre Spatial de Liège (BE) have been signed.

Finance, planning & control operations

Financial Operations

Intense effort was devoted to certifying the accounts and focusing on the accounting dimension of the Agency's payments. This entailed the analysis, validation and approval of all payments to suppliers, representing approximately 40 000 invoices and 35 000 mission settlements.

A second 'procure-to-pay' workshop was held to identify improvements to ease payment to suppliers while ensuring full compliance with accounting standards, thereby improving the efficiency of payment operations. Fifty staff from Finance, Procurement and Programmes were involved. Concrete measures were adopted and an action plan implemented. A valuable 'good practices' manual instructing how to draft contractual clauses with financial and accounting impact was adapted into an Administrative Instruction which, when approved, will be promoted ESA-wide with relevant training. A full set of statistics on time-to-payment to industry was also developed, confirming that, on average, the 30-day payment term is respected.

Corporate Controlling

Corporate planning and control played a crucial role in 2014 in proposing budgets that were commensurate with cost plans approved by the Programme Boards, taking into account the funding capacity of Member States, ensuring balance between cash outflows and available resources and avoiding the need to call for a loan.

In order to improve and strengthen cash management in line with programme and activity needs, a Resolution to Strengthen the General Treasury was adopted by Council in December, improving the contribution call-up scheme by reinstating the scheme in force before 2010 that proved effective for several decades (40% on 31 January, 40% on 30 June and 20% on 31 October).

Significant preparatory work was devoted to preparing the Council meeting at ministerial level in December, including the elaboration of financial input for new programmes, the assessment and optimisation of planning for ongoing programmes to support the subscription to new activities at the 2014 Council at ministerial level and finalisation of the financial part of the 2015–24 ESA Long-Term Plan.

Reporting to the Head of Finance, a Corporate Senior Financial Controller was put in place to ensure better coordination and consolidation of planning and controlling of corporate activities to support informed decision-making.

Cost

ESA's total cost in 2014 amounted to €4416 million¹ (€3831 million relating to ESA programmes and activities, €586 million to third-party activities.)

Mandatory Activities represented 16% (€718 million) whereas 63% was on Optional Programmes (€2795 million), 2% on activities associated to the General Budget (€103 million), 13% on programmes financed by third parties (€586 million)² and 5% others (€215 million).

The core elements of the Agency's Mandatory Activities are the Scientific Programme and Basic Activities including, *inter alia*, the General Studies Programme, Core Technical Expertise, Investments, Technology Research Programme, Earthnet and Long-Term Data Preservation. In 2014, the Scientific Programme cost €484 million (11% of the total). In addition to the development activities, the Scientific Programme funded the operations of 11 satellites in orbit.

The execution of programmes including the study, development, launch and operations of satellites as well as the development, exploitation and maintenance of launchers and other space systems is implemented by means of the Agency's Optional Programmes to which States participate with a voluntary subscription.

In 2014, €771 million, or 17% of the Agency's total cost, was incurred on Launcher Programmes (excluding the Guiana Space Centre, CSG). Ariane exploitation activities supported Ariane 5 exploitation, which resulted in four successful Ariane 5 launches during the year. In development, activities covered Ariane 5 ME, Ariane 5/6 Common Upper Stage and Specific Ariane 6 as well as the Future Launcher Preparatory Programme (Next Generation Launcher, Intermediate eXperimental Vehicle and Period 3).

Earth Observation accounted for 16% of ESA cost, with €699 million accrued. Sentinel-1A was launched on 23 April. Costs in 2014 covered the development of the Sentinels of the GMES Space Component programme, the development of ADM/Aeolus and EarthCare within the family of Earth Explorer missions and the meteorology satellites with activities for Meteosat Third Generation and MetOp Second Generation.

Cost for Human Spaceflight, Microgravity and Transportation, amounting to €675 million (15% of total ESA cost), relates mainly to the exploitation and utilisation of the International Space Station.

¹ Excluding depreciation and provisions

² Difference with 7% income due to fixed asset transfers and cost accrued on EC-delegated agreements

In Telecommunications, which represented 7% (€306 million) of total ESA cost, the main areas were the European Data Relay System programme, telecommunication technology activities and support to industrial competitiveness (covered by ARTES elements 1, 3, 4 and 5), Alphabus subsystem development and SmallGEO.

Robotic Exploration and PRODEX cost of €230 million in 2014 corresponded to 5% of total ESA cost, mainly covering the completion of a number of technical milestones for the ExoMars 2016 and 2018 missions and good progress in technology development for the Mars Robotic Exploration Programme (MREP).

The General Study Technology Programme (GSTP), including Proba-3, Proba-V and In Orbit Demonstration Missions, covered 1.4 % of total ESA cost (€61 million).

The optional Navigation Programmes represented 1.0% of ESA's total cost (€44 million) covering mainly the GNSS Evolution programme.

In the Space Situational Awareness (SSA) Programme, cost amounted to €10 million (0.2% of total ESA cost). SSA architectural design studies activities were completed.

For Third Party activities, €396 million covered Navigation-related Activities funded by the European Commission through Delegation Agreements for Galileo and EGNOS Exploitation. In July, the new 'Delegation Agreement between ESA and the European Union on the Deployment Phase of the European Satellite Radio navigation Programme Galileo' was signed covering €1.77 billion for the deployment phase until 2016. Activities in 2014 focused on preparation of the first two Full Operational Capability (FOC) spacecraft launched on 22 August as well as preparation of the FOC satellite operations and ground system verification activities. In Earth Observation, third-party activities amounted to €131 million, mainly including cooperation with the Eumetsat on Meteosat Second Generation, MetOp and cooperation with the European Commission on Copernicus. The Copernicus agreement between the European Commission and ESA was signed on 28 October.

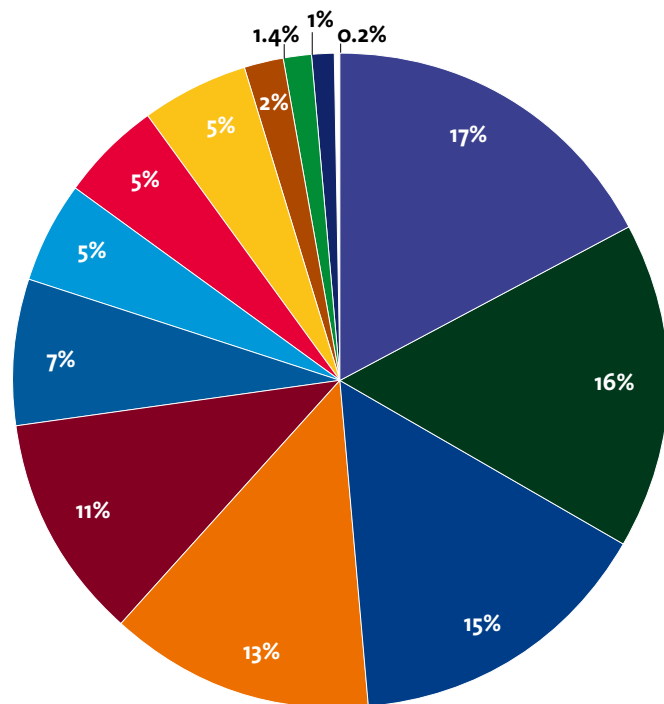
Approximately 83% of the Agency's 2014 cost was accrued on contracts in Member States for research or project-related activities, the running of technical or operational facilities, and financing capital cost and industrial development. The remaining 17% covered operating costs such as workforce and other running costs.

Income

ESA Mandatory Activities (including the General Budget and Scientific Programme) as well as activities associated with the General Budget are financed by Member States' contributions. The ESA Convention dictates that the Agency's Mandatory Programmes are to be financed using a contribution scale based on the national income of the Member States over the last three years for which statistics are available.

ESA Optional Programmes are financed by States' contributions and for some programmes by other funding entities among which the European Union. The Participating States declare a voluntary subscription on a multi-annual basis.

Third-Party Programmes are managed by ESA but totally financed by third parties. In 2014, they represented 7% of total ESA Income.



Cost (€ million)

Long-Term Plan and Key Performance Indicators

The corporate strategic planning process supported the preparation of decisions taken at the Council meeting at ministerial level in December. A noteworthy event was the organisation of the Vision 2030 Workshop in March, at which discussions took place with European space stakeholders on programmes likely to be implemented by European institutional players by 2030, also providing valuable information on how industry perceives ESA and its evolution.

The second annual report on the Agency's performance in 2013, based on selected Key Performance Indicators (KPIs), was presented to the Administrative and Finance Committee and Council.

Corporate Risk Management

The Agency-level Risk Management Framework supporting the strategic planning and decision-making process is in its third operational year. The Network of Risk Coordinators (NoRC) participated actively in the Risk Management process identifying risks to the Agency's strategic objectives and proposing mitigation actions to be implemented and

	2014	
	Mandatory Activities	Optional Programmes
Austria	2.17%	1.49%
Belgium	2.69%	7.10%
Czech Republic	0.99%	0.26%
Denmark	1.74%	0.41%
Finland	1.40%	0.38%
France	15.33%	26.20%
Germany	19.80%	27.43%
Greece	1.87%	0.07%
Ireland	1.22%	0.37%
Italy	11.59%	10.88%
Luxembourg	0.22%	0.71%
The Netherlands	4.55%	3.79%
Norway	2.39%	1.63%
Portugal	2.57%	0.38%
Poland	1.23%	0.28%
Romania	0.99%	0.63%
Spain	8.35%	3.09%
Sweden	2.55%	3.17%
Switzerland	2.88%	4.61%
United Kingdom	15.47%	6.32%
Total MS	100%	99.21%
Canada	3.86%	0.59%
Hungary		0.09%
Estonia		0.06%
Slovenia		0.06%
Total Cooperating	3.86%	0.79%
ESA	103.86%	100%

Contributions for 2014 from Member States to the Agency's Mandatory Activities and Optional Programmes as well as associated budgets are indicated

monitored. The annual report on the major risks to the Agency was presented at the June Council meeting.

In September, both Agency-wide procedures for Performance Management and Risk Management obtained ISO 9001 certification. A study assessing the possible implementation of Performance Risk Management at ESA is ongoing. Benchmarking in other similar organisations will be completed with recommendations to be issued within the year.

Legal Services

In 2014, the ESA Council took a number of decisions and adopted several ESA rules and regulations to enable ESA to carry out its actions, activities and programmes in line with the ESA Convention and all applicable laws, regulations and agreements.

All ESA regulations were revised and released to the public following an ESA Council decision. They can be found on the 'Law at ESA' pages on the ESA website. The set of ESA Regulations encompass the following legal texts:

- ESA Procurement Regulations and related Implementing Instructions
ESA/REG/001, rev.3
- General Clauses and Conditions for ESA Contracts
ESA/REG/002, rev.1
- Security Regulations
ESA/REG/004
- ESA Financial Regulations
ESA/REG/005, rev.1
- Additional Rules on Arbitration
ESA/REG/006
- ESA Staff Regulations
ESA/REG/007
- Rules on Information, Data and Intellectual Property
ESA/REG/008

Nineteen Agreements were concluded by ESA with institutional partners to develop its activities and relations. The list of Agreements signed in 2014 are found in the Annex.

External Relations

Relations with Member States

Coordination with Member States

ESA, as an intergovernmental agency, is duty-bound to pursue the collective interest of its Member States. With 20 Member States in 2014, understanding the drivers and stances of individual Member States was key and required regular follow-up and careful consideration within a well-structured cooperative framework. In addition to the preparation of the regular Council meetings, the 2014 Council meeting at ministerial level resulted in a higher number of meetings, each of which was also preceded by consultations in support of the ensuing decision-making process. The results of the workshops with Member States on National Strategies and Plans were presented at Council highlighting the different priorities and interests that underpin ESA Member States' investment in space activities. Knowing ESA Member States' views and objectives also made it possible to contribute to the preparation of the Resolution on ESA evolution adopted at the Council meeting at ministerial level. Furthermore, leveraging the acquired insight, there was closer interaction with individual delegations on specific matters such as governance and national strategies.

To further promote a common understanding on space strategies and priorities in Europe among legislators, the European Interparliamentary Space Conference (EISC), under French Presidency, discussed the theme of 'Space and Growth' in a workshop in Toulouse on 14–16 April, as well as in a plenary session held in Paris on 19–21 October. Taking stock of the know-how gained in this process and by means of thorough assessment of individual Member State profiles with regard to space activities, a comprehensive database of key figures and facts for each Member State was created, which led to the finalisation of a booklet on each Member State's space governance, capabilities and motivation to invest in space.

An overview was provided of all the celebratory events for the 50 years of European space cooperation that either took place at ESA establishments or were organised by Member States.

Relations with EU Member States not Members of ESA (EU-10)

Since January 2011, the Director General has met twice a year with the EU Member States not members of ESA (EU-10). In 2014, two meetings with EU-10 representatives took place, in June at EAC in Cologne, and in November at ESA HQ in Paris. The objectives of these meetings were to take stock of ESA cooperation with these countries and to share information on the preparation of the Council meeting at ministerial level, which the EU-10 joined as observers.

Negotiations for the accession of Estonia and Hungary to the ESA Convention were carried out throughout the year. Technical visits by an ESA expert team to interview Estonian and Hungarian firms and research institutes took place. Interviews were conducted in order to establish further contact with selected space entities from these countries and to measure their state of preparedness in the light of the future Integration Period and their industrial return.

Represented by its Minister of Economy, Lithuania signed a European Cooperating State (ECS) Agreement with ESA on 7 October at ESA HQ. Following the request for an ECS Agreement by Slovakia, a technical visit to Slovak space entities took place on 12–14 May to jointly assess Slovak space capabilities and to start identifying possible Plan for European Cooperating States (PECS) projects with ESA.

The Malta Council for Science and Technology requested in March to start discussions on an ECS Agreement. ESA co-organised a one-week radar course in Valetta from 10–15 November.

The Bulgarian Minister of Economy and Energy requested in June the initiation of negotiations on an ECS Agreement. A technical visit to assess the capabilities of Bulgarian space entities took place on 20–23 October in Sofia. The Ministry of Science, Education and Sports of Croatia requested in April to start discussions regarding an ESA/Croatia Cooperation Agreement.



Director General Jean-Jacques Dordain and Lithuanian Minister of Economy Valdas Gustas signed the ECS Agreement on 7 October, making Lithuania the eighth ESA European Cooperating State

Relations with EU institutions

In 2014, ESA continued to work closely with the European Union in several ways: on the implementation of the EU's flagship programmes Galileo and Copernicus, as well as securing its future involvement in these programmes until 2021 through

the negotiation and signature of the respective Agreements. It has sought to coordinate with the European Commission, wherever possible, the implementation of the respective research and development (R&D) programmes, and has maintained a continuous dialogue aimed at reinforcing the European Space Policy and ESA/EU institutional relations.

ESA has held discussions with the Commission on cooperation in R&D with a view to finalising a policy for ESA involvement in Horizon 2020.

Through its permanent representation in Brussels, ESA has ensured the coherent and consistent visibility of ESA in EU circles. Following the opening of the new legislature in the summer, ESA strove to consolidate relations with the European Parliament. New entry points were established, most notably with Industry, Research and Energy Committee (ITRE) and the Sky and Space Intergroup. Similarly, with a new College of Commissioners taking up duties, ESA was engaged in reestablishing relations with the individual Commissioners and their Cabinets.

ESA continued work on the evolution of ESA/EU relations, through the ESA Council meeting at ministerial level in December, which took stock of the new status quo in the relationship stemming from the signature of the agreements regarding the GNSS and Copernicus programmes.

International Relations

United States

Several bilateral meetings took place in 2014 between the Director General and the NASA Administrator that enabled the two heads of agencies to review ongoing cooperation developments as well as to define future cooperation activities

Space science remains the main field of ESA/NASA bilateral cooperation. The cooperation agreement on the Hubble Space Telescope has also been extended for a period of five years. Both agencies continued their cooperation on missions in development such as Solar Orbiter and LISA Pathfinder. Concerning the James Webb Space Telescope, a partnership between ESA, NASA and the Canadian Space Agency, the focus was on development activities.

ESA and NASA also began negotiating a Memorandum of Understanding (MoU) to cover the NASA instrument and hardware contributions to the ESA Jupiter Icy Moons Explorer (JUICE) mission, in particular an ultraviolet spectrometer, the Particle Environment package as well as hardware contributions to the Radar for Icy Moon Exploration instrument. Meanwhile, the Letter of Agreement between ESA and NASA concerning the provision of payload elements for the JUICE mission has been extended until the entry into force of the MoU.

ESA and NASA have also agreed on the principle of a joint feasibility study for a potential ESA contribution to a future NASA mission focusing on the exploration of Jupiter's moon, Europa.

In the area of robotic exploration, the agreement with NASA on its contribution to ExoMars 2016 has been finalised, and the agreement on sample return – which allows a dialogue to take place on possible future planetary and asteroid sample



NASA Administrator Charles Bolden and ESA Director General Jean-Jacques Dordain take part in a panel discussion on Future Challenges for Global Space Cooperation at the Berlin Air and Space Show in May

return missions – has been extended until 2020. Multilateral discussions on possible forms of collaboration beyond 2020 have also resumed in the frame of the International Mars Exploration Working Group.

The Rosetta mission attracted a great deal of attention in the US, with articles in the press, requests from schools for related material and even enquiries from ordinary citizens.

Cooperation with NASA continued on ISS operations and utilisation. In this context, ESA and NASA focused on the development of a European Service Module for the NASA Orion Multipurpose Crew Vehicle. Both agencies agreed a delivery date for the European Service Module of March 2017. Two ESA astronauts flew on the ISS in 2014: Alexander Gerst and Samantha Cristoforetti.

Russia

ExoMars project development continued to be the most prominent cooperation activity between ESA and Roscosmos. Collaborative work focused on design and development activities as well as cross-participation in review meetings. In the framework of the ESA/Roscosmos working group on post-ExoMars cooperation, a study for a possible future Phobos Sample Return mission was jointly conducted and preliminary discussions held on the subject.

Opportunities for developing cooperation between ESA and Roscosmos in the area of lunar robotic exploration were identified at the programmatic and technical levels.

Cooperation activities continued on board the International Space Station (ISS) on the Expose-2 experiment and the joint ESA/Roscosmos Plasma-Kristall-4 experiment. Maintenance and repair work for the European Robotic Arm continued. The launch of the Russian Multi-Purpose Module (MLM) is planned for 2017.

Soyuz at CSG cooperation continued with the launches of four Soyuz-ST vehicles.

China

Following the two successful workshops for European and Chinese scientists held in Chengdu and then later in Copenhagen, ESA and the Chinese Academy of Science decided to launch a joint call for a

small space science mission. The main selection criteria for the call have been that the proposals were to be prepared and carried out jointly by European and Chinese scientists.

The 10-year celebration of the Dragon programme was held in Chengdu in China in the presence of Vice-Minister Cao Jianlin, who stated the Chinese Ministry of Science and Technology's strong support for the Dragon programme. The programme, given its success, has been structured around thematic themes. This was necessary in order to allow all the scientists wanting to participate to do so.

ESA and China's Manned Space Agency signed a framework agreement in December. The agreement will permit the two sides to cooperate in the area of human spaceflight. The two sides have identified three areas of potential cooperation, namely scientific utilisation of low-Earth Orbit, astronaut activities and infrastructure.



In Beijing on 29 September, Yves-Louis Desnos, ESA's Directorate of Earth Observation, received the People's Republic of China Friendship Award – the country's highest honour for foreign experts – for his work on the Dragon Programme (SAFEA)

Canada

Priority continues to be given to the implementation of the Cooperation Agreement between ESA and Canada. Beyond Canada's participation in ESA programmes taking place under the Cooperation Agreement, ESA pursued cooperation with the Canadian Space Agency in the context of several multilateral programmes and activities such as the ISS, the James Webb Space Telescope, the Committee on Earth Observation Satellites (CEOS) and the International Space Exploration Coordination Group (ISECG).

Japan

The Director General and the President of the Japanese space agency JAXA met at IAC 2014 in Toronto. The main points of discussion concerned the extension of ISS operations beyond 2020, scientific results from the ISS, the development progress and launch date of BepiColombo (the joint ESA/JAXA mission to Mercury) and EarthCARE (the third ESA Earth Explorer mission for which JAXA will provide a Cloud Profile Radiometer). Also during IAC 2014, the Director General met the Deputy Director General of the Research and Development Bureau of the Ministry of Education, Culture, Sports, Science and Technology of Japan principally to discuss ISS extension beyond 2020 and space exploration.

Argentina

As a result of a meeting between an ESA delegation, the Governor of Mendoza Province and Argentina's Minister of Environment and Natural Resources to discuss various issues relating to the third ESA Deep Space Antenna in Malargüe, a decree relating to the legislation on the status of the science park around the antenna was issued. Other meetings took place in Buenos Aires with the Argentinian space agency CONAE and the Argentinian scientific community to discuss the use of the 10% antenna time by Argentina.

Brazil

The Cooperation Agreement between ESA and Brazil has been extended for a further one-year period, and the Natal Tracking Agreement for five years while discussions on a new Tracking Agreement are ongoing.



International Space Station Agency Heads at ESA Headquarters in Paris, on 4 November. From left: Dr Naoki Okumura, JAXA, Jean-Jacques Dordain, Oleg Ostapenko, Roscosmos, Charles Bolden, NASA, Gilles Leclerc, CSA



On 9 May, ESA renewed its agreement with the Gabonese Republic to use the Libreville tracking station in Gabon. Gabon's Deputy Minister for Foreign Affairs Dieudonné Nzengu and ESA Director of Launchers Gaelle Winters signed the agreement in Libreville

Other developments

ESA renewed its agreement with the Gabonese Republic to use the Libreville tracking station in Gabon. This agreement, signed for the first time in 1986, allows ESA to use the Libreville tracking station for Ariane, Vega and Soyuz and reflects the developments in the legal and regulatory framework in telecommunications in Gabon. The Gabonese Agency for Space Studies and Observations (AGEOS) will coordinate the implementation of this agreement. The agreement was signed by the Director of Launchers and the Minister of Foreign Affairs of Gabon and reflects a common interest and strong will to reinforce and boost cooperation particularly in remote sensing for environmental monitoring, data processing, and Earth observation application projects and training.



50 years of European achievements and space milestones on ESA portal

Communications

2014 was marked, in communication terms, by the 50th anniversary of European cooperation in space, and by spectacular missions in all domains of space, with the Rosetta mission as the flagship.

Rosetta changed the way ESA is perceived, both in Europe and internationally. The success of the communication campaign imposes obligations for the future in order to allow ESA to respond to the new and broader expectations generated.

The strategy to communicate with the public on the inspirational aspects and, at the same time, on the actual benefits of space has led to positive results. There was a continued effort to reach beyond ESA's existing audiences, towards the larger community of Europeans for whom, so far, the Agency and space in general was not so visible.

This involved a number of complementary actions, for example, developing communication products adapted for non-expert audiences, and fostering partnerships with figures or organisations that are in daily contact with the public.

The theme of 'Commemorating 50 years of European Cooperation in Space' was used as a common thread through all major events and communication actions of the year – an endeavour that enhanced the cooperation with Member State delegations that were involved in the preparation of Rosetta and other campaigns, as well as the 50-year celebrations themselves.

Besides Rosetta, other highlights included the Venus Express aerobraking campaign, the first high-resolution data on Earth's



The Rosetta and Philae cartoon series served to engage a wider audience on social media

magnetic field provided by the Swarm mission, and the launches of Sentinel-1A and ATV *Georges Lemaître*. Then there were launches of ESA astronauts Alexander Gerst and Samantha Cristoforetti on their Blue Dot and Futura missions, respectively.

Significant media attention was generated by the signature of the SAT-AIS Phase-B2/C/D contract with LuxSpace (LU), making it a highlight of Space Day at the Farnborough Air Show in July, and with the first laser link between Sentinel-1A and Alphasat in November.

Several new initiatives aimed at reaching out to new audiences in the general public were well received: a video message shared by celebrated musician Mike Oldfield on the occasion of the Sentinel-1A launch, and music written specially for ESA, inspired by the Rosetta mission, by world-renowned composer Vangelis, best known for his Academy Award-winning score for the film *Chariots of Fire*.



Key visual of the ESA Council at ministerial level 2014

Scientist and actress Dr Niamh Shaw of Blackrock Castle Observatory in Cork, Ireland, performed her new To SPACE show with ESA input, and a multi-year partnership with UK-based photographer and artist Edgar Martins yielded a book and an exhibition about ESA that has already toured the UK, France and Portugal and has triggered significant worldwide interest from media and public alike.

A partnership with German publisher TeNeues saw the creation and international distribution of an 'art calendar' for 2015, featuring European cities by night as seen from space. This is the first time such a product promoting ESA has been available to the public. In another



ESA Director of Technology and Quality Management, Head of ESTEC, Mr Franco Ongaro at the inauguration of the 'European cooperation in space' exhibition at the European Parliament, 1 April (EPP Group)

important partnership, Italian toymaker Clementoni is distributing a 'Spaceship to Mars' kit for children in major European countries.

In 2014, ESA TV produced more than 60 high-quality videos and covered all ESA launches and events live, in particular in the context of the key moments of the Rosetta mission. These videos were both exploited by broadcasters and fed other ESA channels, such as on YouTube. ESA TV continued the successful co-production of the 'Space' TV magazine with Euronews in nine languages.

As a result, ESA (and the Rosetta mission in particular) was in the news on the most influential, widely viewed TV channels around the world, leading to a cumulative estimated audience of billions. This visibility was shared with national agencies, in particular DLR and CNES.

During 2014, eight international press trips and 22 international press conferences and media opportunities were organised. News mentioning ESA had an 'advertising equivalent value' of over €35.6 million (in 2013, by comparison, €17 million).

ESA's audiences on all its channels experienced a dramatic increase, positioning ESA as a major and influential actor. This significantly higher visibility and further boost to the Agency's reputation certainly contributed to the success of the ESA Council meeting at ministerial at the end of the year.



At the ESOC event for the 50th anniversary of European space cooperation, German Parliamentary State Secretary Brigitte Zypries joined among other VIPs ESA Director General Jean-Jacques Dordain, ESA Director Thomas Reiter, Head of DLR Jan Woerner and ESA's first astronaut Ulf Merbold



ESA astronaut Tim Peake with UK participants of the Mission-X challenge during Futures Day at the Farnborough International Air Show, 18 July

With 15 million visitors and 53 million page views, ESA's web portal registered an increase of 165% and 128% respectively compared with 2013.

Audience growth on ESA social media channels skyrocketed: an increase of 183% on YouTube (to 17 million views), 127% on ESA's Twitter corporate account (topping 2 million followers overall on Twitter), 543% in Google + (reaching 193 000 followers) and 1514% on Flickr (with 113 million photos viewed). ESA's Facebook accounts, meanwhile, had more than 1 million followers.

Education

The ESA Education Programme, targeting education from primary to university level, continued to prove the importance and success of using space as a powerful, inspirational and content-rich context for the teaching and learning of STEM (science, technology, engineering and mathematics).

At primary and secondary school level, through cross-Directorate cooperation and in close collaboration with the relevant stakeholders within ESA Member and Cooperating States, multiple classroom resources were created, and about 200 teacher-training sessions were offered and attended by more than 2500 primary and secondary teachers, reaching out to more than 75 000 pupils. At least 400 university students benefited directly from the unique hands-on projects and EOP capacity-building opportunities offered by the Agency.

In particular, significant progress was made with the delivery of ESA's offer for primary and secondary school education in Member States, with three new European Space Education Resource Offices (ESEROs) established, in Romania, Poland and the Czech Republic. The ESERO project is now active in 12 ESA Member States.

A number of hands-on educational projects were run with notable success with, for example, for school students, the fourth European CanSat competition (Andøya Rocket Range, Norway), and the Spheres-Zero Robotics and Mission X: Train Like an Astronaut projects. For university students, there was the well-established suborbital REXUS/BEXUS projects (led by SNSB and DLR), the micro- and hypergravity experiment projects Drop Your

Thesis! and Spin Your Thesis!, as well as the continuing European Student Earth Orbiter (ESEO) and CubeSat (Fly Your Satellite!) student satellite projects.

Collaboration was strongly reinforced across ESA Directorates, also drawing in the Communication Department and others, for the benefit of ESA education output towards European educational communities. In particular, the HSO Education Team was integrated with the corporate Education Office as of September.

In line with the Partnership Agreement on the long-term development of the ESA Redu Centre signed in 2014, the Education Office set up the basis for three new elements of its programme: the 'e-robotics lab,' a training facility targeting school teachers; the ESA Training and Learning Centre for university students, to offer a portfolio of courses in space disciplines that complement the academic offer and include a dedicated Concurrent Design Facility (CDF); the European CubeSat Education Centre, an AIV and test facility for education projects, particularly university CubeSats which are part of the ESA 'Fly Your Satellite' educational programme.

Knowledge Management (KM)

The seven Knowledge Management pilot activities identified in 2012 (KM Management Office, Lessons Learned, KM Portal and Search Engine, KM Expertise Directory, KM Capture and Hand-Over, KM Awareness Campaign and Competency Management Tool), were tested and applied operationally to capturing and preserving ATV project know-how in the ATVCAP project. The outcome of the pilots formed the basis for the definition of a new ESA corporate Knowledge Management strategy, to be presented in 2015, aimed at harmonising processes and mechanisms for accessing, capturing, storing and sharing information at ESA. The Knowledge Management strategy foresees a flexible central process supported, across Directorates, by dedicated Knowledge Management technology toolkits and experts.

Sustainable Development

The corporate activity of Sustainable Development reinforced its actions and cooperation with both external and internal stakeholders.

The third ESA Sustainable Development report was prepared, with a view to publication in the first half of 2015. The 2013–14 report highlights the importance of the quality of dialogue, in particular with the Member States. Having developed as from 2008, with its internal stakeholders, a reporting scheme associated with concrete goals and actions, the Agency has been able to benchmark with its partners and share good practices in various formats. Reporting on Sustainable Development also allowed ESA to trigger changes in this area.

The report is, as previously, divided into five chapters:

- Governance
- Space contributing to a sustainable society
- Managing our Environment responsibly
- Relations with partners and suppliers
- Human Resources and Social Responsibility.

There is also an annex listing Key Performance Indicators.

This third report is therefore to be issued in the year the Climate Change Conference COP21 will take place in Paris, providing the opportunity for greater visibility for the role of space towards a more sustainable planet and society. Regarding the Paris 2015 Conference, ESA plans to contribute to its preparation from the societal and citizenship perspective. Taking advantage of its hosting in France and in Paris, connections have been made to allow the Agency to benefit from certain channels of information or opportunities to share with actors from civil society, always with a view to promoting space and ESA.

Regarding dialogue with ESA stakeholders, the conclusion of the first set of the roundtable series (2012–13) was summarised in a first dedicated brochure. The idea of regionally focused events was introduced. In 2014, two were held, one in northern Europe and one in southern Europe (Stockholm and Lisbon). In addition to building a platform among stakeholders, the aim remains to involve local partners and companies in the discussion. Both regional roundtables were planned with a similar set-up and topics thereby making it easier to compare regions.

Each of the regional events was attended by about 50 participants from different sectors of activity. The events covered 'space for sustainability' and 'sustainability in the space sector', and typically involved a field trip to visit companies or buildings related to sustainability.

Internally, jointly with the Facility Management and Human Resources Departments, a Health, Safety, Security and Environment (HSSE) week was organised in September at ESA HQ.

In connection with awareness actions, in particular towards the young generation, the winner of the second edition of the 'EISC-ESA Space for Sustainability Award', Sophie Damy, was announced. Her proposal, 'A Satellite-based Optimisation of the European Freight Transport Network' set out a new sustainable approach to meeting transport needs today while preserving air quality for the next generations. This first prize included attending the launch of a Galileo satellite in Kourou.

At communication and partnership level, the following events were organised:

- May: a panel discussion with pioneers in Sustainable Development (with ESA for space) timed to coincide with the 10th anniversary of the Master's in Sustainable Development of France's HEC business school
- June: the 'The Planetworkshops' think tank on Sustainable Development met at the Abbaye de Fontevraud, in an event which included a dedicated evening session with cross-cutting approaches encompassing space and science, society, ethics, philosophy and religion
- September: a TEDxTalk was organised by the HEC business school on 'Avant-garde thinking' and 'More space in our lives: a surprising look at how space has transformed the way we live'.
- October: the NASA/ESA International Workshop on Environment and Alternative Energy took place at NASA's Kennedy Space Center, creating more opportunities to share concerns and solutions but also best practices in the field of sustainability.

Advisory bodies

HISPAC

ESA's High-level Science Policy Advisory Committee (HISPAC) promotes the vision of a 'science-led one-ESA' – an Agency-wide coherent and comprehensive approach to science and technology as a whole. HISPAC's remit is to reflect upon interdisciplinary science and technology themes and ideas and exploit synergies across the full range of ESA activities, in particular those related to long-term developments beyond the timescale of the currently approved programmes.

In 2014, in addition to its regular duties, HISPAC worked on a series of strategic issues.

At the request of the ESA Director General, it examined the scientific criteria for extensions of ESA's scientific missions. HISPAC concluded that, in general, the various ESA Directorates applied well-considered procedures for deciding on mission extensions. Nevertheless, it considered that these procedures were never being tested to the point of having to make really challenging choices. HISPAC considers it likely that in the near future 'crunch points' will be reached and the need to confront much harder decisions will emerge. ESA and its Science Advisory Structure should be ready for such an eventuality.

In this context, HISPAC provided the Director General with a set of recommendations that take as a starting point that a mission should be considered completed when its scientific objectives have been met. The choice to extend the mission lifetime should, in addition, be seen in a broader context also taking into account competition with other ESA scientific missions (either already operational or in planning). Finally, issues surrounding the use of archived data and clear communication on decisions to terminate missions were identified as being of high importance.

Another topic that occupied HISPAC was identifying ways in which ESA might enhance the way it communicates about scientific achievements. To this end, a two-day 'Science Narrative' workshop was organised in October in ESRIN with the participation of scientists, journalists, narrative experts from academia and ESA staff. The conclusion of the participants at the end of the workshop was that employing personalised serial narrative is an attractive tool to complement the existing channels exploited by ESA to communicate the results of its science missions. Following this, HISPAC formulated a series of recommendations to the ESA Director General on how to implement such a technique and make it available to interested scientists and journalists involved with ESA.

In discussion with the International Space Science Institute (ISSI) in Bern, HISPAC examined the results of the 2015 Forum on 'Understanding Gravity' and decided on the organisation of a scientific workshop on 'High Performance Clocks, with special emphasis on Geodesy and Geophysics and applications to other bodies of the Solar System'. This workshop, which is planned to take place in the second half of 2015, will gather some 40–50 scientists who will discuss, over a five-day period, detailed scientific and technological ideas related to this topic. The results will be a set of recommendations to HISPAC and will be published in the ISSI Scientific Reports series.

The Future Technologies Advisory Panel, FTAP, set up in 2011 at the recommendation of HISPAC, completed its second cycle of assessments based on white papers prepared for a selection of the topics from the original longlist suggested by various ESA science working groups and HISPAC itself. In summary, FTAP recommended studying in more detail the following topics: 'High-throughput Science Data Communication', 'Advanced Navigation Techniques', especially extending the range to geosynchronous Earth orbits and highly elliptical orbits as well as in deep space, and 'Highly Accurate and Ultraprecise Pointing'. In addition, FTAP recommended the development of a research roadmap on 'Torpor and Hibernation', which it considers to be a potential game-changer for future human exploration missions.

HISPAC fully endorsed the FTAP recommendations and, moreover, expressed its great appreciation for the FTAP panel and process, and strongly recommended that the ESA Director General integrate this as a permanent element of the ESA Science Advisory Structure.

History Project & Records Management

Activities conducted in the framework of the ESA History Project focused on the preparation of publications marking 50 years of European cooperation in space.

Half a century after the entry into force of the Conventions of its predecessor organisations ESRO and ELDO, ESA convened a series of events to celebrate with the whole European space sector 50 years of successful cooperation. Showcasing the impact and significance of the impressive results obtained, as well as the reputation gained, these celebrations also mobilised projections in our collective future. Keynote speeches and lectures delivered on these occasions, as well as panel discussions and illustrations, are included in the book *Celebrating Half a Century in Space. Recollections of the 2014 Anniversary Events* published by Beauchesne in the 'Miroir des Savants' series. This book, complementing the scholarly history, is the culmination of a year of celebrations and pays tribute to the expertise, passion and values of all those members of the European space community that made the astonishing record of achievements possible. It is also a guide to the past that points the way to what might happen in the future.

Throughout its 50 years, ESA has developed a strong sense of history and chose to support an independent project to document and critically analyse its achievements. The ten volumes dedicated to the European space endeavour published by Beauchesne in the Explorations collection are among the flagship publications authored in that context. The most recent monograph by Prof. John Krige acknowledges ESA's ability to adjust and take advantage of new opportunities while securing its core mission. *Fifty years of European Cooperation in Space. Building on its past, ESA shapes the future* (Paris, Beauchesne, 2014) details fifty years of European collaboration in space from the origins of the programme in 1964 to the rich complexity we see today.



Fifty years of European cooperation in space: Building on its past, ESA shapes the future, the history book published by Les Editions Beauchesne

The *ESA Bulletin* featured a comprehensive timeline article across three issues published during the year (157–159), introduced by an overview by Prof. Krige, who also delivered in 2014 three different lectures to ESA staff and stakeholders. The history of Europe's presence in space is above all a history of political will, of industrial development, and of the consolidation of extensive scientific, technological and managerial competencies to construct a global space power able to compete and collaborate from a position of strength with the United States and USSR/Russia, while adapting to the changing demands of a new geopolitical world order in the 21st century.

The book by Peter Habison, *Weltraum-Land Österreich. Pioniere der Raumfahrt erzählen*, was also published in 2014 (Sachbuch, Seifert Verlag, Wien). In this book we learn how Austria stepped in and secured an important place in European space cooperation. Seventeen national space pioneers tell us about the beginnings in Paris, Kiruna and New York in the 1960s, the developmental leaps forward in Graz and Vienna in the 1970s and 1980s and the dawn of a new era in science.

These descriptions are also based on interviews conducted in the framework of the Oral History of Europe in Space project run by ESA. It includes over 60 transcripts of historical interviews with people having worked within ministries, space agencies, scientific institutes and industry. These interviews, migrated to a new database system in April, were consulted 160 times by external researchers between 1 May and 2 December.

In addition to the abovementioned books, the following research works were based on ESA Historical Archives deposited with the European University Institute in Florence:

- Dr Frankowski, Jagellonian University, *Space Security*;
- Vittorio Felci, University of Rome, *Relations between the USA and Iran during the Jimmy Carter years*;
- Luc Van den Abeelen, *The Hermes Project*;
- Angeles Gonzales, University of Seville, *Spain, Greece, Portugal before the entry into the EEC*;
- Alexander Geppert, Freie Universität Berlin, *The future of the European Space Age*.

On the website of the Historical Archives of the European Union (HAEU), the ESA Fonds was consulted by 3365 external visitors between 1 February (date of migration to the new database system) and 2 December. About 900 files relating to the year 1999 were transferred to the HAEU, rearranged, properly described and made available to researchers in 2014.

Activities of the Records Management Office were also dedicated in 2014 to the preparation, documentation, and follow-up of the key decisions securing ESA's future tabled to ministers in Luxembourg on 2 December, thus carrying forward into the 21st century the great legacy left by previous generations of visionary scientists, engineers and entrepreneurs. The renewed commitment of Member States shows that the idea of Europe is being sustainably served by ESA and by the way it operates.

Security Office

Nine security inspections were performed at ESA establishment and stations in 2014. The security issues at the corporate level were addressed with bilateral meetings with the responsible departments, namely Facility Management, Human Resources, Information Technology, and Procurement.

This dual-pronged approach (pursuing local and global levels) is going to be consolidated in the future years, as documented in the ESA Master Plan 2015–17, projecting the security activities through the next three years, that was approved first by the Security Committee and then by the DG/Director's Committee.

Reaccreditations of the classified installations related to the Galileo Project were performed at ESTEC (satellite testing facilities, the Public Regulated Service laboratory, Project office premises in the T building), at Redu and at ESOC. The project was supported by participating in several security boards and in the accreditation panel.

The Launcher Directorate was supported in the adoption of the Programme Security Instructions, allowing the projects to handle classified information.

The security measures implemented through the Agency strengthen ESA's ability to undertake security-related programmes. A new activity to develop a space mission security policy applicable to future missions of different categories was initiated, with the aim of ensuring that the security risks, in particular the cyber-related ones, are properly addressed in the planning phase.

Notes to the Financial Statements

The Financial Statements presented in this document are the fifth set drawn up under the new basis of accounting and financial system introduced by the Financial Management Reform (Finref) as of 1 January 2010.

The Finref project launched an entirely new operating model in 2010 based on the SAP ERP tool suite, extensively covering the operational management of all the Agency's projects and activities as well as its financial processes. Those processes were defined by new adopted Financial Regulations, which included the requirement to produce annual Financial Statements in accordance with International Accounting Standards for the Public Sector (IPSAS).

Significant progress has been made since 2010 towards improving the presentation of the financial statements. The Audit Commission has acknowledged the efforts performed so far and was of the opinion that the 2012 and 2013 Financial Statements were compliant with the requirements of IPSAS in all material respects.

In 2014, based on the Audit Commission recommendations, further improvements were made to increase the quality of the accounting data and the presentation of the financial statements.

The emphasis was placed on the review of tangible and intangible fixed assets valuation in view of the end of the transition period provided for under IPSAS 17 by 2015 with the objective:

1. to provide assurance on the quality of the accounting data, starting with a review/cleansing exercise for financial year 2014, to be pursued in 2015;
2. to develop accounting guidelines and 12 corporate procedures for internal control (with a gradual deployment planned throughout 2014 and 2015).

The other major achievements of the year 2014 include:

- Recognition of Ariane 5 and Vega development activities costs as intangible fixed assets;
- Recognition of patents as intangible fixed assets;
- Close monitoring of accruals and provisions (systematic and rigorous review of outstanding contractual milestones planned to be delivered at year-end and goods receipt notes);

- Improved quality check and correction of the accounting data;
- Deployment of 'Essentials of accounting training' for Business Unit Controllers, Project Controllers and Project Managers as a standard ESA-wide training;
- Deployment of an integrated timesheet system;
- Capital Commitment Report which is now disclosed in the Financial Statements as per IPSAS 17 requirements.

The main components of ESA Financial Statements presented herewith are the following:

- Statement of financial position,
- Statement of financial performance,
- Statement of changes in net assets/equity,
- Cash flow statement.

Some items, such as fixed assets, subject to first IPSAS recognition, are presented in these statements with only the counterpart in the reserves, but without impact on the statement of financial performance.

At the end of 2014, final fixed assets amount to €3550.7 million and assets under construction to €3281.3 million net values.

The Net Deficit of 2014 amounts to €1581.9 million, after capitalisation of Assets Under Construction and work-in-progress, as compared to €202 million Net Surplus in 2013 (restated).

The deficit of 2014 is explained by the following two exceptional events:

1. Following the signature of the Copernicus Agreement with the European Commission, according to IPSAS the control of the Sentinel developed satellites has been transferred to EC. Accordingly the GMES assets under construction have been written off for an amount of €1421.3 million;
2. The approaching end of the ATV mission has led to the write-down of the work in progress for an amount of €232.6 million.

Other variations between the 2014 and 2013 accounting results include the following:

- A decrease of operating revenues of €225.5 million;
- A decrease of operating expenses of €85.3 million (write-down of assets excluded).

STATEMENT OF FINANCIAL POSITION

	2014 Million €	2013 restated Million €
TOTAL ASSETS	9987.0	10 701.4
Current assets	1810.1	2026.7
Cash and cash equivalents	739.0	843.8
Unrestricted cash and cash equivalents	101.0	351.9
Restricted cash and cash equivalents	638.0	491.9
Receivable from non-exchange transactions	458.8	459.5
Outstanding contributions Member/Participating States	358.3	361.3
Outstanding contributions EC	4.1	3.7
Member States reimbursable taxes	96.4	94.5
Receivable from exchange transactions	405.5	589.1
Other customers	20.1	30.8
Prepayments and other amounts receivable	385.4	558.3
Inventories	206.8	134.3
Non-current assets	8176.9	8674.7
Long-term receivables from non-exchange transactions	2.8	2.8
Long-term receivables from exchange transactions	780.8	263.1
Financial assets	487.4	426.3
Land and buildings	315.1	294.2
Technical plant and equipment	1270.0	1322.6
Spacecraft & space laboratories	1700.6	1272.2
Administrative equipment and vehicles	13.9	13.9
Intangible fixed assets	251.0	267.5
Assets under construction	3355.3	4812.1
TOTAL LIABILITIES	5770.9	4911.6
Current liabilities	2556.5	2015.5
Payables under exchange transactions	973.5	937.8
Current payables	708.9	860.8
Pre-financing from exchange transactions	249.5	63.0
Deferred income	15.1	14.0
Contributions and other payables	1583.0	1077.7
Prepaid contributions and other amounts payable to Member/Participating States	197.6	137.7
Pre-financing from non-exchange transactions	668.7	379.1
Other payables	265.3	125.7
Employee benefits	193.7	179.6
Provisions	257.7	255.6
Non-current liabilities	3214.4	2896.1
Employee benefits	2944.3	2752.2
Pre-financing	270.1	143.9
NET ASSETS	4216.1	5789.8
Reserves	8214.4	7783.3
Amounts to be called from Member States	-3998.3	-1993.5
Employee benefits and other amounts	-2416.4	-2195.5
Net Surplus/(Deficit) for the period	-1581.9	202.0
TOTAL NET ASSETS	4216.1	5789.8

STATEMENT OF FINANCIAL PERFORMANCE

	2014 Million €	2013 restated Million €
Operating revenues	3680.5	3906.0
Contributions from Member/Participating States	3089.1	3236.1
Other contributions and grants	226.5	324.5
Revenues from exchange transactions	108.7	108.1
Social security	16.5	15.4
Pension schemes	32.5	32.0
Other operating income	207.2	189.9
Operating expenses	5279.1	3721.4
Purchases	2647.3	2447.5
External services	997.6	961.8
Staff expenses	280.9	277.3
Social security	29.0	28.2
Pension schemes	300.6	357.1
Other operating expenses	189.3	181.2
Expenses from non-exchange transactions	101.2	100.9
Expenses transferred to assets	-1521.3	-1424.3
Delegated procurement expenses transferred to amounts receivable	-321.0	-423.6
Expenses transferred to work in progress	-305.8	17.4
Expenses transferred to fixed assets under construction	-894.5	-1018.1
Depreciation	569.6	704.6
Write-downs inventories and fixed assets	1672.4	29.4
Provisions	12.5	57.7
NET SURPLUS/(DEFICIT) FROM OPERATING ACTIVITIES	-1598.6	184.6
Financial activities	16.7	20.3
Financial revenue	28.7	23.0
Financial expenses	12.0	2.7
NET SURPLUS/(DEFICIT) FROM ORDINARY ACTIVITIES	-1581.9	204.9
Extraordinary items	0.0	-2.9
Extraordinary expenses	0.0	2.9
NET SURPLUS/(DEFICIT) FOR THE PERIOD	-1581.9	202.0

STATEMENT OF CHANGES IN NET ASSETS/EQUITY

Million €

			1 January, 2013	Errors – 2013 opening balance impact	1 January, 2013 restated	Movements 2013
RESERVES	A		8383.8	4.4	8388.2	–604.9
Member States capital and reserves			5874.5	–61.1	5813.4	–1095.1
Revaluation differences			5819.2	–52.0	5767.2	–1080.8
Reserves			18.3	–9.1	9.2	–9.2
Currency Exchange Gains Reserve			1.0		1.0	4.3
Sundry reserves			11.0		11.0	–5.9
Other reserves			25.0		25.0	–3.5
Social security reserve			17.7		17.7	2.8
Buffer Fund fair value reserve			10.0		10.0	–4.7
Open Sky Technologies Fund fair value reserve			–2.7		–2.7	–1.6
Accumulated Surplus			2509.3	65.5	2574.8	490.2
AMOUNTS TO BE CALLED FROM						
MEMBER STATES	B		–2503.7	–270.3	–2774.0	803.4
Member States pension scheme receivable			–1820.0	–270.3	–2090.3	–105.1
Accumulated Deficit			–0.1		–0.1	
Net Surplus / (Deficit) for the Period			–683.6		–683.6	908.5
TOTAL NET ASSETS	A+B		5880.1	–265.9	5614.2	198.5
		Errors – 2013 impact	Year 2013 restated	Allocation of 2013 surplus/deficit	Movements 2014	Year 2014
RESERVES	A	0.0	7783.3	422.9	8.2	8214.4
Member States capital and reserves		–146.4	4571.9	–640.3	14.8	3946.4
Revaluation differences		–146.4	4540.0	–642.4		3897.6
Reserves			0.0			0.0
Currency Exchange Gains Reserve			5.3			5.3
Sundry reserves			5.1			5.1
Other reserves			21.5	2.1	14.8	38.4
Social security reserve			20.5	2.1		22.6
Buffer Fund fair value reserve			5.3		10.5	15.8
Open Sky Technologies Fund fair value reserve			–4.3		4.3	0.0
Accumulated Surplus		146.4	3211.4	1063.2	–6.6	4268.0
AMOUNTS TO BE CALLED FROM						
MEMBER STATES	B	–22.9	–1993.5	–422.9	–1581.9	–3998.3
Member States pension scheme receivable			–2195.4	–220.9		–2416.3
Accumulated Deficit			–0.1			–0.1
Net Surplus / (Deficit) for the Period		–22.9	202.0	–202.0	–1581.9	–1581.9
TOTAL NET ASSETS	A+B	–22.9	5789.8	0.0	–1573.7	4216.1

STATEMENT OF CASH FLOW

	2014 Million €	2013 restated Million €
Net cash flows	-104.8	280.2
Net cash flows from operating activities	895.5	1333.0
Surplus/(deficit) from ordinary activities	-1581.9	204.9
Non-cash movements	2231.8	802.4
Depreciation	569.6	704.6
Provisions	-10.2	69.1
Write-downs inventories and fixed assets	1672.4	29.4
Other non-cash movements		-0.7
(Increase)/decrease in inventories	-72.5	17.5
Increase/(decrease) in payable to Member/Participating States	59.8	23.8
Increase/(decrease) in pre-financing	602.4	34.7
Increase/(decrease) in other payables	-32.9	-112.3
Increase/(decrease) in provisions for employee benefits	209.7	275.3
(Increase)/decrease in receivables from Member States	1.1	107.6
(Increase)/decrease in amounts to be called from Member States	-235.7	-112.4
(Increase)/decrease in other receivables	-507.3	-115.3
(Increase)/decrease in other current assets	221.0	206.8
Net cash flows from investing activities	-1006.1	-1058.4
Acquisition of fixed assets	-945.0	-1018.4
(Increase)/decrease in financial investments (portfolio)	-61.1	-40.0
Net cash flows from financing activities	5.8	5.6
Deposits to Member States reserves		4.5
Withdrawals from Member States reserves	-2.5	-15.2
Impact prior years' adjustments on reserves	8.3	16.3
Net increase/(decrease) in cash and cash equivalents	-104.8	280.2
Cash and cash equivalents at the beginning of the year	843.8	563.6
Cash and cash equivalents at year-end	739.0	843.8

Independent auditor's report

To the Council of the European Space Agency

Report on the Financial Statements

We have audited the financial statements of the European Space Agency for 2014, issued under document reference ESA/AF(2015)1, submitted 18 May 2015, which comprise the statement of financial position as at 31 December 2014, the statement of financial performance, the statement of changes in net assets and the statement of cash flow for the year then ended, and a summary of significant accounting policies and other explanatory information.

Director General's Responsibility for the Financial Statements

The Director General of the European Space Agency is responsible for the preparation and fair presentation of these financial statements in accordance with International Public Sector Accounting Standards (IPSAS) and in accordance with the requirements of the Financial Regulations, and for such internal control as he determines is necessary to enable the preparation of financial statements that are free from material misstatement, whether due to fraud or error.

Auditor's Responsibility

Our responsibility is to express an opinion on the financial statements, including the aspect of regularity, based on our audit which has been performed in accordance with article 51 and 52 and Annex II of the Financial Regulations and with International Standards on Auditing (ISA). Those Standards require that we comply with ethical requirements and plan and perform the audit to obtain reasonable assurance on whether the financial statements are free from material misstatement.

An audit involves performing procedures to obtain audit evidence on the amounts and disclosures in the financial statements. The procedures selected depend on the auditor's judgement including the assessment of the risks of material misstatement of the financial statements, whether due to fraud or error. In making those risk assessments, the auditor considers internal control relevant to the entity's preparation and fair presentation of the financial statements in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of the entity's internal control. An audit also includes evaluating the appropriateness of accounting policies used and the reasonableness of accounting estimates made by

management, as well as evaluating the overall presentation of the financial statements.

We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our audit opinion.

Opinion

In our opinion, the financial statements present fairly, in all material respects, the financial position of the European Space Agency as at 31 December 2014, and its financial performance and its cash flows for the year then ended in accordance with IPSAS.

Emphasis of Matter in relation to the use of the transitional provisions of IPSAS 17

As disclosed in notes 1.8 and 2.6 to the financial statements, the Agency has decided to apply the fifth and last year of the transitional period provided by IPSAS 17 Property, Plant and Equipment for the full recognition of fixed assets. As a result, the reported statement of financial position as at 31 December 2014 and associated impact on the statement of financial performance cannot be considered as complete or final. In fact, several significant errors and financial control weaknesses were found in the course of the 2014 audit.

Our opinion is not qualified in respect of this matter.

Report on Other Legal and Regulatory Requirements

In our opinion, the transactions of the European Space Agency that have come to our attention as part of our audit of the financial statements, have, in all significant respects, been in accordance with the Financial Regulations and legislative authority of the Agency.

Further details of our audit of the financial statements, including the aspect of regularity, are set out in our Audit Report on the European Space Agency's financial statements 2014, which we have issued in accordance with article 52 and Annex II of the Financial Regulations.

Signed on 16 June 2015

Frank O'Neill

Chairman

Mónica García Sáenz

Chairman Emeritus

Yvan Pedersen

Member

Jean-Luc Vivet

Member

Events & International Exhibitions

Events and exhibitions either organised by, or with support from, ESA Communications.

Space, die Weltraumausstellung <i>Vienna, Austria</i> 23 October 2013 – June 2014	Exhibition at the Technical Museum
Big Bang Science Fair <i>Birmingham, UK</i> 13–16 March	ESA 'Clean room' laboratory demonstration for teachers and schoolchildren
ESRIN Open Day <i>Frascati, Italy</i> 18–21 March	Annual Open Day with educational events
50 Years of European Cooperation in Space <i>Brussels, Belgium</i> March/April	Exhibition for the European Parliament
Copernicus Science Centre 'ESA Day' <i>Warsaw, Poland</i> 11 April	Exhibition marking the accession of Poland as an ESA Member State
SpaceUP Vienna <i>Vienna, Austria</i> 12 April	Space outreach event at the Technical Museum
50 Years of European Cooperation in Space <i>UNESCO, Paris</i> 7 May	High-level anniversary event
Copernicus - Sentinels serving society and the environment <i>Athens, Greece</i> 12–13 May	Exhibition in collaboration with Greek General Secretariat for Research and Technology and the European Commission
50 Years of European Cooperation in Space <i>Deutsches Historisches Museum, Berlin, Germany</i> 19 May	High-level anniversary event
ILA International Airshow <i>Berlin, Germany</i> 20–25 May	Pavilion of European space industry inaugurated by German Chancellor Angela Merkel
Farnborough International Airshow <i>Farnborough, UK</i> 14–20 July	Exhibition and activities with partners UK Space Agency, Roscosmos and UK space industry

Space Summer School <i>Universidad Politécnica de Madrid, Spain</i> July	ESA support included astronaut Pedro Duque as coordinator and ESA lecturers
50 Years of European Cooperation in Space <i>Brussels, Belgium</i> March/April	Exhibition for the European Parliament
50 Years of European Cooperation in Space <i>Conference Centre of Geneva, Switzerland</i> 12 September	High-level anniversary event
Bucharest Science Festival <i>Bucharest, Romania</i> 25 September	ESA Earth Observation from Space exhibition
Researchers' Night, ESRIN/Demokritos Centre <i>Frascati, Italy</i> <i>Athens, Greece</i> 26 September	
My Planet from Space: Fragility and Beauty <i>Palazzo delle Esposizioni, Rome</i> 30 September – 2 November	Exhibition inaugurated by the ESA Director General with EU, ASI and NASA representatives
ESTEC Open Day <i>Noordwijk, Netherlands</i> 5 October	Large-scale annual open day organised during World Space Week
The Economic Impacts of Space Technologies <i>OECD Conference Centre, Paris</i> 23 October	High-Level OECD Symposium
'SPACE FOR OUR FUTURE' <i>Brussels, Belgium</i> 28 October – 14 November	Exhibition at the European Commission inaugurated by José Manuel Barroso
Graduate Engineer & Science Show <i>Derby, UK</i> 29 October	Educational/careers event organised by world's largest engineering and science organisations
ESA Council meeting at ministerial level <i>NCC Kirchberg, Luxembourg</i> 2 December	
Comunicando, de las señales de humo a los satélites <i>Valencia, Spain</i> Mid-December	Science Museum exhibition on the history of telecommunications

Agreements signed in 2014

Exchange of Letters for the Extension of the Agreement between ESA and DLR concerning the installation and utilisation of certain assets located at Lampoldshausen signed in Paris on 23 December 2014 and in Cologne on 12 January 2015 by ESA Director General Jean-Jacques Dordain and Johann-Dietrich Woerner, Chairman of the DLR Executive Board and Hansjörg Dittus, Member of the DLR Executive Board.
(ESA/LEG/234, add.11)

Amendment No. 4 to the Delegation Agreement between ESA and the European Union on the further implementation of the European Satellite Navigation Programme EGNOS signed in Paris on 16 July 2014 by ESA Director General Jean-Jacques Dordain and Daniel Calleja Crespo, Director General, DG Enterprise and Industry for the European Commission.
(ESA/LEG/355, add.4)

Amendment to the Implementing Arrangement between Eumetsat and ESA concerning Cooperation on GMES Sentinel-3 signed at Darmstadt on 19 December 2013 and at Frascati on 20 January 2014 by Director General of Eumetsat Alain Ratier and Volker Liebig, ESA Director of Earth Observation Programmes.
(ESA/LEG/422)

Implementing Arrangement between ESA and the European Defence Agency (EDA) concerning their cooperation on a Demonstration Project on Satellite-based secure Command and Control Data for RPAS Applications signed in Paris on 24 January 2014 and in Brussels on 5 February 2014 by ESA Director of Telecommunications and Integrated Applications Magali Vaissière and Claude-France Arnould, EDA Chief Executive.
(ESA/LEG/425)

Agreement between ESA and the Chinese Academy of Sciences (CAS) concerning cooperation on scientific space activities signed in Paris and in Beijing on 14 February 2014 by ESA Director General Jean-Jacques Dordain and Chunli Bai, CAS President.
(ESA/LEG/426)

Implementing Arrangement between ESA and the National Space Science Center (NSSC) of the Chinese Academy of Sciences concerning the Flight of the Soret Coefficient in Crude Oil Experiment on the SJ-10 Mission signed in Paris on 4 March 2014 by ESA Director General Jean-Jacques Dordain and in Beijing on 11 March 2014 by Wu Ji, NSSC Director General.
(ESA/LEG/427)

Memorandum of Understanding between ESA and the International Space University (ISU) signed in Paris on 7 May 2014 by ESA Director General Jean-Jacques Dordain and Walter Peeters, President of the ISU.
(ESA/LEG/428)

Memorandum of Understanding between ESA and NASA concerning the 2016 ExoMars Mission signed in Paris on 14 April 2014 and in Washington DC on 29 April 2014 by ESA Director General Jean-Jacques Dordain and Charles F. Bolden Jr., NASA Administrator.
(ESA/LEG/429)

Agreement between ESA and the Government of the Gabonese Republic on the setting up and use of a downrange station in Libreville (Gabon) signed in Libreville on 9 May 2014 by ESA Director of Launchers Gaelle Winters and Dieudonné Nzengue, Minister Delegate for the Government of the Gabonese Republic.
(ESA/LEG/430)

Delegation Agreement between ESA and the European Union, represented by the European Commission, on the Deployment Phase of the European Satellite Radionavigation Programme Galileo signed in Paris on 16 July 2014 by ESA Director General Jean-Jacques Dordain and Daniel Calleja Crespo, Director General, DG Enterprise and Industry for the European Commission.
(ESA/LEG/431)

European Cooperating State Agreement between ESA and the Government of the Republic of Lithuania signed in Vilnius on 7 October 2014 by ESA Director General Jean-Jacques Dordain and Evaldas Gustas, Minister of Economy of the Republic of Lithuania.
(ESA/LEG/432)

Cooperation Agreement between ESA and the European Organization for Nuclear Research (CERN) signed on 28 March 2014 in Geneva, Switzerland, by ESA Director General Jean-Jacques Dordain and Rolf Heuer, Director General of CERN.
(ESA/LEG/433)

Agreement between ESA and the US Department of Defense for the Sharing of On-Orbit Space Objects Data and the Provision of Space Situational Awareness Services and Information signed on 30 October 2014 in Washington DC, by ESA Director General Jean-Jacques Dordain and David D. Thompson, Major General USAF, Director, Plans and Policy, US Strategic Command.
(ESA/LEG/434)

Agreement between the European Union, represented by the European Commission, and ESA on the Implementation of the Copernicus Programme including the Transfer of Ownership of the Sentinels (Copernicus Agreement) signed in Brussels, on 28 October 2014 by ESA Director General Jean-Jacques Dordain and Daniel Calleja-Crespo, Director General, DG Enterprise and Industry for the European Commission.
(ESA/LEG/435)

Multilateral Agreement between ESA, ASI, CNES, Ministerio de Industria, Energía y Turismo, DLR, SNSB and SSO concerning the CHEOPS Mission signed at Villafranca on 19 November 2014 by ESA Director of Science and Robotic Exploration Alvaro Giménez Cañete, Andreas Werthmueller for the SSO, Enrico Flamini for ASI, Christian Sirmain for CNES, Román Fernández Del Pilar for the Ministerio de Industria Energía y Turismo, Thomas Galinski for DLR, and Per Magnusson for the SNSB.
(ESA/LEG/436)

Memorandum of Understanding among ESA, CNES, Airbus Defence & Space, Safran and Airbus Safran Launchers signed on 18 December 2014 by ESA Director General Jean-Jacques Dordain, Jean-Yves Le Gall, President of CNES, Francois Auque, President of Airbus Defence & Space, Jean-Lin Fournereaux, Senior Vice-President Space at Safran and Alain Charmeau, Chief Executive Officer of Airbus Safran Launchers.
(ESA/LEG/437)

Agreement between ESA and the China Manned Space Agency concerning Cooperation on Human Spaceflight Activities signed in Beijing on 10 December 2014 by ESA Director General Jean-Jacques Dordain and Wang Zhaoyao, Director General of the China Manned Space Agency.
(ESA/LEG/438)

Multilateral Agreement between ESA, ASI, CNES, DLR, the Foundation for Science and Technology, Ministerio de Economía y Competitividad, the National Space Institute at the Technical University of Denmark, Nederlandse Onderzoekshoof voor Astronomie, the Norwegian Space Centre, the Romanian Space Agency, the UK Space Agency and the University of Helsinki concerning the Euclid mission signed on 28 November 2013 by ESA, CNES, DLR e.V., Ministerio de Economía y Competitividad, the National Space Institute, the Nederlandse Onderzoekshoof voor Astronomie, NSC, ROSA, UK Space Agency, by the University of Helsinki, on 19 November 2014 by the ASI, and on 19 December 2014 by the Foundation for Science and Technology.
(ESA/LEG/439)

Multilateral Agreement between ESA, ASI, CNES, DLR e.V., the Spanish Ministerio de Economía y Competitividad, the Nederlandse Onderzoekshoof voor Astronomie, the UK Space Agency and the Swedish National Space Board concerning the Gaia Data Processing signed on 28 November 2013 by the ESA, CNES, DLR e.V., the Spanish Ministerio de Economía y Competitividad, the Nederlandse Onderzoekshoof voor Astronomie, UK Space Agency, the Swedish National Space Board and on 19 November 2014 by ASI.
(ESA/LEG/440)

ESA patent applications filed in 2014

As a result of a study conducted in 2013, management of the portfolio of ESA patents was transferred from the Legal Department to the Technology Transfer Programme.

Overall, the final report confirmed the high importance for the Agency (its Member States and industry) to protect, maintain and defend its strategic inventions with a view to attaining the purposes defined in the Convention. A number

of recommendations were made to optimise the process and reduce costs:

- the Patents Group's mandate should be amended to simplify decision-making and provide more flexibility in the management of the portfolio;
- the Patents Group should discuss Intellectual Property policy aspects more than had been done previously;
- the patent function (technical assessments/utilisation and management of portfolio) should be consolidated in a single area;
- the proceeds from royalties should be utilised to recover part of the costs.

As the large majority of the patents originates from the Directorate of Technical and Quality Management, the Directorate agreed to cover the funding of the patents portfolio from 2013 onwards and take over responsibility for both the technical and the administrative management. Responsibility for the signature of legal agreements linked to patents (for instance assignments of rights, non-disclosure agreements, licenses and so on) was also delegated to the Directorate in 2014.

The review process was split in two, with inventions classified as over or under five years old.

There were 82 peer reviews organised for inventions over five years old. Of those, 51 were fully or partially abandoned, resulting in 140 patents being abandoned.

The second type of review put in place was with the inventor(s) of inventions less than five years old. The objectives of these meetings are slightly different as the aim of the Secretariat is to inform inventors of the patenting process and the principal steps/key dates ahead of them, as well as to set up an action plan with them to make sure the maturity of the invention is increased and that the invention is pushed to industry.

There were 16 action plans put in place and resulted in three inventions being abandoned. The size of the portfolio at the end of 2014 was 136 inventions and 456 patents.

The overall savings made in the year amounted to €160 000. This covers only the anticipated costs for those inventions in 2014.

Chairs of Council & subordinate bodies (as of 1 July 2014)

Council

	Chair	Harald Posch (AT)
	Vice-Chair	Bo Andersen (NO)
		Jean-Yves Le Gall (FR)
Science Programme Committee		Roel Gathier (NL)
Administrative and Finance Committee		Helene Körling (SE)
Industrial Policy Committee		Cristina Falvella (IT)
International Relations Committee		Michal Fridrich (CZ)
Security Committee		Kai Knape (FI)
Human Spaceflight, Microgravity and Exploration Programme Board		Pilar Roman Fernandez (ES)
Launchers Programme Board		Daniel Neuenschwander (CH)
Earth Observation Programme Board		Jacques Nijskens (BE)
Programme Board on Satellite Navigation		Catherine Mealing-Jones (GB)
Joint Board on Communication Satellite Programmes		Catherine Mealing-Jones (GB)
Space Situational Awareness Programme Board		Klaus Steinberg (DE)

Statutory bodies

Appeals Board

	Chair	Michael Groepper (DE)
	Vice-Chair	Spyridon Flogaitis (GR)
	Members	Sarah Christie (GB)
		Pierluigi di Palma (IT)
		Fabien Raynaud (FR)

Audit Commission

	Chair	Mónica García Sáenz (ES)	
	Chair emeritus	Jari Sanaskoski	(until September 2014)
	Commissioners	Frank O'Neill (IE)	
		Hans Andersen (DK)	
		Jean-Luc Vivet (FR)	(from October 2014)

Staff Association Central Committee

	Chair	Lothar Winzer (ESA)
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National delegations to Council in 2014

AUSTRIA

Klaus Pseiner
Harald Posch (Council Chair as of 1 July 2014)
Andrea Kleinsasser

BELGIUM

Eric Beka
Jacques Nijskens
Agnès Grandjean

CZECH REPUBLIC

Karel Dobeš
Václav Kobera

DENMARK

Peter Sloth
Gorm Kofoed Petersen

FINLAND

Kari Tilli
Kimmo Kanto
Marjaana Aarnikka

FRANCE

Jean-Yves Le Gall
Joël Barre
Christophe Venet

GERMANY

Johann-Dietrich Wörner
(Council Chair until 30 June 2014)
Gerd-Achim Gruppe
Rolf Densing

GREECE

Christos Vasilakos
Xenophon Tsilimparis
Afrodite Patroni

IRELAND

Michael Davitt
Tony McDonald

ITALY

Roberto Battiston
Augusto Cramarossa
Laura Aghilarre

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Marc Serres
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Jasper Wesseling
Frits von Meijenfeldt
Linda van Duivenbode

NORWAY

Bo Andersen
Geir Hovmork

POLAND

Otylia Trzaskalska-Stroińska
Anna Nałęcz-Kobierzycka

PORTUGAL

Miguel Seabra
Luis Serina

ROMANIA

Marius-Ioan Piso
Oana Maria Neagu

SPAIN

Begoña Cristeto
Elisa Robles-Fraga
Juan Carlos Cortés Pulido

SWEDEN

Olle Norberg
Thorwald Larsson
Maria Nilsson

SWITZERLAND

Daniel Neuenschwander
Raphaël von Roten
Urs Frei

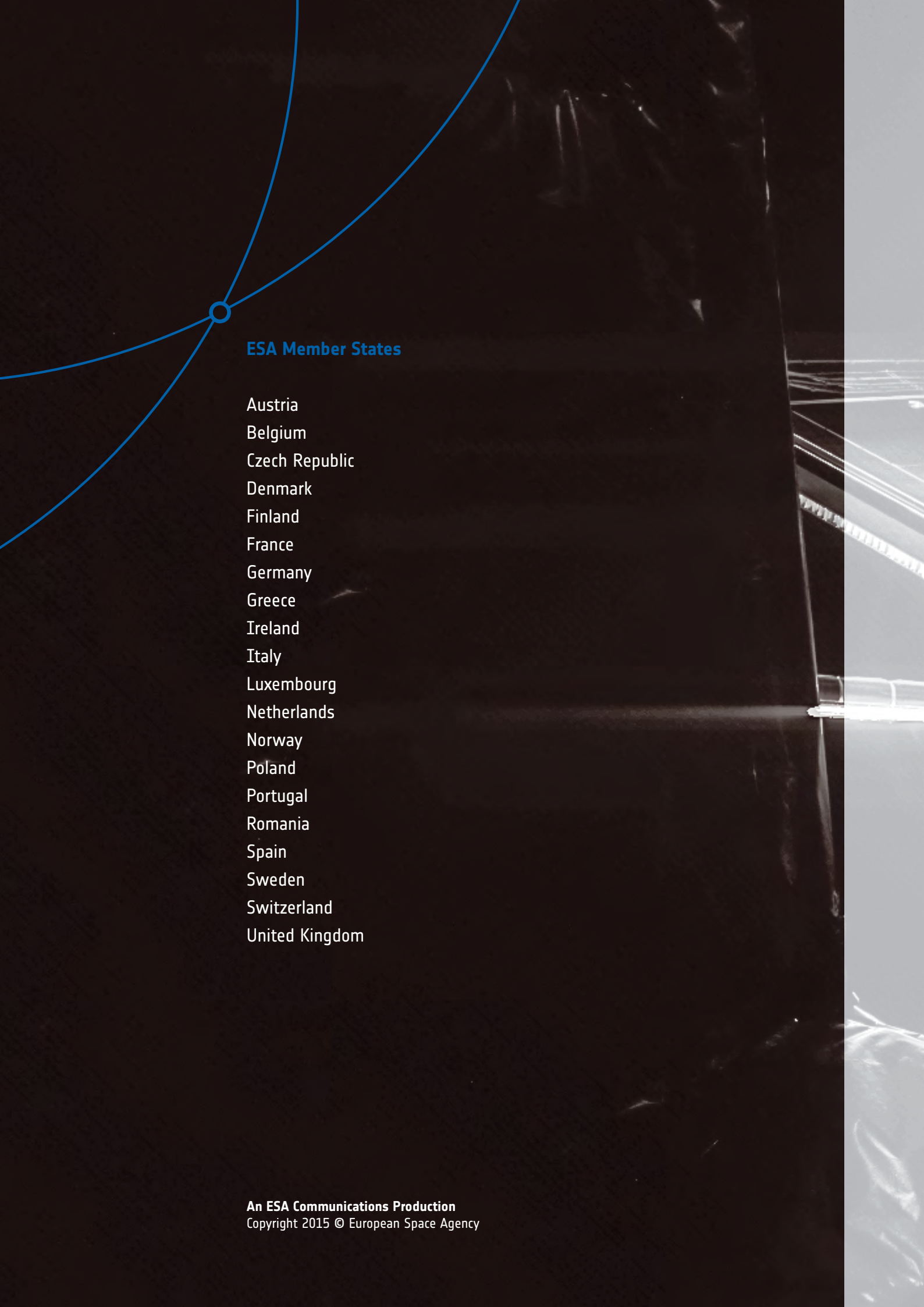
UNITED KINGDOM

David Parker
Alice Bunn

CANADA

Walter Natynczyk
Jean-Claude Piedboeuf
Jean-Pierre Arseneault

[illegible]

The background of the page is a dark, high-contrast image of a satellite or space station structure against a black sky. A bright light source on the right creates a strong lens flare and illuminates parts of the structure. Two thin, light blue curved lines intersect at a small circle on the left side of the page. The text is white, except for the section header which is light blue.

ESA Member States

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